arXiv:2506.01393v4 [cs.LG] 27 Oct 2026

Improved Regret Bounds for Gaussian Process Upper Confidence Bound in Bayesian Optimization

Shogo Iwazaki
LY Corporation
Tokyo, Japan
siwazaki@lycorp.co.jp

Abstract

This paper addresses the Bayesian optimization problem (also referred to as the Bayesian setting of the Gaussian process bandit), where the learner seeks to minimize the regret under a function drawn from a known Gaussian process (GP). Under a Matérn kernel with a certain degree of smoothness, we show that the Gaussian process upper confidence bound (GP-UCB) algorithm achieves $\widetilde{O}(\sqrt{T})$ cumulative regret with high probability. Furthermore, our analysis yields $O(\sqrt{T \ln^2 T})$ regret under a squared exponential kernel. These results fill the gap between the existing regret upper bound for GP-UCB and the best-known bound provided by Scarlett [47]. The key idea in our proof is to capture the concentration behavior of the input sequence realized by GP-UCB, enabling a more refined analysis of the GP's information gain.

1 Introduction

We study the Bayesian optimization (BO) problem, where the learner seeks to minimize the regret under a random function drawn from a known Gaussian process (GP) [19, 20]. Throughout this paper, we focus on the GP-UCB algorithm [53], which combines the posterior distribution of GP with the optimism principle. Due to its simple algorithmic structure and general theoretical framework provided by Srinivas et al. [53], GP-UCB has played an important role in the advancement of the BO field. On the other hand, our theoretical understanding of the performance of GP-UCB has not been improved from [53] in the Bayesian setting, while its frequentist counterpart is studied in several existing works [12, 63]. Specifically, the current regret upper bound for GP-UCB, as provided by Srinivas et al. [53], is known to be worse than that of the algorithm in [47], which achieves state-of-the-art $O(\sqrt{T \ln T})$ cumulative regret. Then, the natural question is whether there is further room for improvement in the existing regret upper bound of GP-UCB. This paper provides an affirmative answer to this question by showing that GP-UCB achieves $\widetilde{O}(\sqrt{T})$ regret with high probability.

Contribution. We summarize our contributions as follows.

- We show that the GP-UCB proposed by Srinivas et al. [53] achieves $\widetilde{O}(\sqrt{T})$ regret with high probability under a Matérn kernel with a certain degree of smoothness (the precise condition is given in Theorem 3). Here, $\widetilde{O}(\cdot)$ is the order notation that hides polylogarithmic dependence. This result is comparable to state-of-the-art $O(\sqrt{T \ln T})$ regret provided by Scarlett [47] up to a polylogarithmic factor and strictly improves upon the existing $\widetilde{O}(T^{\frac{\nu+d}{2\nu+d}})$ upper bound of GP-UCB [53, 60]. Here, d and ν denote the dimension of the input domain and smoothness parameter, respectively.

39th Conference on Neural Information Processing Systems (NeurIPS 2025).

- Furthermore, for a squared exponential kernel, we establish $O\left(\sqrt{T \ln^2 T}\right)$ cumulative regret of GP-UCB. This improves the existing $O\left(\sqrt{T \ln^{d+2} T}\right)$ upper bound provided by Srinivas et al. [53] for any $d \geq 1$.
- The key idea behind our analysis is to refine the existing information gain bounds by leveraging algorithm-dependent behavior and sample path properties of the GP. We also discuss the applicability of this technique to other algorithms and settings in Section 4.

1.1 Related Work

BO has been extensively studied in the past few decades. Several BO algorithms select queries by maximizing an acquisition function defined through the GP posterior, including expected improvement [38], knowledge gradient [18], and the entropy-based algorithms [25]. The theoretical aspect of BO has also been actively studied through the lens of the bandit algorithms, such as GP-UCB [53], Thompson sampling [44], and information directed sampling [45]. In contrast to the noisy observation setting, which these algorithms focus on, algorithms for the noise-free setting form a separate line of research [15, 24, 33]. Extensions of these algorithms to more advanced settings have also been well-studied, e.g., contextual [35], parallel observation [16], high-dimensional [30], time-varying [7], and multi-fidelity settings [31]. Unlike the Bayesian assumption on the objective function adopted in this paper, existing works also extensively study the frequentist assumption of the function, which is also referred to as the frequentist setting of BO or GP bandits [8, 10, 12, 27, 36, 46, 48, 58, 61].

The work of Scarlett [47] is closely related to this paper. It proposes a successive elimination-based algorithm and establishes an $O(\sqrt{T \ln T})$ upper bound and an $\Omega(\sqrt{T})$ lower bound of the cumulative regret for a one-dimensional BO problem. The fundamental theoretical assumptions and the high-level idea of our analysis are built on the proof provided by Scarlett [47]. Following [47], Wang et al. [62] also prove similar regret guarantees under the one-dimensional Brownian motion.

In addition to [47], some parts of our analysis are inspired by the technique leveraged in [9, 29]. Firstly, Cai et al. [9] study the GP-UCB algorithm through a relaxed version of regret, which is called *lenient regret*. In our analysis, the cumulative regret is decomposed into the lenient regret-based term, and we leverage their technique to analyze it. Secondly, Janz et al. [29] proposed the input partitioning-based algorithm for obtaining a superior regret in the frequentist setting. Roughly speaking, the high-level idea of their analysis is based on the fact that tighter information gain bounds can be obtained within a properly shrinking partition of the input. The key idea provided in Section 3.1 is motivated by this fact, while our analysis itself is substantially different from that in [29].

2 Preliminaries

Let $f : \mathcal{X} \to \mathbb{R}$ be a black-box objective function whose input domain $\mathcal{X}$ is $\mathcal{X} \coloneqq [0, r]^d$ with some $r > 0$. At each step $t \in \mathbb{N}_+$, the learner chooses a query point $\boldsymbol{x}_t \in \mathcal{X}$, and then receives a noisy observation $y_t = f(\boldsymbol{x}_t) + \epsilon_t$. Here, ϵ_t is a mean-zero noise random variable. We consider a Bayesian setting, where the objective function f and the noise sequence (ϵ_t) are drawn from a known zero-mean Gaussian process (GP) and a Gaussian distribution, respectively. We formally describe it using the following assumptions.

Assumption 1. *Let $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ be the known positive definite kernel with $\forall \boldsymbol{x} \in \mathcal{X}, k(\boldsymbol{x}, \boldsymbol{x}) \leq 1$. Then, assume $f \sim \mathcal{GP}(0, k)$, where $\mathcal{GP}(0, k)$ denotes the mean-zero GP characterized by the covariance function k.*

Assumption 2. *The noise sequence $(\epsilon_t)_{t \in \mathbb{N}_+}$ is mutually independent. Furthermore, assume $\epsilon_t \sim \mathcal{N}(0, \sigma^2)$, where $\sigma > 0$ is the known constant. Here, $\mathcal{N}(\mu, \sigma^2)$ denotes the Gaussian distribution with mean μ and variance σ^2.*

These are standard sets of assumptions in the existing theory of BO [44, 53]. Specifically, in Assumption 1, we focus on the following squared exponential (SE) kernel k_{SE} and Matérn kernel

Algorithm 1 Gaussian process upper confidence bound

Require: Kernel k, confidence width parameters $(\beta_t^{1/2})_{t\in\mathbb{N}_+}$.
1: **for** $t = 1, 2, \ldots$ **do**
2: $\quad \boldsymbol{x}_t \leftarrow \arg\max_{\boldsymbol{x}\in\mathcal{X}} \mu(\boldsymbol{x}; \mathbf{X}_{t-1}, \boldsymbol{y}_{t-1}) + \beta_t^{1/2}\sigma(\boldsymbol{x}; \mathbf{X}_{t-1})$.
3: $\quad$ Observe y_t and update the posterior mean and variance.
4: **end for**

$k_{\text{Matérn}}$:

$$k_{\text{SE}}(\boldsymbol{x}, \widetilde{\boldsymbol{x}}) = \exp\left(-\frac{\|\boldsymbol{x}-\widetilde{\boldsymbol{x}}\|_2^2}{2\ell^2}\right), \quad k_{\text{Matérn}}(\boldsymbol{x}, \widetilde{\boldsymbol{x}}) = \frac{2^{1-\nu}}{\Gamma(\nu)}\left(\frac{\sqrt{2\nu}\|\boldsymbol{x}-\widetilde{\boldsymbol{x}}\|_2}{\ell}\right)^{\nu} J_\nu\left(\frac{\sqrt{2\nu}\|\boldsymbol{x}-\widetilde{\boldsymbol{x}}\|_2}{\ell}\right), \tag{1}$$

where $\ell > 0$ and $\nu > 0$ are the known lengthscale and smoothness parameters, respectively. In addition, $J_\nu(\cdot)$ and $\Gamma(\cdot)$ denote the modified Bessel function of the second kind and the Gamma function, respectively. Under Assumptions 1 and 2, the learner can infer the function f through the GP posterior distribution. Let $\mathcal{H}_t \coloneqq (\boldsymbol{x}_i, y_i)_{i\le t}$ be the history that the learner obtained up to the end of step t. Given $\mathcal{H}_t$, the posterior distribution of f is again GP, whose posterior mean and variance are respectively defined as

$$\mu(\boldsymbol{x}; \mathbf{X}_t, \boldsymbol{y}_t) = \boldsymbol{k}(\mathbf{X}_t, \boldsymbol{x})^\top (\mathbf{K}(\mathbf{X}_t, \mathbf{X}_t) + \sigma^2 \boldsymbol{I}_t)^{-1} \boldsymbol{y}_t, \tag{2}$$

$$\sigma^2(\boldsymbol{x}; \mathbf{X}_t) = k(\boldsymbol{x}, \boldsymbol{x}) - \boldsymbol{k}(\mathbf{X}_t, \boldsymbol{x})^\top (\mathbf{K}(\mathbf{X}_t, \mathbf{X}_t) + \sigma^2 \boldsymbol{I}_t)^{-1} \boldsymbol{k}(\mathbf{X}_t, \boldsymbol{x}), \tag{3}$$

where $\boldsymbol{k}(\mathbf{X}_t, \boldsymbol{x}) \coloneqq (k(\boldsymbol{x}_i, \boldsymbol{x}))_{i=1}^t$ and $\boldsymbol{y}_t \coloneqq (y_1, \ldots, y_t)^\top$ are the t-dimensional kernel and output vectors, respectively. Here, we set $\mathbf{X}_t = (\boldsymbol{x}_1, \ldots, \boldsymbol{x}_t)$. Furthermore, $\mathbf{K}(\mathbf{X}_t, \mathbf{X}_t) \coloneqq [k(\boldsymbol{x}, \widetilde{\boldsymbol{x}})]_{\boldsymbol{x}, \widetilde{\boldsymbol{x}} \in \mathbf{X}_t}$ and $\boldsymbol{I}_t$ denote the $t \times t$ Gram matrix and the $t \times t$ identity matrix, respectively.

Learner's goal. For a time horizon $T \in \mathbb{N}_+$, the learner's goal is to minimize the cumulative regret $R_T \coloneqq \sum_{t\in[T]} f(\boldsymbol{x}^*) - f(\boldsymbol{x}_t)$, where $\boldsymbol{x}^* \in \operatorname{argmax}_{\boldsymbol{x}\in\mathcal{X}} f(\boldsymbol{x})$ and $[T] = \{1, \ldots, T\}$.

Maximum information gain. To quantify the regret, the existing theory utilizes the following information-theoretic quantity $\gamma_T(\mathcal{X})$ arising from GP:

$$\gamma_T(\mathcal{X}) = \sup_{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_T \in \mathcal{X}} I(\mathbf{X}_T), \text{ where } I(\mathbf{X}_T) = \frac{1}{2}\ln\det(\boldsymbol{I}_T + \sigma^{-2}\mathbf{K}(\mathbf{X}_T, \mathbf{X}_T)). \tag{4}$$

The quantity $\gamma_T(\mathcal{X})$ is referred to as the *maximum information gain* (MIG) over $\mathcal{X}$ [53], since $I(\mathbf{X}_T)$ equals the mutual information between the function values $(f(\boldsymbol{x}_t))_{t\in[T]}$ and the outputs $(y_t)_{t\in[T]}$ under Assumptions 1 and 2, and the input sequence $\mathbf{X}_t = (\boldsymbol{x}_1, \ldots, \boldsymbol{x}_t)$. MIG plays a vital role in the theoretical analysis of BO, and its growth rate is analyzed in several commonly used kernels. For example, $\gamma_T(\mathcal{X}) = O(\ln^{d+1} T)$ as $T \to \infty$ under $k = k_{\text{SE}}$ [53]. For notational convenience, we also define $\gamma_i(\mathcal{X}) = \gamma_{\lceil i \rceil}(\mathcal{X})$ for any non-integer $i > 0$.

Probabilistic property of GP sample path. The existing theory of GP-UCB under the Bayesian setting utilizes the regularity conditions of the realized sample path of the GP. We summarize the existing known properties of the GP sample path in the following lemmas.

Lemma 1 (Lipschitz condition of sample path, e.g., [53])**.** *Suppose $k = k_{\text{SE}}$ or $k = k_{\text{Matérn}}$ with $\nu > 2$. Assume Assumption 1. Then, there exist the constants $a, b > 0$ such that*

$$\forall L > 0, \ \mathbb{P}\left(\forall \boldsymbol{x}, \widetilde{\boldsymbol{x}} \in \mathcal{X}, \ |f(\boldsymbol{x}) - f(\widetilde{\boldsymbol{x}})| \le L\|\boldsymbol{x} - \widetilde{\boldsymbol{x}}\|_1\right) \ge 1 - da\exp\left(-\frac{L^2}{b^2}\right). \tag{5}$$

Lemma 2 (Sample path condition for the global maximizer, e.g., [14, 15, 47])**.** *Suppose $k = k_{\text{SE}}$ or $k = k_{\text{Matérn}}$ with $\nu > 2$. Assume Assumption 1. Then, for any $\delta_{\text{GP}} \in (0, 1)$, there exist the strictly positive constants $c_{\text{gap}}, c_{\text{sup}}, c_{\text{quad}}, \rho_{\text{quad}} > 0$ such that the following statements simultaneously hold with probability at least $1 - \delta_{\text{GP}}$:*

1. *The function f has a unique maximizer $\boldsymbol{x}^* \in \mathcal{X}$ such that $f(\boldsymbol{x}^*) > f(\widetilde{\boldsymbol{x}}^*) + c_{\text{gap}}$ holds for any local (but not global) maximizer $\widetilde{\boldsymbol{x}}^* \in \mathcal{X}$ of f.*

2. *The sup-norm of the sample path is bounded as* $\|f\|_\infty \leq c_{\sup}$.

3. *The function* f *satisfies* $\forall \boldsymbol{x} \in \mathcal{B}_2(\rho_{\text{quad}}; \boldsymbol{x}^*)$, $f(\boldsymbol{x}^*) - c_{\text{quad}}\|\boldsymbol{x}^* - \boldsymbol{x}\|_2^2 \geq f(\boldsymbol{x})$, *where* $\mathcal{B}_2(\rho_{\text{quad}}; \boldsymbol{x}^*) \coloneqq \{\boldsymbol{x} \in \mathcal{X} \mid \|\boldsymbol{x}^* - \boldsymbol{x}\|_2 \leq \rho_{\text{quad}}\}$ *is the L2-ball on* $\mathcal{X}$, *whose radius and center are* ρ_{quad} *and* $\boldsymbol{x}^*$, *respectively*[1]

Lemma 1 states that the sample path f of GP is a Lipschitz function with high probability. This property is leveraged in the theory of GP-UCB to control the discretization error arising from the confidence bound construction in the continuous input domain. As described in [53], Lemma 1 is a direct consequence of Theorem 5 in [22] provided that the kernel has the required fourth-order mixed partial derivatives, as is the case for $k = k_{\text{SE}}$ and $k = k_{\text{Matérn}}$ with $\nu > 2$[2]. Lemma 2 specifies the regularity condition of f related to the maximizer $\boldsymbol{x}^*$. Here, property 1 follows from the fact that the GP-sample path has a unique maximizer almost surely under k_{SE} and $k_{\text{Matérn}}$ [e.g., Lemma 2.6 in 34]. Property 2 follows from, e.g., the compactness of $\mathcal{X}$ and the almost-sure continuity of the sample path under k_{SE} and $k_{\text{Matérn}}$. Property 3 also holds automatically under $k = k_{\text{SE}}$ and $k = k_{\text{Matérn}}$ with $\nu > 2$ and is used in existing works. See Theorem 5 in [14], Assumption 3 in [47], and the discussions provided by them for further details. Note that the properties in Lemma 2 are not used in the existing proof of GP-UCB in [53]. As described in the next section, we analyze the realized input sequence $\mathbf{X}_T$ of GP-UCB by relating it to conditions in Lemma 2.

Summary of existing analysis of GP-UCB. We briefly summarize the existing analysis of GP-UCB (Algorithm 1) provided by Srinivas et al. [53]. Based on Assumptions 1 and 2, we can construct the high-probability confidence bound of the underlying function value $f(\boldsymbol{x})$ for each $\boldsymbol{x}$ and $t \in \mathbb{N}_+$ through the posterior distribution of $f(\boldsymbol{x})$. Specifically, by choosing a properly designed finite representative input set $\mathcal{X}_t \subset \mathcal{X}$ and taking into account the discretization error with Lemma 1, Srinivas et al. [53] showed the following events hold simultaneously with probability at least $1 - \delta$:

1. **Confidence bound.** For any $t \in \mathbb{N}_+$, the function value at the queried point $\boldsymbol{x}_t$ satisfies $\mu(\boldsymbol{x}_t; \mathbf{X}_{t-1}, \boldsymbol{y}_{t-1}) - \beta_t^{1/2}\sigma(\boldsymbol{x}_t; \mathbf{X}_{t-1}) \leq f(\boldsymbol{x}_t)$. Furthermore, for any $t \in \mathbb{N}_+$, any function value $f(\boldsymbol{x})$ on $\mathcal{X}_t$ satisfies $f(\boldsymbol{x}) \leq \mu(\boldsymbol{x}; \mathbf{X}_{t-1}, \boldsymbol{y}_{t-1}) + \beta_t^{1/2}\sigma(\boldsymbol{x}; \mathbf{X}_{t-1})$.
2. **Discretization error.** The discretization error arising from $\mathcal{X}_t$ is at most $1/t^2$. Namely, $|f(\boldsymbol{x}) - f([\boldsymbol{x}]_t)| \leq 1/t^2$ holds for any $\boldsymbol{x} \in \mathcal{X}$ and $t \in \mathbb{N}_+$, where $[\boldsymbol{x}]_t$ denotes one of the closest points of $\boldsymbol{x}$ on $\mathcal{X}_t$.

In the above statements, $\beta_t^{1/2}$ is chosen based on the constants a, b in Lemma 1 and the length r of $\mathcal{X}$, and is defined as

$$\beta_t = 2\ln\frac{2t^2\pi^2}{3\delta} + 2d\ln\left(t^2 dbr\sqrt{\ln\frac{4da}{\delta}}\right). \tag{6}$$

The above two events and the UCB-selection rule for $\boldsymbol{x}_t$ imply

$$R_T = \sum_{t=1}^{T} f(\boldsymbol{x}^*) - f([\boldsymbol{x}^*]_t) + \sum_{t=1}^{T} f([\boldsymbol{x}^*]_t) - f(\boldsymbol{x}_t) \leq \frac{\pi^2}{6} + 2\beta_T^{1/2}\sum_{t=1}^{T}\sigma(\boldsymbol{x}_t; \mathbf{X}_{t-1}). \tag{7}$$

In the above expression, the upper bound $\sum_{t=1}^{T} f(\boldsymbol{x}^*) - f([\boldsymbol{x}^*]_t) \leq \sum_{t=1}^{T} 1/t^2 \leq \pi^2/6$ follows from the second event (discretization error). The inequality $\sum_{t=1}^{T} f([\boldsymbol{x}^*]_t) - f(\boldsymbol{x}_t) \leq 2\beta_T^{1/2}\sum_{t=1}^{T}\sigma(\boldsymbol{x}_t; \mathbf{X}_{t-1})$ also follows from the first event (confidence bound) and the definition of $\boldsymbol{x}_t$. See the proof of Theorem 2 in [53] for details. The above inequality suggests that the regret upper bound of GP-UCB depends on the sum of the posterior standard deviations $\sum_{t=1}^{T}\sigma(\boldsymbol{x}_t; \mathbf{X}_{t-1})$. Srinivas et al. [53] provide the upper bound of this term by leveraging the information gain $I(\mathbf{X}_T)$ as follows:

$$\sum_{t=1}^{T}\sigma(\boldsymbol{x}_t; \mathbf{X}_{t-1}) \leq \sqrt{CTI(\mathbf{X}_T)} \leq \sqrt{CT\gamma_T(\mathcal{X})}, \tag{8}$$

[1] In the earlier version and conference version of the paper, we claim that the linear condition holds when $\boldsymbol{x}^*$ is on the boundary of $\mathcal{X}$. However, the linear condition as in [47] is only valid under $d = 1$, since the partial derivative parallel to the boundary may become zero unless $\boldsymbol{x}^*$ is on the corner of $\mathcal{X}$. Therefore, for $d \geq 2$, even when $\boldsymbol{x}^*$ is on the boundary, we also have to consider the quadratic condition as with the proof of Theorem 5 in [14]. This modification only affects the constant factors in our regret bounds.

[2] Differentiability of $k_{\text{Matérn}}$ is derived in the existing works, e.g., Chapter 2.7 in [54].

where $C = \frac{2}{\ln(1+\sigma^{-2})}$. From Eqs. (7) and (8), we conclude that the regret upper bound of GP-UCB is $O\left(\sqrt{\beta_T T \gamma_T(\mathcal{X})}\right)$ with probability at least $1-\delta$. By combining the explicit upper bound of $\gamma_T(\mathcal{X})$ [53, 60], we also obtain $O\left(\sqrt{T \ln^{d+2} T}\right)$ and $\widetilde{O}\left(T^{\frac{\nu+d}{2\nu+d}}\right)$ regret upper bounds for SE and Matérn kernels, respectively.

3 Improved Regret Bound for GP-UCB

The following theorem presents our main result: a new regret upper bound for GP-UCB.

Theorem 3 (Improved regret upper bound for GP-UCB)**.** *Suppose Assumptions 1 and 2 hold. Set $k = k_{\mathrm{SE}}$ or $k = k_{\text{Matérn}}$ with $\nu > 2$. Furthermore, assume that d, ν, ℓ, r , and σ^2 are fixed constants. Fix any $\delta_{\mathrm{GP}} \in (0,1)$, and set the confidence width parameter β_t of GP-UCB as defined in Eq.* (6) *with any fixed $\delta \in (0, 1-\delta_{\mathrm{GP}})$. Then, with probability at least $1-\delta_{\mathrm{GP}}-\delta$, the cumulative regret of GP-UCB (Algorithm 1) satisfies*

$$R_T = \begin{cases} \widetilde{O}\left(\sqrt{T}\right) & \text{if } k = k_{\text{Matérn}} \text{ with } 2\nu + d \leq \nu^2, \\ O\left(\sqrt{T \ln^2 T}\right) & \text{if } k = k_{\mathrm{SE}}. \end{cases} \tag{9}$$

The hidden constants in the above expressions may depend on $\ln(1/\delta), d, \nu, \ell, r, \sigma^2$, and the constants $c_{\mathrm{sup}}, c_{\mathrm{gap}}, \rho_{\mathrm{quad}}, c_{\mathrm{quad}}$ associated with δ_{GP}, which are guaranteed to exist by Lemma 2.

We would like to note the following three aspects of our results. First, the constants associated with the sample path properties defined in Lemma 2 are used solely for analyzing the regret. On the other hand, the existing algorithm provided by Scarlett [47], which shows the same $\widetilde{O}(\sqrt{T})$ regret as ours, requires prior information about these constants to run the algorithm. This is often unrealistic in practice. Secondly, our result does not imply the upper bound of Bayesian expected regret $\mathbb{E}[R_T]$. The main issue is that the dependence of the constants in Lemma 2 on δ_{GP} is not explicitly known. Removing this limitation is an important direction for future work; however, note that the same limitation exists in the algorithm provided by Scarlett [47]. Thirdly, our results in Theorem 3 only focus on the dependence of the regret on the time horizon T. Therefore, we cannot claim any improvements in the regret with respect to the dependence on the other parameters. For example, compared to the existing $R_T = O(\sqrt{T \ln^{d+2} T})$ regret under $k = k_{\mathrm{SE}}$, our regret upper bound $R_T = O(\sqrt{T \ln^2 T})$ indeed avoids the dependence of d in the logarithmic factor; however, under the joint limit of d and T ($d, T \to \infty$), it easily behaves super-linearly even under the slowly increasing d (e.g., $d = \Theta(\ln T)$) due to the hidden constants in the regret.

3.1 Intuitive Explanation of our Analysis

Before we describe the proof, we provide an intuitive explanation of why GP-UCB admits a tighter regret bound than the existing $O(\sqrt{\beta_T T \gamma_T(\mathcal{X})})$ upper bound. The motivation for our new analysis comes from the observation that the upper bound of the information gain: $I(\mathbf{X}_T) \leq \gamma_T(\mathcal{X})$ in Eq. (8) is not always tight depending on the specific realization of the input sequence $\mathbf{X}_T$. To see this, let us observe the following two simple extreme cases of $\mathbf{X}_T$ where the inequality $I(\mathbf{X}_T) \leq \gamma_T(\mathcal{X})$ is loose and tight:

- **Case I:** $I(\mathbf{X}_T) \leq \gamma_T(\mathcal{X})$ **is loose**: Let us assume all queried inputs coincide with the unique maximizer $\boldsymbol{x}^*$ (namely, $\forall t \in [T], \boldsymbol{x}_t = \boldsymbol{x}^*$). Then, when the kernel function satisfies $\forall \boldsymbol{x} \in \mathcal{X}, k(\boldsymbol{x}, \boldsymbol{x}) = 1$ as with k_{SE} and $k_{\text{Matérn}}$, we have:

$$I(\mathbf{X}_T) = \frac{1}{2} \ln \det(\boldsymbol{I}_T + \sigma^{-2} \mathbf{K}(\mathbf{X}_T, \mathbf{X}_T)) = \frac{1}{2} \sum_{i=1}^{T} \ln(1 + \sigma^{-2} \lambda_i) = \frac{1}{2} \ln(1 + \sigma^{-2} T), \tag{10}$$

  where λ_i is the i-th eigenvalue of $\mathbf{K}(\mathbf{X}_T, \mathbf{X}_T) = \mathbf{1}\mathbf{1}^\top$ with $\mathbf{1} = (1, \ldots, 1)^\top \in \mathbb{R}^T$. The third equation uses the fact that $\mathbf{1}\mathbf{1}^\top$ is rank 1, and its unique non-zero eigenvalue is T.
- **Case II:** $I(\mathbf{X}_T) \leq \gamma_T(\mathcal{X})$ **is tight**: Let us assume that $(\boldsymbol{x}_t)$ is the same as the input sequence generated by the maximum variance reduction (MVR) algorithm (namely, $\forall t \in [T], \boldsymbol{x}_t \in$

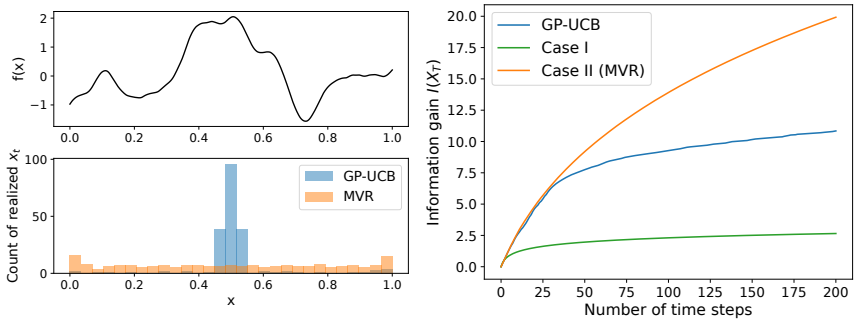


Figure 1: The behavior of the realized input sequence $\mathbf{X}_T$ (left) and the corresponding information gain $I(\mathbf{X}_T)$ (right) in the 1-dimensional BO problem with $\sigma^2 = 3$. The top left figure shows the objective function f realized by GP under $k = k_{\text{Matérn}}$ with $\nu = 5/2$. The bottom left figure shows the histogram of the realized inputs: $(\boldsymbol{x}_t)_{t\in[200]}$ with GP-UCB (blue) and MVR (orange) under f in the top left figure. Furthermore, the right plot shows the corresponding information gain $I(\mathbf{X}_t)$ under GP-UCB or MVR. We also plot the reference curve $0.5\ln(1+\sigma^{-2}t)$ corresponding to Case I described in Section 3.1. We can observe that the inputs selected by GP-UCB are concentrated around the maximizer from the left figure. Then, from the right figure, we also observe that the corresponding information gain increases more slowly than that of MVR, and behaves similarly to Case I on $t \geq 30$. More comprehensive empirical results are also provided in Appendix D.

$\text{argmax}_{\boldsymbol{x}\in\mathcal{X}}\, \sigma(\boldsymbol{x}; \mathbf{X}_{t-1})$) [53, 58]. Then, from the discussion in Sections 2 and 5 in [53], we already know that $\gamma_T(\mathcal{X}) \leq (1-1/e)^{-1} I(\mathbf{X}_T)$. This suggests that $I(\mathbf{X}_T) \leq \gamma_T(\mathcal{X})$ is tight up to a constant factor when $\mathbf{X}_T$ is realized by MVR.

From Case I, we observe that $I(\mathbf{X}_T)$ satisfies $\Theta(\ln T) \leq I(\mathbf{X}_T) \leq \gamma_T(\mathcal{X})$ depending on $\mathbf{X}_T$. Furthermore, by comparing the input sequences in cases I and II, we expect that $I(\mathbf{X}_T)$ becomes small if $\mathbf{X}_T$ concentrates around the neighborhood of $\boldsymbol{x}^*$, while $I(\mathbf{X}_T)$ becomes large if $\mathbf{X}_T$ spreads over the entire input domain $\mathcal{X}$. Then, from the fact that the worst-case regret of GP-UCB increases sub-linearly with the speed of $O(\sqrt{\beta_T T \gamma_T(\mathcal{X})})$, we can deduce that the input sequence $\mathbf{X}_T$ of GP-UCB will eventually concentrate around the maximizer $\boldsymbol{x}^*$ if $\boldsymbol{x}^*$ is unique and $\|f\|_\infty$ is not extremely small[3]. We provide an illustrative image in Figure 1. Our proof is designed so as to capture the above intuition that $I(\mathbf{X}_T)$ could be improved from $\gamma_T(\mathcal{X})$ to $\Theta(\ln T)$ under "favorable" sample path f.

3.2 Proof of Theorem 3

Let $\mathcal{A}$ be an event such that the two high-probability events of the original GP-UCB proof (described in the last paragraph in Section 2) and Lemma 2 with the confidence level δ_{GP} simultaneously hold. Note that event $\mathcal{A}$ occurs with probability at least $1 - \delta_{\text{GP}} - \delta$ from the union bound. Therefore, it is enough to prove our upper bound under $\mathcal{A}$. To encode the high-level idea in the previous section, we need to capture the concentration behavior of the input sequence $\mathbf{X}_T$ around the maximizer $\boldsymbol{x}^*$. From this motivation, given some constant $\varepsilon > 0$, we decompose the regret as $R_T = R_T^{(1)}(\varepsilon) + R_T^{(2)}(\varepsilon)$, where:

$$R_T^{(1)}(\varepsilon) = \sum_{t\in\mathcal{T}(\varepsilon)} f(\boldsymbol{x}^*) - f(\boldsymbol{x}_t),\ R_T^{(2)}(\varepsilon) = \sum_{t\in\mathcal{T}^c(\varepsilon)} f(\boldsymbol{x}^*) - f(\boldsymbol{x}_t). \tag{11}$$

We set $\mathcal{T}(\varepsilon) = \{t \in [T] \mid f(\boldsymbol{x}^*) - f(\boldsymbol{x}_t) > \varepsilon\}$ and $\mathcal{T}^c(\varepsilon) = [T] \setminus \mathcal{T}(\varepsilon)$ in the above definition. A key observation is that, if we set sufficiently small ε depending on the constants in Lemma 2, the inputs $(\boldsymbol{x}_t)$ in $R_T^{(2)}(\varepsilon)$ (namely, inputs $(\boldsymbol{x}_t)$ such that $f(\boldsymbol{x}^*) - f(\boldsymbol{x}_t) \leq \varepsilon$ holds) are on the

[3]Specifically, if $T\|f\|_\infty \leq O(\sqrt{\beta_T T \gamma_T(\mathcal{X})})$, we cannot make any claims about $\mathbf{X}_T$ based on the worst-case bound since any sequence $\mathbf{X}_T$ satisfies the worst-case bound without concentrating around maximizer. This is why our analysis technique does not improve the worst-case regret in the frequentist setting. Indeed, in the proof of the worst-case lower bound for the frequentist setting [48], the existence of the function f with $T\|f\|_\infty = O(\sqrt{\beta_T T \gamma_T(\mathcal{X})})$ is guaranteed.

locally quadratic region around the maximizer $\boldsymbol{x}^*$ due to conditions 1 and 3 in Lemma 2. The formal descriptions are provided in Lemma 20 in Appendix C. This fact is originally leveraged in [47] to analyze the successive elimination-based algorithm. In the analysis of GP-UCB, it enables us to analyze the behavior of the sub-input sequence $\{\boldsymbol{x}_t \mid f(\boldsymbol{x}^*) - f(\boldsymbol{x}_t) \leq \varepsilon\}$ through the regularity constant c_{quad}. Below, we formally give the upper bound for $R_T^{(2)}(\varepsilon)$.

Lemma 4 (General upper bound of $R_T^{(2)}$)**.** *Suppose $(\boldsymbol{x}_t)_{t\in[T]}$ is the input query sequence realized by the GP-UCB algorithm. Furthermore, let $\overline{\gamma}_t$ be an upper bound of MIG $\gamma_t(\mathcal{X})$ such that $\overline{\gamma}_t/t$ is non-increasing on $[\overline{T}, \infty)$ with some $\overline{T} \in \mathbb{N}_+$[4]. Then, under event $\mathcal{A}$, we have*

$$R_T^{(2)}(\varepsilon) \leq 2c_{\mathrm{sup}}\overline{T} + \frac{\pi^2}{3}\left(\log_2 T + 1\right) + \frac{2\sqrt{2C\beta_T T}}{\sqrt{2}-1}\max_{i\in[\bar{i}]}\sqrt{\gamma_{(T/2^{i-1})}\left(\mathcal{B}_2\left(\sqrt{c_{\mathrm{quad}}^{-1}\eta_i}; \boldsymbol{x}^*\right)\right)},$$

where $C = 2/\ln(1+\sigma^{-2})$, $\bar{i} = \lfloor \log_2 \frac{T}{\overline{T}} \rfloor + 1$, $\eta_i = \frac{2\left(2\sqrt{C\beta_T(T/2^{i-1})\overline{\gamma}_{T/2^{i-1}}} + \frac{\pi^2}{6}\right)}{(T/2^{i-1})}$, *and* $\varepsilon = \min\{c_{\mathrm{gap}}, c_{\mathrm{quad}}\rho_{\mathrm{quad}}^2\}$.

We give the full proof in Appendix A.1. Here, the dominant term in the above lemma is given as:

$$R_T^{(2)}(\varepsilon) = \widetilde{O}\left(\max_i \sqrt{T\gamma_{(T/2^{i-1})}\left(\mathcal{B}_2\left(\sqrt{c_{\mathrm{quad}}^{-1}\eta_i}; \boldsymbol{x}^*\right)\right)}\right). \tag{12}$$

Note that η_i is decreasing as the time index $T/2^{i-1}$ of MIG increases. In other words, the input domain $\mathcal{B}_2\left(\sqrt{c_{\mathrm{quad}}^{-1}\eta_i}; \boldsymbol{x}^*\right)$ of MIG shrinks as the time index $T/2^{i-1}$ increases. This property is beneficial for obtaining a tighter upper bound than that from the existing technique. For example, under $k = k_{\text{Matérn}}$ with $2\nu + d \leq \nu^2$, we can confirm that the dominant polynomial term in MIG is canceled out by the shrinking of the input domain in MIG. Namely, we can obtain the following result under $k = k_{\text{Matérn}}$:

$$\max_i \gamma_{(T/2^{i-1})}\left(\mathcal{B}_2\left(\sqrt{c_{\mathrm{quad}}^{-1}\eta_i}; \boldsymbol{x}^*\right)\right) = \widetilde{O}(1) \ \ (\text{as } T \to \infty), \tag{13}$$

which leads to $R_T^{(2)}(\varepsilon) = \widetilde{O}(\sqrt{T})$. This strictly improves the trivial upper bound $R_T^{(2)}(\varepsilon) = \widetilde{O}(\sqrt{T\gamma_T(\mathcal{X})})$ under $k = k_{\text{Matérn}}$. The formal descriptions are given in the next lemma.

Lemma 5 (Upper bound of $R_T^{(2)}$ under k_{SE} and $k_{\text{Matérn}}$)**.** *Suppose $(\boldsymbol{x}_t)_{t\in[T]}$ is the input sequence realized by the GP-UCB algorithm. Furthermore, ε is set as that in Lemma 4. Then, under event $\mathcal{A}$,*

$$R_T^{(2)}(\varepsilon) = \begin{cases} \widetilde{O}(\sqrt{T}) & \text{if } k = k_{\text{Matérn}} \text{ with } 2\nu + d \leq \nu^2, \\ O\left(\sqrt{T\ln^2 T}\right) & \text{if } k = k_{\mathrm{SE}}. \end{cases} \tag{14}$$

The full proof is given in Appendix A.2. The remaining interest is the upper bound of $R_T^{(1)}(\varepsilon)$. The definition of $R_T^{(1)}(\varepsilon)$ is the same as the *lenient regret* [9], which is known to be smaller than the original regret R_T in GP-UCB. Although Cai et al. [9] study the frequentist setting, their proof strategy is also applicable to the Bayesian setting as described in Section 3.4 in [9]. The following lemma provides the formal statement about the upper bound of $R_T^{(1)}(\varepsilon)$.

Lemma 6 (Upper bound of $R_T^{(1)}$, adaptation of the proof of Theorem 1 in [9])**.** *Fix any $\varepsilon > 0$. Suppose $k = k_{\mathrm{SE}}$ or $k = k_{\text{Matérn}}$. Then, when running GP-UCB, $R_T^{(1)}(\varepsilon) = \widetilde{O}(1)$ holds under event $\mathcal{A}$.*

We provide the proof in Appendix A.3 for completeness. For both kernels, $R_T^{(1)}(\varepsilon)$ is dominated by the upper bound of $R_T^{(2)}(\varepsilon)$. Finally, we obtain the desired results by aggregating the inequalities in Lemmas 5 and 6.

[4]Namely, $\forall t \geq \overline{T}, \forall \epsilon \geq 0, \overline{\gamma}_t/t \geq \overline{\gamma}_{t+\epsilon}/(t+\epsilon)$ and $\forall t \geq \overline{T}, \gamma_t(\mathcal{X}) \leq \overline{\gamma}_t$ hold for some $\overline{T} \in \mathbb{N}_+$.

4 Discussions

Below, we discuss the limitations of our results and outline possible directions for future research.

- **Optimality.** Based on the $\Omega(\sqrt{T})$ lower bound on the expected regret provided by Scarlett [47], we conjecture that our $\widetilde{O}(\sqrt{T})$ high-probability regret bound for GP-UCB is near-optimal. However, it is not straightforward to extend the lower bound for the expected regret in [47] to a high probability result. Specifically, the lower bound in [47] is quantified by a mutual information term (Lemma 4 in [47]); however, to our knowledge, the technique used to handle this term appears to be specific to the expected regret setting. We believe that the rigorous optimality argument for the Bayesian high probability regret is an important direction for future research.
- **Smoothness condition.** In our result for the Matérn kernel, we require an additional smoothness constraint to obtain a $\widetilde{O}(\sqrt{T})$ regret bound[5]. To overcome this issue in our proof, we believe that we need stronger regularity conditions on the sample path around the maximizer than those assumed in Lemma 2.
- **Extension to the expected regret.** Our regret bounds involve regularity constants that depend on the sample path. However, to our knowledge, there is no existing research that rigorously analyzes how these constants depend on the confidence level δ_{GP}. This makes it difficult to obtain the expected regret guarantees as with the original GP-UCB, whose expected regret bounds are established by properly decreasing the confidence level as a function of T (e.g., [41, 55]). To overcome this issue, further analysis for Lemma 2, or another idea to quantify the sample path regularities, is required.
- **Extension to other algorithms.** One limitation of our technique is its restricted applicability to other algorithms. To apply our proof, at least the algorithm should satisfy the following two conditions: (i) on any index subset, the sub-linear cumulative regret is obtained with high probability (Lemma 21), and (ii) the high-probability lenient regret bound is provided (Lemma 6). The existing analysis of the other major algorithms in the Bayesian setting (e.g., Thompson sampling [44], information directed sampling [45]) does not provide these properties. Nevertheless, we believe that the high-level ideas in our proof (see Section 3.1) could be beneficial for future refined analyses of other algorithms.
- **Instance-dependent analysis in the frequentist setting.** As described in the footnote in Section 3.1, we believe that our analysis does not improve the worst-case regret upper bound in the frequentist setting. On the other hand, our technique can be applied to the instance-dependent analysis [51] for GP-UCB. We expect that our proof strategy could yield a $\widetilde{O}(\sqrt{T})$ instance-dependent regret for GP-UCB by replacing the sample path condition 3 in Lemma 2 with the *growth condition* (Definition 4 in [51]) of the function. It is an interesting direction for future research.

5 Conclusion

We provide a refined analysis of GP-UCB in the BO problem. For both SE and Matérn kernels, our results improve upon existing regret guarantees and fill the gap between the existing regret of GP-UCB and the current best upper bound in [47]. The core idea of our analysis is to capture the shrinking behavior of the input sequence by relating it to the worst-case upper bound and the sample path regularity conditions. Although our current analysis is limited to GP-UCB in the Bayesian setting, we believe it lays the foundation for several promising future research directions.

Acknowledgments

We thank Jonathan Scarlett and Shion Takeno for their valuable comments on revising the manuscript.

[5]For simplicity, in Theorem 3, we focus on the setting where the resulting regret becomes $\widetilde{O}(\sqrt{T})$ under $k_{\mathrm{Matérn}}$. This gives rise to the requirement of the additional smoothness condition $2\nu + d \leq \nu^2$. On the other hand, we can also apply the same technique even under $2\nu + d > \nu^2$. In this case, the resulting regret becomes strictly larger than $\widetilde{O}(\sqrt{T})$, while it is strictly smaller than $\widetilde{O}(T^{\frac{\nu+d}{2\nu+d}})$ of the original GP-UCB's analysis.

References

[1] Kendall Atkinson and Weimin Han. *Spherical harmonics and approximations on the unit sphere: an introduction*, volume 2044. Springer Science & Business Media, 2012.

[2] Jean-Marc Azaïs and Mario Wschebor. *Level sets and extrema of random processes and fields*. John Wiley & Sons, 2009.

[3] Douglas Azevedo and Valdir Antonio Menegatto. Sharp estimates for eigenvalues of integral operators generated by dot product kernels on the sphere. *Journal of Approximation Theory*, 2014.

[4] Francis Bach. Breaking the curse of dimensionality with convex neural networks. *Journal of Machine Learning Research*, 2017.

[5] Felix Berkenkamp, Angela P Schoellig, and Andreas Krause. No-regret Bayesian optimization with unknown hyperparameters. *Journal of Machine Learning Research*, 2019.

[6] Alberto Bietti and Francis Bach. Deep equals shallow for ReLU networks in kernel regimes. *International Conference on Learning Representations*, 2021.

[7] Ilija Bogunovic, Jonathan Scarlett, and Volkan Cevher. Time-varying Gaussian process bandit optimization. In *Proc. International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2016.

[8] Adam D Bull. Convergence rates of efficient global optimization algorithms. *Journal of Machine Learning Research*, 2011.

[9] Xu Cai, Selwyn Gomes, and Jonathan Scarlett. Lenient regret and good-action identification in Gaussian process bandits. In *International Conference on Machine Learning*, pages 1183–1192. PMLR, 2021.

[10] Romain Camilleri, Kevin Jamieson, and Julian Katz-Samuels. High-dimensional experimental design and kernel bandits. In *Proc. International Conference on Machine Learning (ICML)*, 2021.

[11] Alexandre Capone, Armin Lederer, and Sandra Hirche. Gaussian process uniform error bounds with unknown hyperparameters for safety-critical applications. In *International Conference on Machine Learning*, 2022.

[12] Sayak Ray Chowdhury and Aditya Gopalan. On kernelized multi-armed bandits. In *Proc. International Conference on Machine Learning (ICML)*, 2017.

[13] Andreas Christmann and Ingo Steinwart. *Support vector machines*. Springer, 2008.

[14] Nando de Freitas, Alex Smola, and Masrour Zoghi. Regret bounds for deterministic Gaussian process bandits. *arXiv preprint arXiv:1203.2177*, 2012.

[15] Nando De Freitas, Alex J. Smola, and Masrour Zoghi. Exponential regret bounds for Gaussian process bandits with deterministic observations. In *Proceedings of the 29th International Conference on International Conference on Machine Learning*, pages 955–962. Omnipress, 2012.

[16] Thomas Desautels, Andreas Krause, and Joel W. Burdick. Parallelizing exploration-exploitation tradeoffs in Gaussian process bandit optimization. *Journal of Machine Learning Research*, 2014.

[17] Costas Efthimiou and Christopher Frye. *Spherical harmonics in p dimensions*. World Scientific, 2014.

[18] Peter Frazier, Warren Powell, and Savas Dayanik. The knowledge-gradient policy for correlated normal beliefs. *INFORMS journal on Computing*, 21(4):599–613, 2009.

[19] Peter I Frazier. A tutorial on Bayesian optimization. *arXiv preprint arXiv:1807.02811*, 2018.

[20] Roman Garnett. *Bayesian optimization*. Cambridge University Press, 2023.

[21] Amnon Geifman, Abhay Yadav, Yoni Kasten, Meirav Galun, David Jacobs, and Basri Ronen. On the similarity between the Laplace and neural tangent kernels. *Advances in Neural Information Processing Systems*, 2020.

[22] Subhashis Ghosal and Anindya Roy. Posterior consistency of Gaussian process prior for nonparametric binary regression. 2006.

[23] Andrew Gray and George Ballard Mathews. *A treatise on Bessel functions and their applications to physics*. Macmillan, 1895.

[24] Steffen Grünewälder, Jean-Yves Audibert, Manfred Opper, and John Shawe-Taylor. Regret bounds for Gaussian process bandit problems. In *Proc. International Conference on Artificial Intelligence and Statistics (AISTATS)*. JMLR Workshop and Conference Proceedings, 2010.

[25] Philipp Hennig and Christian J Schuler. Entropy search for information-efficient global optimization. *The Journal of Machine Learning Research*, 13(1):1809–1837, 2012.

[26] Shogo Iwazaki and Shinya Suzumura. No-regret bandit exploration based on soft tree ensemble model. *Advances in Neural Information Processing Systems*, 2024.

[27] Shogo Iwazaki and Shion Takeno. Improved regret analysis in Gaussian process bandits: Optimality for noiseless reward, RKHS norm, and non-stationary variance. In *Proc. International Conference on Machine Learning (ICML)*, 2025.

[28] David Janz. *Sequential decision making with feature-linear models*. PhD thesis, 2021.

[29] David Janz, David Burt, and Javier Gonzalez. Bandit optimisation of functions in the Matérn kernel RKHS. In *Proceedings of the Twenty Third International Conference on Artificial Intelligence and Statistics*, volume 108 of *Proceedings of Machine Learning Research*, pages 2486–2495. PMLR, 2020.

[30] Kirthevasan Kandasamy, Jeff Schneider, and Barnabas Poczos. High dimensional Bayesian optimisation and bandits via additive models. In *Proc. International Conference on Machine Learning (ICML)*, 2015.

[31] Kirthevasan Kandasamy, Gautam Dasarathy, Junier Oliva, Jeff Schneider, and Barnabas Poczos. Multi-fidelity Gaussian process bandit optimisation. *Journal of Artificial Intelligence Research*, 2019.

[32] Parnian Kassraie and Andreas Krause. Neural contextual bandits without regret. In *International Conference on Artificial Intelligence and Statistics*, pages 240–278. PMLR, 2022.

[33] Kenji Kawaguchi, Leslie P Kaelbling, and Tomás Lozano-Pérez. Bayesian optimization with exponential convergence. *Advances in neural information processing systems*, 28, 2015.

[34] Jeankyung Kim and David Pollard. Cube root asymptotics. *The Annals of Statistics*, pages 191–219, 1990.

[35] Andreas Krause and Cheng Ong. Contextual Gaussian process bandit optimization. In *Proc. Neural Information Processing Systems (NeurIPS)*, 2011.

[36] Zihan Li and Jonathan Scarlett. Gaussian process bandit optimization with few batches. In *Proc. International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2022.

[37] Ha Quang Minh, Partha Niyogi, and Yuan Yao. Mercer's theorem, feature maps, and smoothing. In *International Conference on Computational Learning Theory*. Springer, 2006.

[38] Jonas Močkus. On Bayesian methods for seeking the extremum. In *Optimization Techniques IFIP Technical Conference Novosibirsk, July 1–7, 1974 6*, pages 400–404. Springer, 1975.

[39] Francis J Narcowich and Joseph D Ward. Scattered data interpolation on spheres: error estimates and locally supported basis functions. *SIAM Journal on Mathematical Analysis*, 33 (6):1393–1410, 2002.

[40] Francis J Narcowich, Xinping Sun, and Joseph D Ward. Approximation power of RBFs and their associated SBFs: a connection. *Advances in Computational Mathematics*, 2007.

[41] Biswajit Paria, Kirthevasan Kandasamy, and Barnabás Póczos. A flexible framework for multi-objective Bayesian optimization using random scalarizations. In *Uncertainty in Artificial Intelligence*, 2020.

[42] Carl Edward Rasmussen and Christopher K. I. Williams. *Gaussian Processes for Machine Learning (Adaptive Computation and Machine Learning)*. The MIT Press, 2006.

[43] Gabriel Riutort-Mayol, Paul-Christian Bürkner, Michael R Andersen, Arno Solin, and Aki Vehtari. Practical Hilbert space approximate Bayesian Gaussian processes for probabilistic programming. *Statistics and Computing*, 33(1):17, 2023.

[44] Daniel Russo and Benjamin Van Roy. Learning to optimize via posterior sampling. *Mathematics of Operations Research*, 39(4):1221–1243, 2014.

[45] Daniel Russo and Benjamin Van Roy. Learning to optimize via information-directed sampling. *Advances in neural information processing systems*, 27, 2014.

[46] Sudeep Salgia, Sattar Vakili, and Qing Zhao. Random exploration in Bayesian optimization: Order-optimal regret and computational efficiency. In *Proc. International Conference on Machine Learning (ICML)*, 2024.

[47] Jonathan Scarlett. Tight regret bounds for Bayesian optimization in one dimension. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 4500–4508. PMLR, 2018.

[48] Jonathan Scarlett, Ilija Bogunovic, and Volkan Cevher. Lower bounds on regret for noisy Gaussian process bandit optimization. In *Proc. Conference on Learning Theory (COLT)*, 2017.

[49] Meyer Scetbon and Zaid Harchaoui. A spectral analysis of dot-product kernels. In *International conference on artificial intelligence and statistics*, 2021.

[50] Shubhanshu Shekhar and Tara Javidi. Significance of gradient information in Bayesian optimization. In *International Conference on Artificial Intelligence and Statistics*, pages 2836–2844. PMLR, 2021.

[51] Shubhanshu Shekhar and Tara Javidi. Instance dependent regret analysis of kernelized bandits. In *International Conference on Machine Learning*, 2022.

[52] Arno Solin and Simo Särkkä. Hilbert space methods for reduced-rank Gaussian process regression. *Statistics and Computing*, 2020.

[53] Niranjan Srinivas, Andreas Krause, Sham Kakade, and Matthias Seeger. Gaussian process optimization in the bandit setting: No regret and experimental design. In *Proc. International Conference on Machine Learning (ICML)*, 2010.

[54] Michael L Stein. *Interpolation of spatial data: some theory for kriging*. Springer Science & Business Media, 1999.

[55] Shion Takeno, Yu Inatsu, and Masayuki Karasuyama. Randomized Gaussian process upper confidence bound with tighter Bayesian regret bounds. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, pages 33490–33515. PMLR, 2023.

[56] Filip Tronarp, Toni Karvonen, and Simo Särkkä. Mixture representation of the Matérn class with applications in state space approximations and Bayesian quadrature. In *2018 IEEE 28th International Workshop on Machine Learning for Signal Processing (MLSP)*, 2018.

[57] Sattar Vakili and Julia Olkhovskaya. Kernelized reinforcement learning with order optimal regret bounds. *Advances in Neural Information Processing Systems*, 2023.

[58] Sattar Vakili, Nacime Bouziani, Sepehr Jalali, Alberto Bernacchia, and Da shan Shiu. Optimal order simple regret for Gaussian process bandits. In *Proc. Neural Information Processing Systems (NeurIPS)*, 2021.

[59] Sattar Vakili, Michael Bromberg, Jezabel Garcia, Da-shan Shiu, and Alberto Bernacchia. Uniform generalization bounds for overparameterized neural networks. *arXiv preprint arXiv:2109.06099*, 2021.

[60] Sattar Vakili, Kia Khezeli, and Victor Picheny. On information gain and regret bounds in Gaussian process bandits. In *Proc. International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2021.

[61] Michal Valko, Nathan Korda, Rémi Munos, Ilias Flaounas, and Nello Cristianini. Finite-time analysis of kernelised contextual bandits. In *Proceedings of the Twenty-Ninth Conference on Uncertainty in Artificial Intelligence*, UAI'13, page 654–663. AUAI Press, 2013.

[62] Zexin Wang, Vincent YF Tan, and Jonathan Scarlett. Tight regret bounds for noisy optimization of a Brownian motion. *IEEE Transactions on Signal Processing*, 70:1072–1087, 2022.

[63] Justin Whitehouse, Zhiwei Steven Wu, and Aaditya Ramdas. Improved self-normalized concentration in Hilbert spaces: Sublinear regret for GP-UCB. *Proc. Neural Information Processing Systems (NeurIPS)*, 2023.

[64] Yun Yang, Anirban Bhattacharya, and Debdeep Pati. Frequentist coverage and sup-norm convergence rate in Gaussian process regression. *arXiv preprint arXiv:1708.04753*, 2017.

[65] Fuzhen Zhang. *Matrix theory: basic results and techniques*. Springer Science & Business Media, 2011.

A Proofs in Section 3

A.1 Proof of Lemma 4

Proof. From Lemma 21, we have the following upper bound for any index set $\mathcal{T} \subset [T]$ under $\mathcal{A}$:

$$\sum_{t \in \mathcal{T}} f(\boldsymbol{x}^*) - f(\boldsymbol{x}_t) \leq 2\sqrt{C\beta_T |\mathcal{T}| \overline{\gamma}_{|\mathcal{T}|}} + \frac{\pi^2}{6}. \tag{15}$$

Here, for any i such that $T/2^{i-1} \geq \overline{T}$, we set (η_i) as

$$\eta_i = \frac{2\left(2\sqrt{C\beta_T (T/2^{i-1}) \overline{\gamma}_{T/2^{i-1}}} + \frac{\pi^2}{6}\right)}{(T/2^{i-1})}. \tag{16}$$

As described in the proof below, these (η_i) are designed so that we can obtain the upper bound of $|\mathcal{T}(\eta_i)|$ in a dyadic manner. Here, we consider the upper bound of $|\mathcal{T}(\eta_i)|$ based on the worst-case upper bound in Eq. (15). From the definition of $\mathcal{T}(\eta)$ and Eq. (15) with $\mathcal{T} = [T]$, the condition $|\mathcal{T}(\eta_1)|\eta_1 \leq 2\sqrt{C\beta_T T \overline{\gamma}_T} + \pi^2/6$ must be satisfied; otherwise, we have $\sum_{t \in [T]} f(\boldsymbol{x}^*) - f(\boldsymbol{x}_t) \geq \sum_{t \in \mathcal{T}(\eta_1)} f(\boldsymbol{x}^*) - f(\boldsymbol{x}_t) \geq |\mathcal{T}(\eta_1)|\eta_1 > 2\sqrt{C\beta_T T \overline{\gamma}_T} + \pi^2/6$, which contradicts worst-case upper bound in Eq. (15). Therefore, we can obtain the following upper bound:

$$|\mathcal{T}(\eta_1)| \leq \max\left\{t \leq T \mid t\eta_1 \leq 2\sqrt{C\beta_T T \overline{\gamma}_T} + \frac{\pi^2}{6}\right\} = \frac{T}{2}. \tag{17}$$

Furthermore, since η_i is monotonic due to the condition about $\overline{\gamma}_t$, we have $\eta_1 \leq \eta_2$, which implies $\mathcal{T}(\eta_2) \subset \mathcal{T}(\eta_1)$. From Eq. (15) with $\mathcal{T} = \mathcal{T}(\eta_1)$, Eq. (17), and $\mathcal{T}(\eta_2) \subset \mathcal{T}(\eta_1)$, we further obtain

$$|\mathcal{T}(\eta_2)| \leq \max\left\{t \leq T/2 \mid t\eta_2 \leq 2\sqrt{C\beta_T (T/2) \overline{\gamma}_{(T/2)}} + \frac{\pi^2}{6}\right\} = \frac{T}{4}. \tag{18}$$

Similarly to $|\mathcal{T}(\eta_2)|$, we have $\mathcal{T}(\eta_3) \subset \mathcal{T}(\eta_2)$ and

$$|\mathcal{T}(\eta_3)| \leq \max\left\{t \leq T/4 \mid t\eta_3 \leq 2\sqrt{C\beta_T (T/4) \overline{\gamma}_{(T/4)}} + \frac{\pi^2}{6}\right\} = \frac{T}{8}. \tag{19}$$

By repeating this argument i times while $T/2^{i-1} \geq \overline{T}$ holds, we have the following inequality for any $i \leq \lfloor \log_2 \frac{T}{\overline{T}} \rfloor + 1$:

$$|\mathcal{T}(\eta_i)| \leq \max\left\{t \leq T/2^{i-1} \mid t\eta_i \leq 2\sqrt{C\beta_T (T/2^{i-1}) \overline{\gamma}_{(T/2^{i-1})}} + \frac{\pi^2}{6}\right\} = \frac{T}{2^i}. \tag{20}$$

Then, we have

$$\begin{aligned}
R_T^{(2)}(\varepsilon) &= \sum_{t \in \mathcal{T}^c(\varepsilon)} f(\boldsymbol{x}^*) - f(\boldsymbol{x}_t) && (21)\\
&= \sum_{t \in \mathcal{T}^c(\varepsilon) \cap \mathcal{T}(\eta_1)} f(\boldsymbol{x}^*) - f(\boldsymbol{x}_t) + \sum_{t \in \mathcal{T}^c(\varepsilon) \cap \mathcal{T}^c(\eta_1)} f(\boldsymbol{x}^*) - f(\boldsymbol{x}_t) && (22)\\
&= \sum_{t \in \mathcal{T}^c(\varepsilon) \cap \mathcal{T}(\eta_1) \cap \mathcal{T}(\eta_2)} f(\boldsymbol{x}^*) - f(\boldsymbol{x}_t) + \sum_{t \in \mathcal{T}^c(\varepsilon) \cap \mathcal{T}(\eta_1) \cap \mathcal{T}^c(\eta_2)} f(\boldsymbol{x}^*) - f(\boldsymbol{x}_t) \\
&\quad + \sum_{t \in \mathcal{T}^c(\varepsilon) \cap \mathcal{T}^c(\eta_1)} f(\boldsymbol{x}^*) - f(\boldsymbol{x}_t) && (23)\\
&= \sum_{t \in \mathcal{T}^c(\varepsilon) \cap \mathcal{T}(\eta_2)} f(\boldsymbol{x}^*) - f(\boldsymbol{x}_t) + \sum_{i=1}^{2} \sum_{t \in \mathcal{T}^c(\varepsilon) \cap \mathcal{T}(\eta_{i-1}) \cap \mathcal{T}^c(\eta_i)} f(\boldsymbol{x}^*) - f(\boldsymbol{x}_t), && (24)
\end{aligned}$$

where the last line follows from $\mathcal{T}(\eta_2) \subset \mathcal{T}(\eta_1)$. In the above inequality, we define $\mathcal{T}(\eta_0)$ as $\mathcal{T}(\eta_0) = [T]$ for notational convenience. By repeatedly applying the above decomposition, we

obtain

$$\sum_{t\in\mathcal{T}^c(\varepsilon)\cap\mathcal{T}(\eta_2)} f(\boldsymbol{x}^*) - f(\boldsymbol{x}_t) + \sum_{i=1}^{2} \sum_{t\in\mathcal{T}^c(\varepsilon)\cap\mathcal{T}(\eta_{i-1})\cap\mathcal{T}^c(\eta_i)} f(\boldsymbol{x}^*) - f(\boldsymbol{x}_t) \tag{25}$$

$$= \sum_{t\in\mathcal{T}^c(\varepsilon)\cap\mathcal{T}(\eta_3)} f(\boldsymbol{x}^*) - f(\boldsymbol{x}_t) + \sum_{i=1}^{3} \sum_{t\in\mathcal{T}^c(\varepsilon)\cap\mathcal{T}(\eta_{i-1})\cap\mathcal{T}^c(\eta_i)} f(\boldsymbol{x}^*) - f(\boldsymbol{x}_t) \tag{26}$$

$$\vdots$$

$$= \sum_{t\in\mathcal{T}^c(\varepsilon)\cap\mathcal{T}(\eta_{\bar{i}})} f(\boldsymbol{x}^*) - f(\boldsymbol{x}_t) + \sum_{i=1}^{\bar{i}} \sum_{t\in\mathcal{T}^c(\varepsilon)\cap\mathcal{T}(\eta_{i-1})\cap\mathcal{T}^c(\eta_i)} f(\boldsymbol{x}^*) - f(\boldsymbol{x}_t), \tag{27}$$

where $\bar{i} = \lfloor \log_2 \frac{T}{\overline{T}} \rfloor + 1$. Regarding the first term in Eq. (27), we have

$$\sum_{t\in\mathcal{T}^c(\varepsilon)\cap\mathcal{T}(\eta_{\bar{i}})} f(\boldsymbol{x}^*) - f(\boldsymbol{x}_t) \le 2c_{\sup}|\mathcal{T}(\eta_{\bar{i}})| \le 2c_{\sup}\overline{T}, \tag{28}$$

where the last inequality follows from $|\mathcal{T}(\eta_{\bar{i}})| \le \overline{T}$, which is implied by $|\mathcal{T}(\eta_{\bar{i}})| \le T/2^{\bar{i}}$ from Eq. (20) and the definition of $\bar{i}$. Next, regarding the second term in Eq. (27), we first define $\mathcal{T}_i$ and $\mathcal{X}^{(i)}$ as $\mathcal{T}_i = \mathcal{T}^c(\varepsilon) \cap \mathcal{T}(\eta_{i-1}) \cap \mathcal{T}^c(\eta_i)$ and $\mathcal{X}^{(i)} = \{\boldsymbol{x}_t \mid t \in \mathcal{T}_i\}$, respectively. Then, by applying Lemma 21 with $\mathcal{T} = \mathcal{T}_i$, we have

$$\sum_{t\in\mathcal{T}^c(\varepsilon)\cap\mathcal{T}(\eta_{i-1})\cap\mathcal{T}^c(\eta_i)} f(\boldsymbol{x}^*) - f(\boldsymbol{x}_t) = \sum_{t\in\mathcal{T}_i} f(\boldsymbol{x}^*) - f(\boldsymbol{x}_t) \tag{29}$$

$$\le 2\sqrt{C\beta_T|\mathcal{T}_i| I(\mathcal{X}^{(i)})} + \frac{\pi^2}{6} \tag{30}$$

$$\le 2\sqrt{C\beta_T|\mathcal{T}_i|\gamma_{|\mathcal{T}_i|}(\mathcal{X}^{(i)})} + \frac{\pi^2}{6} \tag{31}$$

$$\le 2\sqrt{C\beta_T|\mathcal{T}(\eta_{i-1})|\gamma_{|\mathcal{T}(\eta_{i-1})|}(\mathcal{X}^{(i)})} + \frac{\pi^2}{6} \tag{32}$$

$$\le 2\sqrt{C\beta_T(T/2^{i-1})\gamma_{(T/2^{i-1})}(\mathcal{X}^{(i)})} + \frac{\pi^2}{6}, \tag{33}$$

where the third inequality follows from $|\mathcal{T}_i| \le |\mathcal{T}(\eta_{i-1})|$, and the last inequality follows from Eq. (20). By aggregating Eqs. (27), (28), and (33), we obtain the following inequality under $\mathcal{A}$:

$$R_T^{(2)}(\varepsilon) \le 2c_{\sup}\overline{T} + 2\sum_{i=1}^{\bar{i}} \left[\sqrt{C\beta_T(T/2^{i-1})\gamma_{(T/2^{i-1})}(\mathcal{X}^{(i)})} + \frac{\pi^2}{6} \right] \tag{34}$$

$$\le 2c_{\sup}\overline{T} + \frac{\pi^2}{3}\left(\log_2 T + 1\right) + 2\sqrt{C\beta_T T}\sum_{i=1}^{\bar{i}} \frac{1}{2^{(i-1)/2}} \sqrt{\gamma_{(T/2^{i-1})}(\mathcal{X}^{(i)})} \tag{35}$$

$$\le 2c_{\sup}\overline{T} + \frac{\pi^2}{3}\left(\log_2 T + 1\right) + \frac{2\sqrt{2C\beta_T T}}{\sqrt{2}-1} \max_{i\in[\bar{i}]} \sqrt{\gamma_{(T/2^{i-1})}(\mathcal{X}^{(i)})}. \tag{36}$$

The last line follows from $\sum_{i=1}^{\bar{i}} \frac{1}{2^{(i-1)/2}} \le \sum_{i=1}^{\infty} \frac{1}{2^{(i-1)/2}} = \frac{1}{1-1/\sqrt{2}} = \frac{\sqrt{2}}{\sqrt{2}-1}$. The last part of the proof is to specify the radius of the ball $\mathcal{B}_2(\cdot;\boldsymbol{x}^*)$ such that $\mathcal{X}^{(i)}$ is included in it.

Conversion of the sub-optimality gap into the upper bound input radius. From condition 3 in Lemma 2, the definition of $\mathcal{T}^c(\varepsilon)$, ε, and Lemma 20, we have $\boldsymbol{x} \in \mathcal{B}_2(\rho_{\text{quad}};\boldsymbol{x}^*)$ for any $\boldsymbol{x} \in \mathcal{X}^{(i)}$. This implies $\forall \boldsymbol{x} \in \mathcal{X}^{(i)}$, $f(\boldsymbol{x}^*) - f(\boldsymbol{x}) \ge c_{\text{quad}}\|\boldsymbol{x}-\boldsymbol{x}^*\|_2^2$ from condition 3 in Lemma 2. Since $\forall \boldsymbol{x} \in \mathcal{X}^{(i)}$, $f(\boldsymbol{x}^*) - f(\boldsymbol{x}) \le \eta_i$ from $\mathcal{T}_i \subset \mathcal{T}^c(\eta_i)$, we have $\eta_i \ge c_{\text{quad}}\|\boldsymbol{x}-\boldsymbol{x}^*\|_2^2 \Leftrightarrow \sqrt{c_{\text{quad}}^{-1}\eta_i} \ge \|\boldsymbol{x}-\boldsymbol{x}^*\|_2$, which implies $\mathcal{X}^{(i)} \subset \mathcal{B}_2(\sqrt{c_{\text{quad}}^{-1}\eta_i};\boldsymbol{x}^*)$. Therefore, we have

$$\gamma_{(T/2^{i-1})}(\mathcal{X}^{(i)}) \le \gamma_{(T/2^{i-1})}\left(\mathcal{B}_2\left(\sqrt{c_{\text{quad}}^{-1}\eta_i};\boldsymbol{x}^*\right)\right). \tag{37}$$

Finally, combining Eq. (35) with Eq. (37), we have

$$R_T^{(2)}(\varepsilon) \leq 2c_{\sup}\overline{T} + \frac{\pi^2}{3}\left(\log_2 T + 1\right) + \frac{2\sqrt{2C\beta_T T}}{\sqrt{2}-1}\max_{i\in[\bar{i}]}\sqrt{\gamma_{(T/2^{i-1})}\left(\mathcal{B}_2\left(\sqrt{c_{\text{quad}}^{-1}\eta_i};\boldsymbol{x}^*\right)\right)}. \tag{38}$$

□

A.2 Proof of Lemma 5

To prove Lemma 5, we require the upper bound of MIG with the explicit dependence on the radius of the input domain. In Corollary 8 in Appendix B, we provide it with a full proof. Below, we establish the proof of Lemma 5 based on Corollary 8.

When $k = k_{\text{Matérn}}$. Set $C_{\text{Mat}} > 0$ as the constant such that the following inequalities hold:

$$\forall t \geq 2, \gamma_t(\mathcal{X}) \leq C_{\text{Mat}} t^{\frac{d}{2\nu+d}} \ln^{\frac{4\nu+d}{2\nu+d}} t, \tag{39}$$

$$\forall t \geq 2, \forall \eta > 0, \gamma_t\left(\{\boldsymbol{x} \in \mathbb{R}^d \mid \|\boldsymbol{x}\|_2 \leq \eta\}\right) \leq C_{\text{Mat}}\left(\eta^{\frac{2\nu d}{2\nu+d}} t^{\frac{d}{2\nu+d}} \ln^{\frac{4\nu+d}{2\nu+d}} t + \ln^2 t\right). \tag{40}$$

The existence of C_{Mat} is guaranteed by the upper bound of MIG established in Corollary 8[6]. Note that C_{Mat} is the constant that may depend on d, ℓ, ν, r, and σ^2. Furthermore, we set $\overline{\gamma}_t = C_{\text{Mat}} t^{\frac{d}{2\nu+d}} \ln^{\frac{4\nu+d}{2\nu+d}} t$. For function $g(t) \coloneqq \overline{\gamma}_t / t$, we have

$$g'(t) = -\frac{2\nu C_{\text{Mat}}}{2\nu+d} t^{-\frac{2\nu}{2\nu+d}-1} \ln^{\frac{4\nu+d}{2\nu+d}} t + C_{\text{Mat}} \frac{4\nu+d}{2\nu+d} t^{-\frac{2\nu}{2\nu+d}-1} \ln^{\frac{2\nu}{2\nu+d}} t \tag{41}$$

$$= \frac{C_{\text{Mat}}}{2\nu+d} t^{-\frac{2\nu}{2\nu+d}-1} (\ln^{\frac{2\nu}{2\nu+d}} t)\left(-2\nu \ln t + 4\nu + d\right). \tag{42}$$

From the above expression, if $2\nu \ln t \geq 4\nu + d \Leftrightarrow t \geq \exp(2 + d/(2\nu))$, $\overline{\gamma}_t / t$ is non-increasing. Therefore, we set $\overline{T} = \lceil \exp(2 + d/(2\nu)) \rceil$, which is independent of T. Here, for any $\eta > 0$ and $t \geq 2$, we have

$$\gamma_t\left(\mathcal{B}_2(\eta; \boldsymbol{x}^*)\right) \leq \gamma_t\left(\{\boldsymbol{x} \in \mathbb{R}^d \mid \|\boldsymbol{x} - \boldsymbol{x}^*\|_2 \leq \eta\}\right) \tag{43}$$

$$= \gamma_t\left(\{\boldsymbol{x} \in \mathbb{R}^d \mid \|\boldsymbol{x}\|_2 \leq \eta\}\right) \tag{44}$$

$$\leq C_{\text{Mat}}\left(\eta^{\frac{2\nu d}{2\nu+d}} t^{\frac{d}{2\nu+d}} \ln^{\frac{4\nu+d}{2\nu+d}} t + \ln^2 t\right), \tag{45}$$

where the second line follows from the fact that $k_{\text{Matérn}}$ is the stationary kernel (namely, $k_{\text{Matérn}}$ is translation invariant against any shift of inputs). Regarding η_i in Lemma 4, by setting T_i as $T_i = T/2^{i-1}$, we have

$$\eta_i = \frac{2\left(2\sqrt{C\beta_T T_i \overline{\gamma}_{T_i}} + \frac{\pi^2}{6}\right)}{T_i} \tag{46}$$

$$= \frac{4\sqrt{C\beta_T T_i \overline{\gamma}_{T_i}}}{T_i} + \frac{\pi^2}{3T_i} \tag{47}$$

$$= \frac{4\sqrt{C\beta_T T_i C_{\text{Mat}} T_i^{\frac{d}{2\nu+d}} \ln^{\frac{4\nu+d}{2\nu+d}} T_i}}{T_i} + \frac{\pi^2}{3T_i} \tag{48}$$

$$= 4\sqrt{CC_{\text{Mat}}\beta_T}\left(T_i^{-\frac{\nu}{2\nu+d}} \ln^{\frac{4\nu+d}{4\nu+2d}} T_i\right) + \frac{\pi^2}{3T_i} \tag{49}$$

$$\leq \widetilde{C}_{\text{Mat}}\sqrt{\beta_T}\left(T_i^{-\frac{\nu}{2\nu+d}} \ln^{\frac{4\nu+d}{4\nu+2d}} T_i\right), \tag{50}$$

[6] If we rely on the result in [60], we can tighten the logarithmic term from $\ln^{\frac{4\nu+d}{2\nu+d}} t$ to $\ln^{\frac{2\nu}{2\nu+d}} t$; however, due to the technical issue of [60] described in Appendix B, we base our proof on Corollary 8.

where $\widetilde{C}_{\mathrm{Mat}} > 0$ is a sufficiently large constant such that $\widetilde{C}_{\mathrm{Mat}}\sqrt{\beta_T}\left(T_i^{-\frac{\nu}{2\nu+d}}\ln^{\frac{4\nu+d}{4\nu+2d}} T_i\right) \geq 4\sqrt{CC_{\mathrm{Mat}}\beta_T}\left(T_i^{-\frac{\nu}{2\nu+d}}\ln^{\frac{4\nu+d}{4\nu+2d}} T_i\right) + \frac{\pi^2}{3T_i}$ for any $T_i \geq 2$. Note that we can choose $\widetilde{C}_{\mathrm{Mat}} > 0$ without depending on T. From Eqs. (45) and (50), for any i, we have

$$\gamma_{T/2^{i-1}}\left(\mathcal{B}_2\left(\sqrt{c_{\mathrm{quad}}^{-1}\eta_i};\boldsymbol{x}^*\right)\right) \tag{51}$$

$$\leq C_{\mathrm{Mat}}\left(c_{\mathrm{quad}}^{-\frac{\nu d}{2\nu+d}}\eta_i^{\frac{\nu d}{2\nu+d}}T_i^{\frac{d}{2\nu+d}}\ln^{\frac{4\nu+d}{2\nu+d}} T_i + \ln^2 T_i\right) \tag{52}$$

$$\leq C_{\mathrm{Mat}}\left[c_{\mathrm{quad}}^{-\frac{\nu d}{2\nu+d}}\widetilde{C}_{\mathrm{Mat}}^{\frac{\nu d}{2\nu+d}}\beta_T^{\frac{\nu d}{2(2\nu+d)}}\left(T_i^{-\frac{\nu}{2\nu+d}}\ln^{\frac{4\nu+d}{4\nu+2d}} T_i\right)^{\frac{\nu d}{2\nu+d}}T_i^{\frac{d}{2\nu+d}}\ln^{\frac{4\nu+d}{2\nu+d}} T_i + \ln^2 T\right]. \tag{53}$$

Furthermore, by noting condition $2\nu + d \leq \nu^2$, we have

$$\left(T_i^{-\frac{\nu}{2\nu+d}}\ln^{\frac{4\nu+d}{4\nu+2d}} T_i\right)^{\frac{\nu d}{2\nu+d}}T_i^{\frac{d}{2\nu+d}}\ln^{\frac{4\nu+d}{2\nu+d}} T_i = \widetilde{O}\left(T_i^{-\frac{\nu^2 d}{(2\nu+d)^2}+\frac{d}{2\nu+d}}\right) \tag{54}$$

$$= \widetilde{O}\left(T_i^{\frac{d(2\nu+d)-\nu^2 d}{(2\nu+d)^2}}\right) \tag{55}$$

$$= \widetilde{O}\left(T_i^{\frac{d(2\nu+d-\nu^2)}{(2\nu+d)^2}}\right) \tag{56}$$

$$= \widetilde{O}(1). \tag{57}$$

From the above inequalities, we have $\gamma_{T/2^{i-1}}\left(\mathcal{B}_2\left(\sqrt{c_{\mathrm{quad}}^{-1}\eta_i};\boldsymbol{x}^*\right)\right) = \widetilde{O}(1)$. Therefore, Lemma 4 implies

$$R_T^{(2)}(\varepsilon) \leq 2c_{\mathrm{sup}}\overline{T} + \frac{\pi^2}{3}\left(\log_2 T + 1\right) + \frac{2\sqrt{2C\beta_T T}}{\sqrt{2}-1}\times\widetilde{O}(1) \tag{58}$$

$$= \widetilde{O}(\sqrt{T}). \tag{59}$$

When $k = k_{\mathrm{SE}}$. The proof for $k = k_{\mathrm{SE}}$ is not as straightforward as the proof for $k = k_{\text{Matérn}}$. Specifically, we have to choose a proper $\overline{T}$ so as to obtain an $O(\ln T)$ upper bound of MIG. Let $C_{\mathrm{SE}} > 0$ be the constant such that the following inequalities hold:

$$\forall t \geq 2, \gamma_t(\mathcal{X}) \leq C_{\mathrm{SE}}\ln^{d+1} t, \tag{60}$$

$$\forall t \geq 2, \forall \eta \in (0, \sqrt{\frac{2\ell^2}{e^2 c_d}}), \gamma_t(\{\boldsymbol{x} \in \mathbb{R}^d \mid \|\boldsymbol{x}\|_2 \leq \eta\}) \leq C_{\mathrm{SE}}\left(\frac{\ln^{d+1} t}{\ln^d\left(\frac{2\ell^2}{\eta^2 e c_d}\right)} + \ln t\right). \tag{61}$$

The existence of such C_{SE} is guaranteed by Corollary 8. In the above inequalities, c_d is the constant defined in Corollary 8. We also set $\overline{\gamma}_t$ as $\overline{\gamma}_t = C_{\mathrm{SE}}\ln^{d+1} t$. We choose $\overline{T}$ later such that we can leverage the second statement in the above inequalities. Under $k = k_{\mathrm{SE}}$, we have

$$\eta_i = \frac{2\left(2\sqrt{C\beta_T T_i\overline{\gamma}_{T_i}} + \frac{\pi^2}{6}\right)}{T_i} \tag{62}$$

$$= \frac{4\sqrt{C\beta_T T_i\overline{\gamma}_{T_i}}}{T_i} + \frac{\pi^2}{3T_i} \tag{63}$$

$$= \frac{4\sqrt{C\beta_T T_i C_{\mathrm{SE}}\ln^{d+1} T_i}}{T_i} + \frac{\pi^2}{3T_i} \tag{64}$$

$$= 4\sqrt{CC_{\mathrm{SE}}\beta_T}\left(T_i^{-\frac{1}{2}}\ln^{\frac{d+1}{2}} T_i\right) + \frac{\pi^2}{3T_i} \tag{65}$$

$$\leq \widetilde{C}_{\mathrm{SE}}\sqrt{\beta_T}\left(T_i^{-\frac{1}{2}}\ln^{\frac{d+1}{2}} T_i\right), \tag{66}$$

where $\widetilde{C}_{\mathrm{SE}} > 0$ is a sufficiently large constant such that $\widetilde{C}_{\mathrm{SE}}\sqrt{\beta_T}\left(T_i^{-\frac{1}{2}}\ln^{\frac{d+1}{2}} T_i\right) \geq 4\sqrt{CC_{\mathrm{SE}}\beta_T}\left(T_i^{-\frac{1}{2}}\ln^{\frac{d+1}{2}} T_i\right) + \frac{\pi^2}{3T_i}$ for any $T_i \geq 2$. Hereafter, we define $\overline{\eta}_i \coloneqq \widetilde{C}_{\mathrm{SE}}\sqrt{\beta_T}\left(T_i^{-\frac{1}{2}}\ln^{\frac{d+1}{2}} T_i\right)$. Then, to apply Eq. (61), we consider the lower bound of T_i such that $\sqrt{c_{\mathrm{quad}}^{-1}\overline{\eta}_i} < \sqrt{2\ell^2/(e^2c_d)}$ holds. From the condition $\sqrt{c_{\mathrm{quad}}^{-1}\overline{\eta}_i} < \sqrt{2\ell^2/(e^2c_d)}$, we have

$$\sqrt{c_{\mathrm{quad}}^{-1}\overline{\eta}_i} < \sqrt{\frac{2\ell^2}{e^2c_d}} \Leftrightarrow c_{\mathrm{quad}}^{-1}\frac{e^2c_d}{2\ell^2}\widetilde{C}_{\mathrm{SE}}\sqrt{\beta_T}\ln^{\frac{d+1}{2}} T_i < T_i^{\frac{1}{2}} \tag{67}$$

$$\Leftarrow c_{\mathrm{quad}}^{-1}\frac{e^2c_d}{2\ell^2}\widetilde{C}_{\mathrm{SE}}\sqrt{\beta_T}\ln^{\frac{d+1}{2}} T < T_i^{\frac{1}{2}} \tag{68}$$

$$\Leftrightarrow \left(\frac{e^2c_d\widetilde{C}_{\mathrm{SE}}}{2\ell^2c_{\mathrm{quad}}}\right)^2 \beta_T \ln^{d+1} T < T_i. \tag{69}$$

From the above inequality, we set $\overline{T}$ such that

$$\left(\frac{e^2c_d\widetilde{C}_{\mathrm{SE}}}{2\ell^2c_{\mathrm{quad}}}\right)^2 \beta_T \ln^{d+1} T < \overline{T}. \tag{70}$$

Then, from $T_i \geq \overline{T}$ and Eqs. (67), and (70),

$$\gamma_{T/2^{i-1}}\left(\mathcal{B}_2\left(\sqrt{c_{\mathrm{quad}}^{-1}\eta_i}; \boldsymbol{x}^*\right)\right) \leq \gamma_{T_i}\left(\left\{\boldsymbol{x} \in \mathbb{R}^d \mid \|\boldsymbol{x}\|_2 \leq \sqrt{c_{\mathrm{quad}}^{-1}\eta_i}\right\}\right) \tag{71}$$

$$\leq C_{\mathrm{SE}}\left(\frac{\ln^{d+1} T_i}{\ln^d\left(\frac{2c_{\mathrm{quad}}\ell^2}{\overline{\eta}_i e c_d}\right)} + \ln T\right). \tag{72}$$

Based on Eq. (72), we further consider the lower bound of T_i such that

$$\frac{\ln^{d+1} T_i}{\ln^d\left(\frac{2c_{\mathrm{quad}}\ell^2}{\overline{\eta}_i e c_d}\right)} = O(\ln T). \tag{73}$$

For the condition in Eq. (73), we have

$$\frac{2c_{\mathrm{quad}}\ell^2}{\overline{\eta}_i e c_d} \geq T_i^{1/4} \Leftrightarrow \frac{2c_{\mathrm{quad}}\ell^2}{e c_d\widetilde{C}_{\mathrm{SE}}\sqrt{\beta_T}T_i^{-1/2}\ln^{\frac{d+1}{2}} T_i} \geq T_i^{1/4} \tag{74}$$

$$\Leftrightarrow T_i^{1/4} \geq \frac{e c_d\widetilde{C}_{\mathrm{SE}}\sqrt{\beta_T}\ln^{\frac{d+1}{2}} T_i}{2c_{\mathrm{quad}}\ell^2} \tag{75}$$

$$\Leftarrow T_i^{1/4} \geq \frac{e c_d\widetilde{C}_{\mathrm{SE}}\sqrt{\beta_T}\ln^{\frac{d+1}{2}} T}{2c_{\mathrm{quad}}\ell^2} \tag{76}$$

$$\Leftrightarrow T_i \geq \left(\frac{e c_d\widetilde{C}_{\mathrm{SE}}\sqrt{\beta_T}\ln^{\frac{d+1}{2}} T}{2c_{\mathrm{quad}}\ell^2}\right)^4. \tag{77}$$

Hence, if $\overline{T} \geq \left(\frac{e c_d\widetilde{C}_{\mathrm{SE}}\sqrt{\beta_T}\ln^{\frac{d+1}{2}} T}{2c_{\mathrm{quad}}\ell^2}\right)^4$, we have

$$C_{\mathrm{SE}}\left(\frac{\ln^{d+1} T_i}{\ln^d\left(\frac{2c_{\mathrm{quad}}\ell^2}{\overline{\eta}_i e c_d}\right)} + \ln T\right) \leq C_{\mathrm{SE}}\left(\frac{\ln^{d+1} T_i}{4^{-d}\ln^d T_i} + \ln T\right) \tag{78}$$

$$\leq C_{\mathrm{SE}}\left(4^d \ln T + \ln T\right). \tag{79}$$

By aggregating the conditions (70) and (77), we set $\overline{T}$ as the smallest natural number such that the following inequalities hold:

$$\overline{T} \geq \left(\frac{e^2 c_d \widetilde{C}_{\mathrm{SE}}}{2\ell^2 c_{\mathrm{quad}}}\right)^2 \beta_T \ln^{d+1} T, \quad \text{and} \quad \overline{T} \geq \left(\frac{e c_d}{2 c_{\mathrm{quad}} \ell^2}\right)^4 \widetilde{C}_{\mathrm{SE}}^4 \beta_T^2 \ln^{2(d+1)} T. \tag{80}$$

Then, from Eqs. (72) and (79), we have

$$\sqrt{\gamma_{(T/2^{i-1})}\left(\mathcal{B}_2\left(\sqrt{c_{\mathrm{quad}}^{-1}}\eta_i; \boldsymbol{x}^*\right)\right)} = O(\sqrt{\ln T}). \tag{81}$$

Finally, by noting $\overline{T} = O(\ln^{2d+4} T)$, we obtain the following result from Lemma 4:

$$R_T^{(2)}(\varepsilon) = O\left(\ln^{2d+4} T + \sqrt{T \ln^2 T}\right). \tag{82}$$

Since d is a fixed constant, the above equation implies $R_T^{(2)}(\varepsilon) = O(\sqrt{T \ln^2 T})$. □

A.3 Proof of Lemma 6

Proof. From the upper bound of the discretization error in event $\mathcal{A}$, we have $\forall t \geq \sqrt{2/\varepsilon}, \forall \boldsymbol{x} \in \mathcal{X}, |f(\boldsymbol{x}) - f([\boldsymbol{x}]_t)| \leq \varepsilon/2$. Here, we set $\underline{\mathcal{T}}(\varepsilon) = \{t \in \mathbb{N}_+ \mid t \geq \sqrt{2/\varepsilon}\}$. By relying on the standard argument of MIG [53], we observe the following inequality for any realizations and $\varepsilon > 0$:

$$\min_{t \in \mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon)} \sigma(\boldsymbol{x}_t; \mathbf{X}_{t-1}) \leq \sqrt{\frac{C \gamma_{|\mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon)|}(\mathcal{X})}{|\mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon)|}}, \tag{83}$$

where $\mathcal{T}(\varepsilon) = \{t \in [T] \mid f(\boldsymbol{x}^*) - f(\boldsymbol{x}_t) > \varepsilon\}$ and $C = 2/\ln(1 + \sigma^{-2})$. Under $\mathcal{A}$, we further have the following inequalities for any $\widetilde{t} \in \operatorname{argmin}_{t \in \mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon)} \sigma(\boldsymbol{x}_t; \mathbf{X}_{t-1})$[7]:

$$\begin{aligned}
&\mu(\boldsymbol{x}_{\widetilde{t}}; \mathbf{X}_{\widetilde{t}-1}, \boldsymbol{y}_{\widetilde{t}-1}) + \beta_{\widetilde{t}}^{1/2} \sigma(\boldsymbol{x}_{\widetilde{t}}; \mathbf{X}_{\widetilde{t}-1}) && (84)\\
&= \mu(\boldsymbol{x}_{\widetilde{t}}; \mathbf{X}_{\widetilde{t}-1}, \boldsymbol{y}_{\widetilde{t}-1}) - \beta_{\widetilde{t}}^{1/2} \sigma(\boldsymbol{x}_{\widetilde{t}}; \mathbf{X}_{\widetilde{t}-1}) + 2\beta_{\widetilde{t}}^{1/2} \sigma(\boldsymbol{x}_{\widetilde{t}}; \mathbf{X}_{\widetilde{t}-1}) && (85)\\
&\leq f(\boldsymbol{x}_{\widetilde{t}}) + 2\beta_{\widetilde{t}}^{1/2} \sigma(\boldsymbol{x}_{\widetilde{t}}; \mathbf{X}_{\widetilde{t}-1}) && (86)\\
&< f(\boldsymbol{x}^*) - \varepsilon + 2\sqrt{\frac{C \beta_{\widetilde{t}} \gamma_{|\mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon)|}(\mathcal{X})}{|\mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon)|}} && (87)\\
&\leq |f(\boldsymbol{x}^*) - f([\boldsymbol{x}^*]_{\widetilde{t}})| + f([\boldsymbol{x}^*]_{\widetilde{t}}) - \varepsilon + 2\sqrt{\frac{C \beta_{\widetilde{t}} \gamma_{|\mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon)|}(\mathcal{X})}{|\mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon)|}} && (88)\\
&\leq \mu([\boldsymbol{x}^*]_{\widetilde{t}}; \mathbf{X}_{\widetilde{t}-1}, \boldsymbol{y}_{\widetilde{t}-1}) + \beta_{\widetilde{t}}^{1/2} \sigma([\boldsymbol{x}^*]_{\widetilde{t}}; \mathbf{X}_{\widetilde{t}-1}) - \frac{\varepsilon}{2} + 2\sqrt{\frac{C \beta_{\widetilde{t}} \gamma_{|\mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon)|}(\mathcal{X})}{|\mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon)|}}, && (89)
\end{aligned}$$

where the second inequality follows from the definition of $\mathcal{T}(\varepsilon)$, and the last inequality follows from $\widetilde{t} \in \underline{\mathcal{T}}(\varepsilon)$ and event $\mathcal{A}$. Therefore, under $\mathcal{A}$, the inequality $-\frac{\varepsilon}{2} + 2\sqrt{\frac{C \beta_{\widetilde{t}} \gamma_{|\mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon)|}(\mathcal{X})}{|\mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon)|}} \geq 0$ must hold; otherwise, $\mu(\boldsymbol{x}_{\widetilde{t}}; \mathbf{X}_{\widetilde{t}-1}, \boldsymbol{y}_{\widetilde{t}-1}) + \beta_{\widetilde{t}}^{1/2} \sigma(\boldsymbol{x}_{\widetilde{t}}; \mathbf{X}_{\widetilde{t}-1}) < \mu([\boldsymbol{x}^*]_{\widetilde{t}}; \mathbf{X}_{\widetilde{t}-1}, \boldsymbol{y}_{\widetilde{t}-1}) + \beta_{\widetilde{t}}^{1/2} \sigma([\boldsymbol{x}^*]_{\widetilde{t}}; \mathbf{X}_{\widetilde{t}-1})$, which contradicts $\boldsymbol{x}_{\widetilde{t}} \in \operatorname{argmax}_{\boldsymbol{x} \in \mathcal{X}} \mu(\boldsymbol{x}; \mathbf{X}_{\widetilde{t}-1}, \boldsymbol{y}_{\widetilde{t}-1}) + \beta_{\widetilde{t}}^{1/2} \sigma(\boldsymbol{x}; \mathbf{X}_{\widetilde{t}-1})$. This further implies

$$|\mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon)| \leq \frac{16 C \beta_{\widetilde{t}} \gamma_{|\mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon)|}(\mathcal{X})}{\varepsilon^2} \leq \frac{16 C \beta_T \gamma_{|\mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon)|}(\mathcal{X})}{\varepsilon^2} \tag{90}$$

[7] If $\mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon) = \emptyset$, the theorem's statement clearly holds; therefore, we suppose $\mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon) \neq \emptyset$ in this proof.

for any $\varepsilon > 0$. Furthermore,

$$R_T^{(1)}(\varepsilon) = \sum_{t \in \mathcal{T}(\varepsilon)} f(\boldsymbol{x}^*) - f(\boldsymbol{x}_t) \tag{91}$$

$$\leq 2c_{\sup}\sqrt{\frac{2}{\varepsilon}} + \sum_{t \in \mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon)} f(\boldsymbol{x}^*) - f(\boldsymbol{x}_t) \tag{92}$$

$$\leq 2c_{\sup}\sqrt{\frac{2}{\varepsilon}} + \frac{\pi^2}{6} + 2\sqrt{C\beta_T |\mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon)| \gamma_{|\mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon)|}(\mathcal{X})} \tag{93}$$

for any $\varepsilon > 0$. In the above expressions, the last inequality follows from Lemma 21. The remaining part of the proof is to substitute the quantity $|\mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon)|$ in Eq. (93) into its upper bound, which is deduced from Eq. (90) depending on the kernel.

For $k = k_{\mathrm{SE}}$. Under $k = k_{\mathrm{SE}}$, we crudely take the upper bound of $|\mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon)|$ as

$$|\mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon)| \leq \frac{16C\beta_T \gamma_{|\mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon)|}(\mathcal{X})}{\varepsilon^2} \leq \frac{16C\beta_T \gamma_T(\mathcal{X})}{\varepsilon^2}. \tag{94}$$

The above upper bound implies $|\mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon)| = O\left(\beta_T \gamma_T(\mathcal{X})\right)$. Since $\gamma_T(\mathcal{X}) = O(\ln^{d+1} T)$ under $k = k_{\mathrm{SE}}$, Eq. (93) implies

$$R_T^{(1)}(\varepsilon) \leq 2c_{\sup}\sqrt{\frac{2}{\varepsilon}} + \frac{\pi^2}{6} + O\left(\sqrt{\beta_T (\beta_T \gamma_T(\mathcal{X})) \ln^{d+1}(\beta_T \gamma_T(\mathcal{X}))}\right) \tag{95}$$

$$= O\left(\beta_T \sqrt{(\ln^{d+1} T) \ln^{d+1}(\ln^{d+2} T)}\right) \tag{96}$$

$$= O\left(\sqrt{(\ln T)^{d+3} (\ln \ln T)^{d+1}}\right) \tag{97}$$

$$= \widetilde{O}(1). \tag{98}$$

For $k = k_{\text{Matérn}}$. Set $C_{\mathrm{Mat}} > 0$ as the constant such that the following inequality holds:

$$\forall t \geq 2, \gamma_t(\mathcal{X}) \leq C_{\mathrm{Mat}} t^{\frac{d}{2\nu+d}} \ln^{\frac{4\nu+d}{2\nu+d}} t. \tag{99}$$

The existence of C_{Mat} is guaranteed by the upper bound of MIG established in Corollary 8. Then, if $|\mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon)| \geq 2$ holds, Eq. (90) implies

$$|\mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon)| \leq \frac{16C\beta_T C_{\mathrm{Mat}} |\mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon)|^{\frac{d}{2\nu+d}} \ln^{\frac{4\nu+d}{2\nu+d}} |\mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon)|}{\varepsilon^2} \tag{100}$$

$$\Rightarrow |\mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon)| \leq \frac{16C\beta_T C_{\mathrm{Mat}} |\mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon)|^{\frac{d}{2\nu+d}} \ln^{\frac{4\nu+d}{2\nu+d}} T}{\varepsilon^2} \tag{101}$$

$$\Leftrightarrow |\mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon)|^{\frac{2\nu}{2\nu+d}} \leq \frac{16C\beta_T C_{\mathrm{Mat}} \ln^{\frac{4\nu+d}{2\nu+d}} T}{\varepsilon^2} \tag{102}$$

$$\Leftrightarrow |\mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon)| \leq \left(\frac{16C\beta_T C_{\mathrm{Mat}} \ln^{\frac{4\nu+d}{2\nu+d}} T}{\varepsilon^2}\right)^{1+\frac{d}{2\nu}}. \tag{103}$$

Therefore, we have $|\mathcal{T}(\varepsilon) \cap \underline{\mathcal{T}}(\varepsilon)| = \widetilde{O}(1)$ under fixed ε, d, and ν. Hence, from Eq. (93), we obtain $R_T^{(1)}(\varepsilon) = \widetilde{O}(1)$. □

B Information Gain Upper Bound

Our analysis requires the upper bound of MIG with explicit dependence on the radius of the input domain. Several existing works [5, 28, 29] established such a result by extending the proof in [53]. However, the proof strategy in [53] results in $\widetilde{O}(T^{\frac{d(d+1)}{2\nu+d(d+1)}})$ upper bound of MIG in Matérn kernel, which is strictly worse than the best achievable $\widetilde{O}(T^{\frac{d}{2\nu+d}})$ upper bound. Vakili et al. [60] show

$\widetilde{O}(T^{\frac{d}{2\nu+d}})$ upper bound of MIG with $\nu > 1/2$ under the uniform boundedness assumption of the eigenfunctions. Furthermore, the following work [57] shows $\gamma_T(\{\boldsymbol{x} \in \mathbb{R}^d \mid \|\boldsymbol{x}\|_2 \leq \eta\}) = \widetilde{O}(\eta^{\frac{2\nu d}{2\nu+d}} T^{\frac{d}{2\nu+d}})$ for any radius $\eta > 0$ if there exist eigenfunctions uniformly bounded without depending on $\eta > 0$. Some of the related results support the uniform boundedness assumption under $d = 1$ [28, 64], or under the approximated version of the original Matérn kernel [43, 52]; however, to our knowledge, we are not aware of any literature that rigorously supports the uniform boundedness assumption under the general compact input domain with $d \geq 2$ and $\nu > 1/2$. See Chapter 4.4 in [28] for the detailed discussion. Therefore, this section's goal is twofold: (i) prove $\widetilde{O}(T^{\frac{d}{2\nu+d}})$ upper bound as claimed in [60] without relying on the uniform boundedness assumption, and (ii) clarify the explicit dependence on the input radius in the upper bound proved in (i).

Below, we formally describe our MIG upper bound.

Theorem 7. *Fix any $d \in \mathbb{N}_+$, $\sigma^2 > 0$, and $T \in \mathbb{N}_+$. Let us assume $\mathcal{X} = \{\boldsymbol{x} \in \mathbb{R}^d \mid \|\boldsymbol{x}\|_2 \leq 1\}$. Then,*

- *For $k = k_{\mathrm{SE}}$, $\gamma_T(\mathcal{X})$ satisfies*

$$\gamma_T(\mathcal{X}) \leq \frac{C_d^{(1)}}{\theta^d} \ln^{d+1}\left(1 + \frac{T}{\sigma^2}\right) + \ln\left(1 + \frac{T}{\sigma^2}\right) + C_d^{(2)} \exp\left(-\frac{2}{\theta} + \frac{1}{\theta^2}\right) \tag{104}$$

  if $\theta \leq e^2 c_d$ and $T/(e-1) \geq \sigma^2$. Furthermore, for any $\theta > e^2 c_d$, we have

$$\gamma_T(\mathcal{X}) \leq \frac{C_d^{(3)}}{\ln^d\left(\frac{\theta}{e c_d}\right)} \ln^{d+1}\left(1 + \frac{T}{\sigma^2}\right) + C_d^{(4)} \ln\left(1 + \frac{T}{\sigma^2}\right) + C_d^{(5)}. \tag{105}$$

  Here, we set $\theta = 2\ell^2$ and $c_d = \max\left\{1, \exp\left(\frac{1}{e}\left(\frac{d}{2} - 1\right)\right)\right\}$. Furthermore, $C_d^{(1)}, C_d^{(2)}, C_d^{(3)}, C_d^{(4)}, C_d^{(5)} > 0$ are the constants only depending on d.

- *For $k = k_{\text{Matérn}}$ with $\nu > 1/2$, $\gamma_T(\mathcal{X})$ satisfies*

$$\gamma_T(\mathcal{X}) \leq C(T, \nu, \sigma^2)\overline{\gamma}_T + C \tag{106}$$

  where $C(T, \nu, \sigma^2) = \max\left\{1, \log_2 \max\left\{1, 2\ln\frac{2C_\nu T^2}{\Gamma(\nu)\sigma^2}\right\} + \frac{1}{\nu}\log_2\left(\frac{T^2}{\nu\Gamma(\nu)\sigma^2}\right) + 1\right\}$. Here, $C_\nu > 0$ and $C > 0$ are, respectively, a constant depending only on $\nu > 0$ and an absolute constant. Furthermore, $\overline{\gamma}_T$ is defined as

$$\overline{\gamma}_T = C_{d,\nu}^{(1)} \ln\left(1 + \frac{2T}{\sigma^2}\right) + C_{d,\nu}^{(2)} \left(\frac{T}{\sigma^2 \ell^{2\nu}}\right)^{\frac{d}{2\nu+d}} \ln^{\frac{2\nu}{2\nu+d}}\left(1 + \frac{2T}{\sigma^2}\right), \tag{107}$$

  where $C_{d,\nu}^{(1)}, C_{d,\nu}^{(2)} > 0$ are the constants only depending on d and ν.

We also obtain the following corollary by adjusting the lengthscale parameter $\ell > 0$ based on the radius of the input domain.

Corollary 8. *Fix any $d \in \mathbb{N}_+$, $\sigma^2 > 0$, $T \in \mathbb{N}_+$, $\eta > 0$. Let us assume $\mathcal{X} = \{\boldsymbol{x} \in \mathbb{R}^d \mid \|\boldsymbol{x}\|_2 \leq \eta\}$. Then,*

- *For $k = k_{\mathrm{SE}}$, $\gamma_T(\mathcal{X})$ satisfies*

$$\gamma_T(\mathcal{X}) \leq \frac{C_d^{(3)}}{\ln^d\left(\frac{2\ell^2}{\eta^2 e c_d}\right)} \ln^{d+1}\left(1 + \frac{T}{\sigma^2}\right) + C_d^{(4)} \ln\left(1 + \frac{T}{\sigma^2}\right) + C_d^{(5)} \tag{108}$$

  if $2\ell^2/\eta^2 > e^2 c_d$.

- *For $k = k_{\text{Matérn}}$ with $\nu > 1/2$, $\gamma_T(\mathcal{X})$ satisfies Eq. (106), with*

$$\overline{\gamma}_T = C_{d,\nu}^{(1)} \ln\left(1 + \frac{2T}{\sigma^2}\right) + C_{d,\nu}^{(2)} \eta^{\frac{2\nu d}{2\nu+d}} \left(\frac{T}{\sigma^2 \ell^{2\nu}}\right)^{\frac{d}{2\nu+d}} \ln^{\frac{2\nu}{2\nu+d}}\left(1 + \frac{2T}{\sigma^2}\right). \tag{109}$$

The constants in the above statements are the same as those in Theorem 7.

While the above results ignore the explicit dependence on d and ν, all the other parameters are explicitly stated in our upper bound of MIG. We would like to emphasize that we do not rely on the uniform boundedness assumption of the eigenfunctions to prove the above results. In the above results for $k = k_{\mathrm{SE}}$, we obtain the same $O(\ln^{d+1} T)$ upper bound as that in [60] except for the constant factor. For $k = k_{\text{Matérn}}$, we also obtain the same $\widetilde{O}(T^{\frac{d}{2\nu+d}})$ upper bound as that in [60], while the logarithmic dependence becomes worse from $O\left(\ln^{(2\nu)/(2\nu+d)} T\right)$ to $O\left(\ln^{(4\nu+d)/(2\nu+d)} T\right)$. Furthermore, the above result reveals the explicit dependence on the radius η of the input domain. Regarding the case $k = k_{\text{Matérn}}$, our result suggests $\widetilde{O}(\eta^{\frac{2\nu d}{2\nu+d}} T^{\frac{d}{2\nu+d}})$-upper bound of MIG, which is consistent with that in [57] with a uniform boundedness assumption.

Proof overview. The basic proof strategy follows that in [60], which leverages the Mercer decomposition of the kernel. To bypass the uniform boundedness assumption in the proof of [60], we must resort to other specific properties of the eigenfunctions. However, except for some exceptional cases, the eigenfunction of the kernel on the general compact domain is difficult to specify in an analytical form and complex to analyze. To avoid this issue, instead of studying the original definition of the MIG on $\mathbb{R}^d$, we consider reducing the original MIG on $\mathcal{X} \coloneqq \{\boldsymbol{x} \in \mathbb{R}^d \mid \|\boldsymbol{x}\|_2 \leq 1\}$ to that on a hypersphere $\mathbb{S}^d \coloneqq \{\boldsymbol{x} \in \mathbb{R}^{d+1} \mid \|\boldsymbol{x}\|_2 = 1\}$ defined in $\mathbb{R}^{d+1}$. The eigensystems on $\mathbb{S}^d$ are one of the exceptional cases, whose eigenfunctions are given by *spherical harmonics* on $\mathbb{S}^d$ [1, 17]. Indeed, by using the addition theorem of the spherical harmonics (Lemma 14), the existing works [26, 32, 59] already demonstrated that the upper bound of MIG on $\mathbb{S}^d$ is proved as in [60] without the uniform boundedness assumption. We use their technique to show the upper bound of MIG under SE and Matérn kernels on $\mathbb{R}^d$, while the original motivation of these existing works is to study the MIG under the neural tangent kernel on a hypersphere. The remaining parts of this section are constructed as follows:

- In Section B.1, we show our core result (Lemma 9) that guarantees the MIG on $\{\boldsymbol{x} \in \mathbb{R}^d \mid \|\boldsymbol{x}\|_2 \leq 1\}$ is bounded from above by that on $\mathbb{S}^d$ up to logarithmic factor.
- In Section B.2, we summarize the basic known results about Mercer decomposition on $\mathbb{S}^d$, which is the foundation of the following subsections.
- In Section B.3, we provide the general upper bound of the information gain on $\mathbb{S}^d$ (Lemma 15), represented by the kernel function's eigenvalues. The argument is essentially the same as in [32, 59]; we include it for completeness.
- In Section B.4, we provide the upper bound of the decaying rate of the eigenvalues in SE and Matérn kernels.
- In Section B.5, we establish the full proof of Theorem 7 based on the results in Sections B.1–B.4.

B.1 Reduction of the MIG on $\mathbb{R}^d$ to $\mathbb{S}^d$

Lemma 9 (Reduction to the hypersphere in $\mathbb{R}^{d+1}$)**.** *Fix any $d \in \mathbb{N}_+$, $\sigma^2 > 0$, and $T \in \mathbb{N}_+$. Suppose $\mathcal{X} = \{(x_1, \ldots, x_d, 0)^\top \in \mathbb{R}^{d+1} \mid \sum_{i=1}^d x_i^2 \leq 1\}$, and define $\mathbb{S}^d$ as $\mathbb{S}^d = \{\boldsymbol{x} \in \mathbb{R}^{d+1} \mid \|\boldsymbol{x}\|_2 = 1\}$. Then,*

- *For $k = k_{\mathrm{SE}}$, we have*

$$\max_{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_T \in \mathcal{X}} \ln \det(\boldsymbol{I}_T + \sigma^{-2}\mathbf{K}(\mathbf{X}_T, \mathbf{X}_T)) \leq \max_{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_T \in \mathbb{S}^d} \ln \det(\boldsymbol{I}_T + \sigma^{-2}\mathbf{K}(\mathbf{X}_T, \mathbf{X}_T)). \quad (110)$$

- *For $k = k_{\text{Matérn}}$, we have*

$$\max_{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_T \in \mathcal{X}} \ln \det(\boldsymbol{I}_T + \sigma^{-2}\mathbf{K}(\mathbf{X}_T, \mathbf{X}_T)) \quad (111)$$

$$\leq C(T, \nu, \sigma^2) \max_{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_T \in \mathbb{S}^d} \ln \det(\boldsymbol{I}_T + 2\sigma^{-2}\mathbf{K}(\mathbf{X}_T, \mathbf{X}_T)) + C, \quad (112)$$

where $C(T, \nu, \sigma^2) = \max\left\{1, \log_2 \max\left\{1, 2\ln \frac{2C_\nu T^2}{\Gamma(\nu)\sigma^2}\right\} + \frac{1}{\nu}\log_2\left(\frac{T^2}{\nu\Gamma(\nu)\sigma^2}\right) + 1\right\}$. Here, $C_\nu > 0$ is the constant that only depends on $\nu > 0$. Furthermore, $C > 0$ is an absolute constant.

Proof. For any $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_T \in \mathcal{X}$, we construct the new input sequence $\widetilde{\boldsymbol{x}}_1, \ldots, \widetilde{\boldsymbol{x}}_T$ on $\mathbb{S}^d$, where $\widetilde{\boldsymbol{x}}_i = \left(x_{i,1}, \ldots, x_{i,d}, \sqrt{1 - \sum_{j=1}^d x_{i,j}^2}\right)^\top$.

Under $k = k_{\mathrm{SE}}$. It is enough to show the following inequality:

$$\det(\boldsymbol{I}_T + \sigma^{-2}\mathbf{K}(\mathbf{X}_T, \mathbf{X}_T)) \leq \det(\boldsymbol{I}_T + \sigma^{-2}\mathbf{K}(\widetilde{\mathbf{X}}_T, \widetilde{\mathbf{X}}_T)), \tag{113}$$

where $\widetilde{\mathbf{X}}_T = (\widetilde{\boldsymbol{x}}_1, \ldots, \widetilde{\boldsymbol{x}}_T)$. From the definition of $\widetilde{\boldsymbol{x}}_i$, we rewrite R.H.S. in the above inequality as

$$\det(\boldsymbol{I}_T + \sigma^{-2}\mathbf{K}(\widetilde{\mathbf{X}}_T, \widetilde{\mathbf{X}}_T)) = \det(\widetilde{\mathbf{K}} \odot (\boldsymbol{I}_T + \sigma^{-2}\mathbf{K}(\mathbf{X}_T, \mathbf{X}_T))), \tag{114}$$

where $[\widetilde{\mathbf{K}}]_{i,j} = k(\widetilde{\boldsymbol{x}}_i, \widetilde{\boldsymbol{x}}_j)/k(\boldsymbol{x}_i, \boldsymbol{x}_j)$. Here, $A \odot B$ denotes the Hadamard product of the matrices A and B. Then, Oppenheim inequality (e.g., Theorem 7.27 in [65]) implies $\det(A \odot B) \geq \det(B) \prod_i A_{ii}$ for any positive semi-definite matrices A and B. Therefore, if $\widetilde{\mathbf{K}}$ is a positive semi-definite matrix, Eq. (114) immediately implies

$$\det(\boldsymbol{I}_T + \sigma^{-2}\mathbf{K}(\widetilde{\mathbf{X}}_T, \widetilde{\mathbf{X}}_T)) \geq \det(\boldsymbol{I}_T + \sigma^{-2}\mathbf{K}(\mathbf{X}_T, \mathbf{X}_T)) \prod_{i \in [T]} \widetilde{\mathbf{K}}_{ii} = \det(\boldsymbol{I}_T + \sigma^{-2}\mathbf{K}(\mathbf{X}_T, \mathbf{X}_T)). \tag{115}$$

From the definition of k_{SE} and $\widetilde{\boldsymbol{x}}_i$, we have

$$\frac{k(\widetilde{\boldsymbol{x}}_i, \widetilde{\boldsymbol{x}}_j)}{k(\boldsymbol{x}_i, \boldsymbol{x}_j)} = \exp\left(-\frac{\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2^2 + \left(\sqrt{1 - \|\boldsymbol{x}_i\|_2^2} - \sqrt{1 - \|\boldsymbol{x}_j\|_2^2}\right)^2}{2\ell^2} + \frac{\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2^2}{2\ell^2}\right) \tag{116}$$

$$= \exp\left(-\frac{\left(\sqrt{1 - \|\boldsymbol{x}_i\|_2^2} - \sqrt{1 - \|\boldsymbol{x}_j\|_2^2}\right)^2}{2\ell^2}\right). \tag{117}$$

The above equation suggests that $\widetilde{\mathbf{K}}$ is equal to the kernel matrix of the one-dimensional SE-kernel, whose inputs are transformed by $\sqrt{1 - \|\cdot\|_2^2}$. Since the SE kernel is positive definite, the matrix $\widetilde{\mathbf{K}}$ is also positive semi-definite, and we complete the proof for $k = k_{\mathrm{SE}}$.

Under $k = k_{\text{Matérn}}$. Similarly to the proof for $k = k_{\mathrm{SE}}$, we consider the application of the Oppenheim inequality; however, the positive semi-definiteness of the element-wise quotient matrix $\widetilde{\mathbf{K}}$ is unknown for $k = k_{\text{Matérn}}$. To avoid this problem, we leverage the following representation of $k_{\text{Matérn}}$, which is given as the form of the lengthscale mixture of the SE kernel [56]:

$$k(\boldsymbol{x}, \widetilde{\boldsymbol{x}}) = \frac{1}{\Gamma(\nu)} \int_0^\infty z^{\nu-1} e^{-z} \exp\left(-\frac{\|\boldsymbol{x} - \widetilde{\boldsymbol{x}}\|_2^2}{2\ell^2 z \nu^{-1}}\right) \mathrm{d}z. \tag{118}$$

Based on the above representation, we decompose the original kernel function k into the following three components:

$$k(\boldsymbol{x}, \widetilde{\boldsymbol{x}}) = k_1(\boldsymbol{x}, \widetilde{\boldsymbol{x}}) + k_2(\boldsymbol{x}, \widetilde{\boldsymbol{x}}) + k_3(\boldsymbol{x}, \widetilde{\boldsymbol{x}}), \tag{119}$$

where:

$$k_1(\boldsymbol{x}, \widetilde{\boldsymbol{x}}) = \frac{1}{\Gamma(\nu)} \int_0^{\eta_1} z^{\nu-1} e^{-z} \exp\left(-\frac{\|\boldsymbol{x} - \widetilde{\boldsymbol{x}}\|_2^2}{2\ell^2 z \nu^{-1}}\right) \mathrm{d}z, \tag{120}$$

$$k_2(\boldsymbol{x}, \widetilde{\boldsymbol{x}}) = \frac{1}{\Gamma(\nu)} \int_{\eta_1}^{\eta_2} z^{\nu-1} e^{-z} \exp\left(-\frac{\|\boldsymbol{x} - \widetilde{\boldsymbol{x}}\|_2^2}{2\ell^2 z \nu^{-1}}\right) \mathrm{d}z, \tag{121}$$

$$k_3(\boldsymbol{x}, \widetilde{\boldsymbol{x}}) = \frac{1}{\Gamma(\nu)} \int_{\eta_2}^{\infty} z^{\nu-1} e^{-z} \exp\left(-\frac{\|\boldsymbol{x} - \widetilde{\boldsymbol{x}}\|_2^2}{2\ell^2 z \nu^{-1}}\right) \mathrm{d}z \tag{122}$$

with some $\eta_2 > \eta_1 > 0$[8]. Then, as with the proof of Theorem 3 in [35], we have

$$\begin{aligned}
&\ln\det(\boldsymbol{I}_T + \sigma^{-2}\mathbf{K}(\mathbf{X}_T, \mathbf{X}_T)) \\
&\leq \underbrace{\ln\det(\boldsymbol{I}_T + \sigma^{-2}\mathbf{K}_1(\mathbf{X}_T, \mathbf{X}_T))}_{(i)} + \underbrace{\ln\det(\boldsymbol{I}_T + \sigma^{-2}\mathbf{K}_2(\mathbf{X}_T, \mathbf{X}_T))}_{(ii)} + \underbrace{\ln\det(\boldsymbol{I}_T + \sigma^{-2}\mathbf{K}_3(\mathbf{X}_T, \mathbf{X}_T))}_{(iii)},
\end{aligned} \tag{123}$$

where $\mathbf{K}_1(\mathbf{X}_T, \mathbf{X}_T)$, $\mathbf{K}_2(\mathbf{X}_T, \mathbf{X}_T)$, and $\mathbf{K}_3(\mathbf{X}_T, \mathbf{X}_T)$ are the kernel matrices of k_1, k_2, and k_3, respectively. We set sufficiently small η_1 and large η_2 so that the crude upper bounds of the first term (i) and the third term (iii) are sufficiently small. For this purpose, the following settings of η_1 and η_2 are sufficient (we confirm in the next paragraphs):

$$\eta_1 = \left(\frac{\nu\Gamma(\nu)\sigma^2}{T^2}\right)^{\frac{1}{\nu}}, \ \eta_2 = \max\left\{1, 2\ln\frac{2C_\nu T^2}{\Gamma(\nu)\sigma^2}\right\}, \tag{124}$$

where $C_\nu > 0$ is the constant such that $\forall z \geq 1, \ z^{\nu-1}e^{-z} \leq C_\nu e^{-z/2}$. Hereafter, we suppose that $\eta_1 < \eta_2$ holds with the above definition. The case $\eta_1 \geq \eta_2$ is considered in the last parts of the proof.

Upper bound for the first term (i). From the definition of k_1, we have

$$\begin{aligned}
|k_1(\boldsymbol{x}, \widetilde{\boldsymbol{x}})| &= \frac{1}{\Gamma(\nu)}\int_0^{\eta_1} z^{\nu-1}e^{-z}\exp\left(-\frac{\|\boldsymbol{x} - \widetilde{\boldsymbol{x}}\|_2^2}{2\ell^2 z\nu^{-1}}\right)\mathrm{d}z && (125)\\
&\leq \frac{1}{\Gamma(\nu)}\int_0^{\eta_1} z^{\nu-1}e^{-z}\mathrm{d}z && (126)\\
&\leq \frac{1}{\Gamma(\nu)}\int_0^{\eta_1} z^{\nu-1}\mathrm{d}z && (127)\\
&= \frac{1}{\nu\Gamma(\nu)}[z^\nu]_0^{\eta_1} && (128)\\
&= \frac{1}{\nu\Gamma(\nu)}\eta_1^\nu. && (129)
\end{aligned}$$

Then, from the definition of η_1, we have

$$\frac{1}{\nu\Gamma(\nu)}\eta_1^\nu = \frac{1}{\nu\Gamma(\nu)}\left(\frac{\nu\Gamma(\nu)\sigma^2}{T^2}\right) = \frac{\sigma^2}{T^2}. \tag{130}$$

Therefore, by denoting the eigenvalues of $\mathbf{K}_1(\mathbf{X}_T, \mathbf{X}_T)$ with decreasing order as $(\lambda_i)_{i\in[T]}$[9], we have

$$\begin{aligned}
\ln\det(\boldsymbol{I}_T + \sigma^{-2}\mathbf{K}_1(\mathbf{X}_T, \mathbf{X}_T)) &= \ln\prod_{i=1}^{T}(1 + \sigma^{-2}\lambda_i) && (131)\\
&\leq \ln(1 + \sigma^{-2}\lambda_1)^T && (132)\\
&\leq \ln(1 + T^{-1})^T, && (133)
\end{aligned}$$

where the last inequality follows from $\lambda_1 \leq \sqrt{\sum_{i=1}^T \lambda_i^2} = \|\mathbf{K}_1(\mathbf{X}_T, \mathbf{X}_T)\|_F = \sqrt{\sum_{i,j} k_1(\boldsymbol{x}_i, \boldsymbol{x}_j)^2} \leq \sigma^2/T$. Since $\ln(1 + T^{-1})^T \to 1$ as $T \to \infty$, there exists a constant $C > 0$ such that $\ln\det(\boldsymbol{I}_T + \sigma^{-2}\mathbf{K}_1(\mathbf{X}_T, \mathbf{X}_T)) \leq C$ for all $T \in \mathbb{N}_+$.

[8]Note that the linear combination of the positive definite kernel with non-negative coefficients and its limit are also positive definite. Therefore, k_1, k_2, and k_3 are also positive definite as far as $\eta_2 > \eta_1$.

[9]Note that λ_i is non-negative from the positive semi-definiteness of $\mathbf{K}_1(\mathbf{X}_T, \mathbf{X}_T)$.

Upper bound for the third term (iii). From the definition of k_3, we have

$$|k_3(\boldsymbol{x}, \widetilde{\boldsymbol{x}})| = \frac{1}{\Gamma(\nu)} \int_{\eta_2}^{\infty} z^{\nu-1} e^{-z} \exp\left(-\frac{\|\boldsymbol{x} - \widetilde{\boldsymbol{x}}\|_2^2}{2\ell^2 z \nu^{-1}}\right) \mathrm{d}z \tag{134}$$

$$\leq \frac{1}{\Gamma(\nu)} \int_{\eta_2}^{\infty} z^{\nu-1} e^{-z} \mathrm{d}z \tag{135}$$

$$\leq \frac{C_\nu}{\Gamma(\nu)} \int_{\eta_2}^{\infty} e^{-z/2} \mathrm{d}z \tag{136}$$

$$= \frac{-2C_\nu}{\Gamma(\nu)} [e^{-z/2}]_{\eta_2}^{\infty} \tag{137}$$

$$= \frac{2C_\nu}{\Gamma(\nu)} \exp\left(-\frac{\eta_2}{2}\right). \tag{138}$$

Then, from the definition of η_2, we have

$$\frac{2C_\nu}{\Gamma(\nu)} \exp\left(-\frac{\eta_2}{2}\right) \leq \frac{2C_\nu}{\Gamma(\nu)} \exp\left(-\ln\left(\frac{2C_\nu T^2}{\Gamma(\nu)\sigma^2}\right)\right) = \frac{\sigma^2}{T^2}. \tag{139}$$

By following the same arguments as after Eq. (130) in the upper bound of the first term (i), we conclude that there exists a constant $C > 0$ such that $\ln\det(\boldsymbol{I}_T + \sigma^{-2}\mathbf{K}_3(\mathbf{X}_T, \mathbf{X}_T)) \leq C$ for all $T \in \mathbb{N}_+$.

Upper bound for the second term (ii). We further decompose k_2 by partitioning the integration range into dyadic intervals:

$$k_2(\boldsymbol{x}, \widetilde{\boldsymbol{x}}) = \sum_{q=1}^{Q} k_2^{(q)}(\boldsymbol{x}, \widetilde{\boldsymbol{x}}), \tag{140}$$

where:

$$k_2^{(q)}(\boldsymbol{x}, \widetilde{\boldsymbol{x}}) = \frac{1}{\Gamma(\nu)} \int_{\eta_1 2^{q-1}}^{\min\{\eta_1 2^q, \eta_2\}} z^{\nu-1} e^{-z} \exp\left(-\frac{\|\boldsymbol{x} - \widetilde{\boldsymbol{x}}\|_2^2}{2\ell^2 z \nu^{-1}}\right) \mathrm{d}z. \tag{141}$$

Here, $Q \in \mathbb{N}_+$ is the minimum number such that $\eta_1 2^Q \geq \eta_2$ holds. Then, as with Eq. (123), we have

$$\ln\det(\boldsymbol{I}_T + \sigma^{-2}\mathbf{K}_2(\mathbf{X}_T, \mathbf{X}_T)) \leq \sum_{q=1}^{Q} \ln\det\left(\boldsymbol{I}_T + \sigma^{-2}\mathbf{K}_2^{(q)}(\mathbf{X}_T, \mathbf{X}_T)\right), \tag{142}$$

where $\mathbf{K}_2^{(q)}(\mathbf{X}_T, \mathbf{X}_T)$ is the kernel matrix of $k_2^{(q)}$. Next, for any q, we define a new kernel function $\widetilde{k}_2^{(q)}(\boldsymbol{x}, \widetilde{\boldsymbol{x}})$ as

$$\widetilde{k}_2^{(q)}(\boldsymbol{x}, \widetilde{\boldsymbol{x}}) = k_2^{(q)}(\boldsymbol{x}, \widetilde{\boldsymbol{x}}) \exp\left(-\frac{\left(\sqrt{1 - \|\boldsymbol{x}\|_2^2} - \sqrt{1 - \|\widetilde{\boldsymbol{x}}\|_2^2}\right)^2}{2\ell^2 \nu^{-1} \min\{\eta_1 2^q, \eta_2\}}\right). \tag{143}$$

We further denote the kernel matrix of $\widetilde{k}_2^{(q)}$ by $\widetilde{\mathbf{K}}_2^{(q)}(\mathbf{X}_T, \mathbf{X}_T)$. Then, from Oppenheim's inequality, we have

$$\ln\det\left(\boldsymbol{I}_T + \sigma^{-2}\mathbf{K}_2^{(q)}(\mathbf{X}_T, \mathbf{X}_T)\right) \leq \ln\det\left(\boldsymbol{I}_T + \sigma^{-2}\widetilde{\mathbf{K}}_2^{(q)}(\mathbf{X}_T, \mathbf{X}_T)\right). \tag{144}$$

Furthermore, for any $z \in [\eta_1 2^{q-1}, \min\{\eta_1 2^q, \eta_2\}]$, the following kernel function $\widehat{k}_2^{(q)}(\boldsymbol{x}, \widetilde{\boldsymbol{x}}; z)$ is positive definite (e.g., Lemma A.5 in [11]):

$$\widehat{k}_2^{(q)}(\boldsymbol{x}, \widetilde{\boldsymbol{x}}; z) = 2\exp\left(-\frac{\left(\sqrt{1 - \|\boldsymbol{x}\|_2^2} - \sqrt{1 - \|\widetilde{\boldsymbol{x}}\|_2^2}\right)^2}{2\ell^2 \nu^{-1} z}\right) - \exp\left(-\frac{\left(\sqrt{1 - \|\boldsymbol{x}\|_2^2} - \sqrt{1 - \|\widetilde{\boldsymbol{x}}\|_2^2}\right)^2}{2\ell^2 \nu^{-1} \min\{\eta_1 2^q, \eta_2\}}\right). \tag{145}$$

Note that $2k_2^{(q)}(\widetilde{\boldsymbol{x}}_i, \widetilde{\boldsymbol{x}}_j) - \widetilde{k}_2^{(q)}(\boldsymbol{x}_i, \boldsymbol{x}_j)$ is represented as

$$2k_2^{(q)}(\widetilde{\boldsymbol{x}}_i, \widetilde{\boldsymbol{x}}_j) - \widetilde{k}_2^{(q)}(\boldsymbol{x}_i, \boldsymbol{x}_j) \tag{146}$$

$$= \frac{1}{\Gamma(\nu)} \int_{\eta_1 2^{q-1}}^{\min\{\eta_1 2^q, \eta_2\}} z^{\nu-1} e^{-z} \exp\left(-\frac{\|\boldsymbol{x}_i - \boldsymbol{x}_j\|_2^2}{2\ell^2 z \nu^{-1}}\right) \widehat{k}_2^{(q)}(\boldsymbol{x}_i, \boldsymbol{x}_j; z) \mathrm{d}z. \tag{147}$$

By noting that the product of two positive definite kernels is also positive definite, the above expression implies that $2\mathbf{K}_2^{(q)}(\widetilde{\mathbf{X}}_T, \widetilde{\mathbf{X}}_T) - \widetilde{\mathbf{K}}_2^{(q)}(\mathbf{X}_T, \mathbf{X}_T)$ is the positive semi-definite matrix. Therefore, we have[10]

$$\sum_{q=1}^{Q} \ln\det\left(\boldsymbol{I}_T + \sigma^{-2}\widetilde{\mathbf{K}}_2^{(q)}(\mathbf{X}_T, \mathbf{X}_T)\right) \le \sum_{q=1}^{Q} \ln\det\left(\boldsymbol{I}_T + 2\sigma^{-2}\mathbf{K}_2^{(q)}(\widetilde{\mathbf{X}}_T, \widetilde{\mathbf{X}}_T)\right) \tag{148}$$

$$\le Q \ln\det\left(\boldsymbol{I}_T + 2\sigma^{-2}\mathbf{K}(\widetilde{\mathbf{X}}_T, \widetilde{\mathbf{X}}_T)\right), \tag{149}$$

where the second inequality follows from the fact that $\mathbf{K}(\widetilde{\mathbf{X}}_T, \widetilde{\mathbf{X}}_T) - \mathbf{K}_2^{(q)}(\widetilde{\mathbf{X}}_T, \widetilde{\mathbf{X}}_T)$ is positive semi-definite. From the definition of Q, we have

$$Q \le \log_2\left(\frac{\eta_2}{\eta_1}\right) + 1 \tag{150}$$

$$= \log_2 \eta_2 - \log_2 \eta_1 + 1 \tag{151}$$

$$= \log_2 \max\left\{1, 2\ln\frac{2C_\nu T^2}{\Gamma(\nu)\sigma^2}\right\} - \log_2\left(\frac{\nu\Gamma(\nu)\sigma^2}{T^2}\right)^{\frac{1}{\nu}} + 1 \tag{152}$$

$$\le C(T, \nu, \sigma^2). \tag{153}$$

By combining Eqs. (142), (144), (149), and (153), we conclude

$$\ln\det(\boldsymbol{I}_T + \sigma^{-2}\mathbf{K}_2(\mathbf{X}_T, \mathbf{X}_T)) \tag{154}$$

$$\le C(T, \nu, \sigma^2) \ln\det\left(\boldsymbol{I}_T + 2\sigma^{-2}\mathbf{K}(\widetilde{\mathbf{X}}_T, \widetilde{\mathbf{X}}_T)\right). \tag{155}$$

By aggregating the upper bounds of (i), (ii), and (iii), we have the following inequality under $\eta_1 < \eta_2$:

$$\ln\det(\boldsymbol{I}_T + \sigma^{-2}\mathbf{K}(\mathbf{X}_T, \mathbf{X}_T)) \le C(T, \nu, \sigma^2) \ln\det\left(\boldsymbol{I}_T + 2\sigma^{-2}\mathbf{K}(\widetilde{\mathbf{X}}_T, \widetilde{\mathbf{X}}_T)\right) + 2C. \tag{156}$$

Finally, if $\eta_1 \ge \eta_2$, we have

$$\ln\det(\boldsymbol{I}_T + \sigma^{-2}\mathbf{K}(\mathbf{X}_T, \mathbf{X}_T)) \tag{157}$$

$$\le \ln\det(\boldsymbol{I}_T + \sigma^{-2}(\mathbf{K}_1(\mathbf{X}_T, \mathbf{X}_T) + \mathbf{K}_3(\mathbf{X}_T, \mathbf{X}_T))) \tag{158}$$

$$\le \ln\det(\boldsymbol{I}_T + \sigma^{-2}\mathbf{K}_1(\mathbf{X}_T, \mathbf{X}_T)) + \ln\det(\boldsymbol{I}_T + \sigma^{-2}\mathbf{K}_3(\mathbf{X}_T, \mathbf{X}_T)) \tag{159}$$

$$\le 2C \tag{160}$$

$$\le C(T, \nu, \sigma^2) \ln\det\left(\boldsymbol{I}_T + 2\sigma^{-2}\mathbf{K}(\widetilde{\mathbf{X}}_T, \widetilde{\mathbf{X}}_T)\right) + 2C, \tag{161}$$

where the last inequality follows from $\ln\det\left(\boldsymbol{I}_T + 2\sigma^{-2}\mathbf{K}(\widetilde{\mathbf{X}}_T, \widetilde{\mathbf{X}}_T)\right) \ge 0$ and $C(T, \nu, \sigma^2) \ge 0$. The desired result is obtained by setting a new absolute constant C to $2C$ in the above inequality. □

B.2 Summary of Mercer Decomposition for Dot-Product Kernel on Sphere

In this subsection, we summarize the basic known results of the Mercer decomposition on $\mathbb{S}^d$. The content of this subsection is related to the analysis of the spherical harmonics. We refer to [1, 17]

[10] For any positive semi-definite matrices A, B such that $A - B$ is positive semi-definite, we have $\lambda_i^{(A)} \ge \lambda_i^{(B)}$, where $(\lambda_i^{(A)})$ and $(\lambda_i^{(B)})$ are the nonnegative eigenvalues of A and B in nonincreasing order. (This is a consequence of Courant–Fischer's min-max theorem.) Therefore, we have $\det(A) = \prod_i \lambda_i^{(A)} \ge \prod_i \lambda_i^{(B)} = \det(B)$ for such A and B.

as the basic textbooks. In the kernel method literature, the Mercer decomposition of the dot-product kernel and its eigendecay have been studied. See, e.g., [3, 37, 49]. Furthermore, the existing analysis of the neural tangent kernel also leverages the Mercer decomposition based on the spherical harmonics. We also refer to the appendix of [4, 6] as the related works of this subsection.

We first describe Mercer's theorem. Let $L^2(\mathcal{X}, \mu) \coloneqq \{f : \mathcal{X} \to \mathbb{R} \mid \int_{\mathcal{X}} f^2(\boldsymbol{x})\mu(\mathrm{d}\boldsymbol{x}) < \infty\}$ denote the space of square-integrable functions on $\mathcal{X}$ under the measure μ. Furthermore, let us define the kernel integral operator $\mathcal{T}_k : L^2(\mathcal{X}, \mu) \to L^2(\mathcal{X}, \mu)$ of a square-integrable kernel function $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ as $(\mathcal{T}_k f)(\cdot) = \int_{\mathcal{X}} k(\cdot, \boldsymbol{x}) f(\boldsymbol{x})\mu(\mathrm{d}\boldsymbol{x})$. Then, Mercer's theorem guarantees that the positive kernel k is decomposed based on the eigenvalues and eigenfunctions sequence of $\mathcal{T}_k$ with absolute and uniform convergence on $\mathcal{X} \times \mathcal{X}$. We give the formal statement below.

Theorem 10 (Mercer's theorem, e.g., Theorem 4.49 in [13])**.** *Let $\mathcal{X}$ be a compact metric space, μ be a finite Borel measure whose support is $\mathcal{X}$, and $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ be a continuous and square-integrable, positive-definite kernel on $(\mathcal{X}, \mu)$. Suppose that $(\phi_i)_{i\in\mathbb{N}}$ and $(\lambda_i)_{i\in\mathbb{N}}$ are eigenfunctions and eigenvalues of the kernel integral operator $\mathcal{T}_k$, respectively. Namely, $(\phi_i)_{i\in\mathbb{N}}$ is an orthonormal basis of $\{\mathcal{T}_k f \mid f \in L^2(\mathcal{X}, \mu)\}$ such that $\mathcal{T}_k \phi_i(\cdot) = \lambda_i \phi_i(\cdot)$ for all $i \in \mathbb{N}$. Then, we have*

$$k(\boldsymbol{x}, \widetilde{\boldsymbol{x}}) = \sum_{i\in\mathbb{N}} \lambda_i \phi_i(\boldsymbol{x})\phi_i(\widetilde{\boldsymbol{x}}), \tag{162}$$

where the convergence is absolute and uniform on $\mathcal{X} \times \mathcal{X}$.

Specifically, our interest is the Mercer decomposition of the kernel on $\mathbb{S}^d$. This is given as spherical harmonics on $\mathbb{S}^d$, which we define below.

Definition 1 (Spherical harmonics, e.g., Definition 2.7 in [1])**.** Fix any $d \geq 1$ and $m \in \mathbb{N}$. Let $\mathbb{Y}_m(\mathbb{R}^{d+1})$ be the space of all homogeneous polynomials of degree m in $\mathbb{R}^{d+1}$ that are also harmonic[11]. The space $\mathbb{Y}_m^{d+1} = \mathbb{Y}_m(\mathbb{R}^{d+1}) \mid_{\mathbb{S}^d}$ is called the spherical harmonic space of order m in $d + 1$ dimensions. Any function in $\mathbb{Y}_m^{d+1}$ is called a spherical harmonic of order m in $d + 1$ dimensions.

The following lemmas provide the properties of the spherical harmonics, which guarantee that the Mercer decomposition of the continuous dot-product kernel on $\mathbb{S}^d$ is defined based on spherical harmonics.

Lemma 11 (Dimension and completeness of spherical harmonics, e.g., Chapter 2.1.3, Corollary 2.15, and Theorem 2.38 in [1])**.** *Fix any $d \geq 1$. Then, the following statements hold:*

- *For any $m \in \mathbb{N}$, we have $\dim(\mathbb{Y}_m^{d+1}) = N_{d+1,m}$ with $N_{d+1,m} = \frac{(2m+d-1)(m+d-2)!}{m!(d-1)!}$. Furthermore, for any $m, n \in \mathbb{N}$ with $m \neq n$, we have $\mathbb{Y}_m^{d+1} \perp \mathbb{Y}_n^{d+1}$*[12].
- *Let $(Y_{m,j})_{j\in[N_{d+1,m}]}$ be an orthonormal basis of $\mathbb{Y}_m^{d+1}$. Then, $\cup_{m\in\mathbb{N}}(Y_{m,j})_{j\in[N_{d+1,m}]}$ becomes an orthonormal basis of $L^2(\mathbb{S}^d, \sigma)$, where $\sigma(\cdot)$ is the induced Lebesgue measure on $\mathbb{S}^d$.*

Lemma 12 (Funk-Hecke Formula, e.g., Theorem 2.22 in [1] or Theorem 4.24 in [17])**.** *Fix any $d \geq 1$. Let $f : [-1, 1] \to \mathbb{R}$ be a continuous function. Define $|\mathbb{S}^{d-1}| \coloneqq \frac{2\pi^{d/2}}{\Gamma(d/2)}$ as the surface area of $\mathbb{S}^{d-1}$. Then, for any $m \in \mathbb{N}$ and $Y_m \in \mathbb{Y}_m^{d+1}$, we have*

$$\int_{\mathbb{S}^d} f(\boldsymbol{z}^\top \boldsymbol{\eta}) Y_m(\boldsymbol{\eta})\sigma(\mathrm{d}\boldsymbol{\eta}) = \lambda_m Y_m(\boldsymbol{z}), \tag{163}$$

where $\sigma(\cdot)$ is the induced Lebesgue measure on $\mathbb{S}^d$. Furthermore, λ_m is defined as

$$\lambda_m = |\mathbb{S}^{d-1}| \int_{-1}^{1} P_{m,d+1}(t) f(t)(1 - t^2)^{\frac{d-2}{2}} \mathrm{d}t, \tag{164}$$

[11] A polynomial $H(x_1, \ldots, x_{d+1})$ is called homogeneous of degree m if $H(tx_1, \ldots, tx_{d+1}) = t^m H(x_1, \ldots, x_{d+1})$. Furthermore, a polynomial $H(x_1, \ldots, x_{d+1})$ is called harmonic if $\Delta_{d+1} H = 0$, where Δ_{d+1} is the Laplace operator. See Chapter 4 in [17] or Chapter 2 in [1].

[12] Here, as with the second statement, we consider $L^2(\mathbb{S}^d, \sigma)$. Therefore, the inner product for any $f, g : \mathbb{S}^d \to \mathbb{R}$ is defined on $L^2(\mathbb{S}^d, \sigma)$ as $\int_{\mathbb{S}^d} f(\boldsymbol{x})g(\boldsymbol{x})\sigma(\mathrm{d}\boldsymbol{x})$.

where $P_{m,d+1}(t)$ is the Legendre polynomial of degree m in $d+1$ dimensions, which is defined as

$$P_{m,d+1}(t) = m!\Gamma\left(\frac{d}{2}\right)\sum_{k=0}^{\lfloor m/2\rfloor}(-1)^k\frac{(1-t^2)^k t^{m-2k}}{4^k k!(m-2k)!\Gamma\left(k+\frac{d}{2}\right)}. \tag{165}$$

Lemma 12 suggests that the spherical harmonics are eigenfunctions of the continuous dot-product kernel $k(\boldsymbol{x},\widetilde{\boldsymbol{x}}) = \widetilde{k}(\boldsymbol{x}^\top\widetilde{\boldsymbol{x}})$ on $\mathbb{S}^d$. Furthermore, Lemma 11 guarantees that $\cup_{m\in\mathbb{N}}(Y_{m,j})_{j\in[N_{d+1,m}]}$ forms an orthonormal basis of $L^2(\mathbb{S}^d,\sigma)$, which implies that they are an orthonormal basis of the eigenspace of $\mathcal{T}_k$. These facts give the following explicit form of Mercer decomposition for a continuous dot-product kernel.

Corollary 13. *Fix any $d\in\mathbb{N}_+$. Suppose $\mathcal{X}=\mathbb{S}^d$. Furthermore, assume the kernel function $k:\mathcal{X}\times\mathcal{X}\to\mathbb{R}$ is the positive definite kernel such that $\forall\boldsymbol{x},\widetilde{\boldsymbol{x}}\in\mathcal{X}, k(\boldsymbol{x},\widetilde{\boldsymbol{x}}) = \widetilde{k}(\boldsymbol{x}^\top\widetilde{\boldsymbol{x}})$ with some continuous function $\widetilde{k}:[-1,1]\to\mathbb{R}$. Then, we have the following Mercer decomposition of k:*

$$k(\boldsymbol{x},\widetilde{\boldsymbol{x}}) = \sum_{m=0}^{\infty}\lambda_m\sum_{j=1}^{N_{d+1,m}}Y_{m,j}(\boldsymbol{x})Y_{m,j}(\widetilde{\boldsymbol{x}}), \tag{166}$$

where $(Y_{m,j}(\cdot))_{j\in[N_{d+1,m}]}$ denotes the spherical harmonics, which form an orthonormal basis of $\mathbb{Y}_m^{d+1}$. Furthermore, $\lambda_m\geq 0$ is defined as

$$\lambda_m = |\mathbb{S}^{d-1}|\int_{-1}^{1}P_{m,d+1}(t)\widetilde{k}(t)(1-t^2)^{\frac{d-2}{2}}\mathrm{d}t. \tag{167}$$

Note that $\|\boldsymbol{x}-\widetilde{\boldsymbol{x}}\|_2 = \sqrt{2-2\boldsymbol{x}^\top\widetilde{\boldsymbol{x}}}$ for any $\boldsymbol{x},\widetilde{\boldsymbol{x}}\in\mathbb{S}^d$. Therefore, we can represent k_{SE} and $k_{\text{Matérn}}$ on $\mathbb{S}^d$ by Eq. (166). Finally, we describe the following addition theorem of the spherical harmonics, which plays a central role in avoiding the uniform boundedness assumption in the existing proof of MIG on $\mathbb{S}^d$.

Lemma 14 (Addition theorem, e.g., Theorem 2.9 in [1] or Theorem 4.11 in [17])**.** *Fix any $d\geq 1$ and $m\in\mathbb{N}$. Let $(Y_{m,j})_{j\in[N_{d+1,m}]}$ be an orthonormal basis of $\mathbb{Y}_m^{d+1}$. Then, we have*

$$\forall\boldsymbol{x},\widetilde{\boldsymbol{x}}\in\mathbb{S}^d,\quad\sum_{j=1}^{N_{d+1,m}}Y_{m,j}(\boldsymbol{x})Y_{m,j}(\widetilde{\boldsymbol{x}}) = \frac{N_{d+1,m}}{|\mathbb{S}^d|}P_{m,d+1}(\boldsymbol{x}^\top\widetilde{\boldsymbol{x}}), \tag{168}$$

where $P_{m,d+1}(t)$ is the Legendre polynomial of degree m in $d+1$ dimensions, which is defined in Eq. (165)*.*

B.3 Upper Bound of MIG with Mercer Decomposition

By using Corollary 13 and Lemma 14 in the previous subsection, we can derive the following general form of the upper bound of MIG.

Lemma 15 (Adapted from [59])**.** *Suppose the kernel function k satisfies the condition in Corollary 13. Furthermore, assume $|k(\boldsymbol{x},\widetilde{\boldsymbol{x}})|\leq 1$ for all $\boldsymbol{x},\widetilde{\boldsymbol{x}}\in\mathcal{X}$. Then, for any $M\in\mathbb{N}$, MIG on $\mathbb{S}^d$ satisfies*

$$\frac{1}{2}\max_{\boldsymbol{x}_1,\ldots,\boldsymbol{x}_T\in\mathbb{S}^d}\ln\det(\boldsymbol{I}_T+\sigma^{-2}\mathbf{K}(\mathbf{X}_T,\mathbf{X}_T))\leq N_M\ln\left(1+\frac{T}{\sigma^2}\right)+\frac{T}{|\mathbb{S}^d|\sigma^2}\sum_{m=M+1}^{\infty}\lambda_m N_{d+1,m}, \tag{169}$$

where $N_M=\sum_{m=0}^{M}N_{d+1,m}$.

The proof almost directly follows from [60], while a minor modification is required to deal with the lack of a uniform bound across eigenfunctions through the addition theorem. The same proof strategy is already provided in [32, 59] for analyzing the MIG of the neural tangent kernel on the sphere. Although our proof has no intrinsic change from their proof, we give the details below for completeness of our paper.

Proof. We first decompose the kernel matrix as $\mathbf{K}(\mathbf{X}_T, \mathbf{X}_T) = \mathbf{K}_{\text{head}} + \mathbf{K}_{\text{tail}}$, where $[\mathbf{K}_{\text{head}}]_{i,l} = \sum_{m=0}^{M} \lambda_m \sum_{j=1}^{N_{d+1,m}} Y_{m,j}(\boldsymbol{x}_i) Y_{m,j}(\boldsymbol{x}_l)$ and $[\mathbf{K}_{\text{tail}}]_{i,l} = \sum_{m=M+1}^{\infty} \lambda_m \sum_{j=1}^{N_{d+1,m}} Y_{m,j}(\boldsymbol{x}_i) Y_{m,j}(\boldsymbol{x}_l)$. Then, as with the proof in [60], the MIG is decomposed as

$$\frac{1}{2} \max_{\boldsymbol{x}_1, \ldots, \boldsymbol{x}_T \in \mathbb{S}^d} \ln \det(\boldsymbol{I}_T + \sigma^{-2} \mathbf{K}(\mathbf{X}_T, \mathbf{X}_T)) \tag{170}$$

$$= \frac{1}{2} \ln \det \left(\boldsymbol{I}_T + \frac{1}{\sigma^2} \mathbf{K}_{\text{head}} \right) + \frac{1}{2} \ln \det \left(\boldsymbol{I}_T + \frac{1}{\sigma^2} \left(\boldsymbol{I}_T + \frac{1}{\sigma^2} \mathbf{K}_{\text{head}} \right)^{-1} \mathbf{K}_{\text{tail}} \right). \tag{171}$$

Based on the feature representation of the kernel, the first term is further bounded from above as follows (see [59, 60]):

$$\frac{1}{2} \ln \det \left(\boldsymbol{I}_T + \frac{1}{\sigma^2} \mathbf{K}_{\text{head}} \right) \leq N_M \ln \left(1 + \frac{T}{\sigma^2 N_M} \right) \leq N_M \ln \left(1 + \frac{T}{\sigma^2} \right), \tag{172}$$

where the second inequality follows from $N_M \geq 1$. Regarding the second term, as with [60], we have

$$\frac{1}{2} \ln \det \left(\boldsymbol{I}_T + \frac{1}{\sigma^2} \left(\boldsymbol{I}_T + \frac{1}{\sigma^2} \mathbf{K}_{\text{head}} \right)^{-1} \mathbf{K}_{\text{tail}} \right) \tag{173}$$

$$\leq T \ln \left(T^{-1} \mathrm{Tr} \left(\boldsymbol{I}_T + \frac{1}{\sigma^2} \left(\boldsymbol{I}_T + \frac{1}{\sigma^2} \mathbf{K}_{\text{head}} \right)^{-1} \mathbf{K}_{\text{tail}} \right) \right) \tag{174}$$

$$\leq T \ln \left(T^{-1} \left(T + \frac{1}{\sigma^2} \mathrm{Tr} \left(\mathbf{K}_{\text{tail}} \right) \right) \right), \tag{175}$$

where the first inequality follows from $\ln \det(A) \leq T \ln(\mathrm{Tr}(A)/T)$ for any positive definite matrix $A \in \mathbb{R}^{T \times T}$ (e.g., Lemma 1 in [60]). Then, from addition theorem (Lemma 14), we have

$$\mathrm{Tr} \left(\mathbf{K}_{\text{tail}} \right) = \sum_{t=1}^{T} \sum_{m=M+1}^{\infty} \lambda_m \sum_{j=1}^{N_{d+1,m}} Y_{m,j}(\boldsymbol{x}_t) Y_{m,j}(\boldsymbol{x}_t) \tag{176}$$

$$= \sum_{t=1}^{T} \sum_{m=M+1}^{\infty} \lambda_m \frac{N_{d+1,m}}{|\mathbb{S}^d|} P_{m,d+1}(\boldsymbol{x}_t^\top \boldsymbol{x}_t) \tag{177}$$

$$= \frac{T}{|\mathbb{S}^d|} \sum_{m=M+1}^{\infty} \lambda_m N_{d+1,m}, \tag{178}$$

where the last line uses $P_{m,d+1}(\boldsymbol{x}_t^\top \boldsymbol{x}_t) = P_{m,d+1}(1) = 1$. By combining the above equation with Eq. (175), we have

$$\frac{1}{2} \ln \det \left(\boldsymbol{I}_T + \frac{1}{\sigma^2} \left(\boldsymbol{I}_T + \frac{1}{\sigma^2} \mathbf{K}_{\text{head}} \right)^{-1} \mathbf{K}_{\text{tail}} \right) \leq T \ln \left(1 + \frac{1}{\sigma^2 |\mathbb{S}^d|} \sum_{m=M+1}^{\infty} \lambda_m N_{d+1,m} \right) \tag{179}$$

$$\leq \frac{T}{\sigma^2 |\mathbb{S}^d|} \sum_{m=M+1}^{\infty} \lambda_m N_{d+1,m}, \tag{180}$$

where the last line uses $\forall z > -1, \ln(1 + z) \leq z$. □

To obtain the explicit upper bound of Eq. (169), we introduce the following lemma.

Lemma 16 (Upper bound of $N_{d+1,m}$ and N_M)**.** *Fix any $d \in \mathbb{N}_+$. Then, for any $m \in \mathbb{N}_+$, we have*

$$N_{d+1,m} \leq (d+1) e^{d-1} m^{d-1}. \tag{181}$$

Furthermore, for any $M \in \mathbb{N}$, we have

$$N_M \leq 1 + (d+1) e^{d-1} M^d. \tag{182}$$

Proof. Recall $N_{d+1,m} = \frac{(2m+d-1)(m+d-2)!}{m!(d-1)!}$. Under $d = 1$, we have

$$N_{d+1,m} = \frac{(2m)(m-1)!}{m!} = 2 = (d+1)e^{d-1}m^{d-1} \tag{183}$$

for any $m \in \mathbb{N}_+$. Under $d \geq 2$, since $N_{d+1,m} = \frac{(2m+d-1)(m+d-2)!}{m!(d-1)!} = \frac{2m+d-1}{m}\binom{m+d-2}{d-1}$ and $\binom{m+d-2}{d-1} \leq \left(\frac{(m+d-2)e}{d-1}\right)^{d-1}$, we have

$$N_{d+1,m} \leq \frac{2m+d-1}{m}\left(\frac{(m+d-2)e}{d-1}\right)^{d-1} \tag{184}$$

$$\leq (2+d-1)e^{d-1}\left(\frac{m+d-2}{d-1}\right)^{d-1} \tag{185}$$

$$\leq (d+1)e^{d-1}m^{d-1}. \tag{186}$$

Finally, since $N_{d+1,0} = 1$, we have

$$N_M = 1 + \sum_{m=1}^{M} N_{d+1,m} \leq 1 + (d+1)e^{d-1}\sum_{m=1}^{M} m^{d-1} \leq 1 + (d+1)e^{d-1}M^d. \tag{187}$$

□

B.4 Eigendecay of SE and Matérn Kernel

To obtain the explicit upper bound of Eq. (169), we need the upper bound of the eigenvalue in Eq. (167) under the SE and Matérn kernels. Regarding the SE kernel, several existing works have already studied it [37, 40]. We formally provide the following lemma from [37].

Lemma 17 (Eigendecay for $k = k_{\mathrm{SE}}$ on $\mathbb{S}^d$, Theorem 2 in [37])**.** *Fix any $d \in \mathbb{N}_+$, $\theta > 0$, and define $\mathcal{X} = \mathbb{S}^d$. Suppose that $k : \mathcal{X}\times\mathcal{X} \to \mathbb{R}$ is defined as $k(\boldsymbol{x}, \widetilde{\boldsymbol{x}}) = \exp\left(-\frac{\|\boldsymbol{x}-\widetilde{\boldsymbol{x}}\|_2^2}{\theta}\right)$. Then, the eigenvalues $(\lambda_m)_{m\in\mathbb{N}_+}$ defined in* (167) *satisfy*

$$\lambda_m < |\mathbb{S}^d|\left(\frac{2e}{\theta}\right)^m \frac{(2e)^{\frac{d+1}{2}}\Gamma\left(\frac{d+1}{2}\right)}{\sqrt{\pi}(2m+d-1)^{m+\frac{d}{2}}}\exp\left(-\frac{2}{\theta}+\frac{1}{\theta^2}\right). \tag{188}$$

Regarding Matérn kernel, we provide the upper bound of λ_m for $\nu > 1/2$ by extending the proof in [21], which studies λ_m for Laplace kernel (Matérn with $\nu = 1/2$). As with the proof in [21], we leverage the following lemma, which relates the spectral density of the kernel to λ_m.

Lemma 18 (Eigenvalues and spectral density, Theorem 4.1 in [39])**.** *Fix any $d \in \mathbb{N}_+$. Suppose that $k : \mathbb{R}^{d+1}\times\mathbb{R}^{d+1} \to \mathbb{R}$ is a positive definite, stationary, and isotropic kernel function on $\mathbb{R}^{d+1}$ such that $\forall \boldsymbol{x}, \widetilde{\boldsymbol{x}}, k(\boldsymbol{x}, \widetilde{\boldsymbol{x}}) = \Phi(\boldsymbol{x} - \widetilde{\boldsymbol{x}})$ for some function $\Phi(\cdot)$. Furthermore, suppose $\Phi(\cdot)$ is represented as*

$$\Phi(\boldsymbol{x}) = \frac{1}{(2\pi)^{d+1}}\int_{\mathbb{R}^{d+1}} \widehat{\Phi}(\|\boldsymbol{\eta}\|_2)e^{i\boldsymbol{\eta}^\top\boldsymbol{x}}\mathrm{d}\boldsymbol{\eta}, \tag{189}$$

for some function $\widehat{\Phi}$ such that $\forall a \geq 0, \widehat{\Phi}(a) \geq 0$ and $\int_{\mathbb{R}^{d+1}} \widehat{\Phi}(\|\boldsymbol{\eta}\|_2)\mathrm{d}\boldsymbol{\eta} < \infty$. Then, there exists a function $\widetilde{k} : [-1, 1] \to \mathbb{R}$ such that $\forall \boldsymbol{x}, \widetilde{\boldsymbol{x}} \in \mathbb{S}^d, \widetilde{k}(\boldsymbol{x}^\top\widetilde{\boldsymbol{x}}) = k(\boldsymbol{x}, \widetilde{\boldsymbol{x}})$. Furthermore, λ_m in Eq. (167) *is given by*

$$\lambda_m = \int_0^\infty t\widehat{\Phi}(t)B^2_{m+\frac{d-1}{2}}(t)\mathrm{d}t, \tag{190}$$

where $B_{m+\frac{d-1}{2}}(\cdot)$ is the usual Bessel function of the first kind and of order $m + \frac{d-1}{2}$.

In the Matérn kernel, the spectral density that satisfies the conditions in the lemma is defined when $\nu > 1/2$. Then, the explicit form of $\widehat{\Phi}(t)$ is given as:

$$\widehat{\Phi}(t) = \frac{C_{d,\nu}}{\ell^{2\nu}}\left(\frac{2\nu}{\ell^2}+t^2\right)^{-\left(\nu+\frac{d+1}{2}\right)}, \tag{191}$$

where

$$C_{d,\nu} = \frac{2^{d+1}\pi^{(d+1)/2}\Gamma\left(\nu + \frac{d+1}{2}\right)(2\nu)^{\nu}}{\Gamma(\nu)}. \tag{192}$$

See, Chapter 4.2 in [42]. By using Lemma 18, we obtain the following lemma.

Lemma 19 (Eigendecay for $k = k_{\text{Matérn}}$ on $\mathbb{S}^d$)**.** *Fix any $d \in \mathbb{N}_+$, $\ell > 0$, and define $\mathcal{X} = \mathbb{S}^d$. Suppose that $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ is defined as $k(\boldsymbol{x}, \widetilde{\boldsymbol{x}}) = \frac{2^{1-\nu}}{\Gamma(\nu)}\left(\frac{\sqrt{2\nu}\|\boldsymbol{x}-\widetilde{\boldsymbol{x}}\|_2}{\ell}\right)^{\nu} J_{\nu}\left(\frac{\sqrt{2\nu}\|\boldsymbol{x}-\widetilde{\boldsymbol{x}}\|_2}{\ell}\right)$. Then, the eigenvalues $(\lambda_m)_{m\in\mathbb{N}_+}$ defined in* (167) *satisfy*

$$\lambda_m \leq \frac{\widetilde{C}_{d,\nu}}{\ell^{2\nu}} m^{-2\nu-d} \tag{193}$$

if $m > 2\nu$ and $\nu > 1/2$. Here, $\widetilde{C}_{d,\nu}$ is defined as

$$\widetilde{C}_{d,\nu} = C_{d,\nu}\frac{\Gamma(2\nu+d)}{\Gamma^2\left(\nu + \frac{d+1}{2}\right)}\exp\left(2\nu + d + \frac{1}{6}\right). \tag{194}$$

Proof. From Lemma 18, we have

$$\lambda_m = \int_0^{\infty} t\widehat{\Phi}(t)B^2_{m+\frac{d-1}{2}}(t)\mathrm{d}t \tag{195}$$

$$= \frac{C_{d,\nu}}{\ell^{2\nu}}\int_0^{\infty} t\left(\frac{2\nu}{\ell^2} + t^2\right)^{-\left(\nu+\frac{d+1}{2}\right)} B^2_{m+\frac{d-1}{2}}(t)\mathrm{d}t \tag{196}$$

$$\leq \frac{C_{d,\nu}}{\ell^{2\nu}}\int_0^{\infty} t^{-2\nu-d}B^2_{m+\frac{d-1}{2}}(t)\mathrm{d}t. \tag{197}$$

As with the proof of Theorem 7 in [21], we evaluate the integral $\int_0^{\infty} t^{-2\nu-d}B^2_{m+\frac{d-1}{2}}(t)\mathrm{d}t$ by using the following identity (Chapter 13.4.1 in [23]):

$$\int_0^{\infty}\frac{B_p(at)B_q(at)}{t^z}\mathrm{d}t = \frac{\left(\frac{1}{2}a\right)^{z-1}\Gamma(z)\Gamma\left(\frac{1}{2}p + \frac{1}{2}q - \frac{1}{2}z + \frac{1}{2}\right)}{2\Gamma\left(\frac{1}{2}z + \frac{1}{2}q - \frac{1}{2}p + \frac{1}{2}\right)\Gamma\left(\frac{1}{2}z + \frac{1}{2}p + \frac{1}{2}q + \frac{1}{2}\right)\Gamma\left(\frac{1}{2}z + \frac{1}{2}p - \frac{1}{2}q + \frac{1}{2}\right)}, \tag{198}$$

where $p + q + 1 > z > 0$. By setting $p = q = m + \frac{d-1}{2}$, $z = 2\nu + d$, and $a = 1$, we have $p + q + 1 > z \Leftrightarrow m > \nu$. Hence, for any $m > \nu$, we have

$$\int_0^{\infty} t^{-2\nu-d}B^2_{m+\frac{d-1}{2}}(t)\mathrm{d}t = \left(\frac{1}{2}\right)^{2\nu+d-1}\frac{\Gamma(2\nu+d)\Gamma(m-\nu)}{2\Gamma^2\left(\nu+\frac{d+1}{2}\right)\Gamma(m+\nu+d)}. \tag{199}$$

Stirling's formula implies that there exists a constant $C > 0$ such that

$$\Gamma(m-\nu) \leq C(m-\nu)^{m-\nu-\frac{1}{2}}\exp(-m+\nu)\exp\left(\frac{1}{12(m-\nu)}\right), \tag{200}$$

$$\Gamma(m+\nu+d) \geq C(m+\nu+d)^{m+\nu+d-\frac{1}{2}}\exp\left(-(m+\nu+d)\right). \tag{201}$$

Therefore, for any $m \geq 2\nu$ with $\nu > 1/2$, we have

$$\frac{\Gamma(m-\nu)}{\Gamma(m+\nu+d)} \leq \frac{(m-\nu)^{m-\nu-\frac{1}{2}} \exp(-m+\nu) \exp\left(\frac{1}{12(m-\nu)}\right)}{(m+\nu+d)^{m+\nu+d-\frac{1}{2}} \exp\left(-(m+\nu+d)\right)} \tag{202}$$

$$\leq \frac{(m-\nu)^{m-\nu-\frac{1}{2}}}{(m+\nu+d)^{m+\nu+d-\frac{1}{2}}} \exp\left(2\nu+d+\frac{1}{6}\right) \tag{203}$$

$$\leq \frac{(m-\nu)^{m-\nu-\frac{1}{2}}}{(m-\nu)^{m+\nu+d-\frac{1}{2}}} \exp\left(2\nu+d+\frac{1}{6}\right) \tag{204}$$

$$= (m-\nu)^{-2\nu-d} \exp\left(2\nu+d+\frac{1}{6}\right) \tag{205}$$

$$\leq 2^{2\nu+d} m^{-2\nu-d} \exp\left(2\nu+d+\frac{1}{6}\right), \tag{206}$$

where the second inequality follows from $m-\nu \geq 1/2$ due to $m \geq 2\nu \Leftrightarrow m-\nu \geq \nu$, the third inequality follows from $m+\nu+d \geq m-\nu$, and the last inequality follows from $m-\nu \geq m-m/2 \geq m/2$. Combining Eqs. (197), (199), and (206), we have

$$\lambda_m \leq \frac{C_{d,\nu}}{\ell^{2\nu}} \left(\frac{1}{2}\right)^{2\nu+d-1} \frac{\Gamma(2\nu+d)}{2\Gamma^2\left(\nu+\frac{d+1}{2}\right)} 2^{2\nu+d} m^{-2\nu-d} \exp\left(2\nu+d+\frac{1}{6}\right) \tag{207}$$

$$= \frac{\widetilde{C}_{d,\nu}}{\ell^{2\nu}} m^{-2\nu-d}. \tag{208}$$

□

B.5 Proof of Theorem 7

Squared exponential kernel. From Lemma 17, we have

$$\lambda_m < |\mathbb{S}^d| \left(\frac{2e}{\theta}\right)^m \frac{(2e)^{\frac{d+1}{2}} \Gamma\left(\frac{d+1}{2}\right)}{\sqrt{\pi}(2m+d-1)^{m+\frac{d}{2}}} \exp\left(-\frac{2}{\theta}+\frac{1}{\theta^2}\right) \tag{209}$$

$$\leq |\mathbb{S}^d| \frac{(2e)^{\frac{d+1}{2}} \Gamma\left(\frac{d+1}{2}\right)}{\sqrt{\pi}} \exp\left(-\frac{2}{\theta}+\frac{1}{\theta^2}\right) \left(\frac{2e}{\theta}\right)^m (2m)^{-m-\frac{d}{2}} \tag{210}$$

$$\leq |\mathbb{S}^d| \frac{(2e)^{\frac{d+1}{2}} \Gamma\left(\frac{d+1}{2}\right)}{\sqrt{\pi}} \exp\left(-\frac{2}{\theta}+\frac{1}{\theta^2}\right) \left(\frac{e}{\theta}\right)^m m^{-m-\frac{d}{2}}. \tag{211}$$

Here, we set $C_{d,\theta}$ as

$$C_{d,\theta} = \frac{(2e)^{\frac{d+1}{2}} \Gamma\left(\frac{d+1}{2}\right)}{\sqrt{\pi}} \exp\left(-\frac{2}{\theta}+\frac{1}{\theta^2}\right). \tag{212}$$

Then,

$$\gamma_T(\mathcal{X}) \leq N_M \ln\left(1+\frac{T}{\sigma^2}\right) + \frac{T}{|\mathbb{S}^d|\sigma^2} \sum_{m=M+1}^{\infty} \lambda_m N_{d+1,m} \tag{213}$$

$$\leq N_M \ln\left(1+\frac{T}{\sigma^2}\right) + \frac{C_{d,\theta} T}{\sigma^2} \sum_{m=M+1}^{\infty} \left(\frac{e}{\theta}\right)^m m^{-m-\frac{d}{2}} N_{d+1,m} \tag{214}$$

$$\leq \left[1+(d+1)e^{d-1}M^d\right] \ln\left(1+\frac{T}{\sigma^2}\right) + \frac{C_{d,\theta} T}{\sigma^2}(d+1)e^{d-1} \sum_{m=M+1}^{\infty} \left(\frac{e}{\theta m}\right)^m m^{\frac{d}{2}-1}, \tag{215}$$

where the first and last inequalities follow from Lemmas 9, 15 and Lemma 16, respectively. Here, for any $d \in \mathbb{N}_+$ and $m \in \mathbb{N}_+$, we have $m^{\frac{d}{2}-1} \leq c_d^m$ with $c_d = \max\left\{1, \exp\left(\frac{1}{e}\left(\frac{d}{2}-1\right)\right)\right\}$. Indeed, when

$d \leq 2$, we have $m^{d/2-1} \leq 1 = c_d$. When $d \geq 3$, the function $g(m) = m^{\frac{1}{m}\left(\frac{d}{2}-1\right)}$ attains maximum at $m = e$ on $[1, \infty)$, which implies $g(m) \leq \exp\left(\frac{1}{e}\left(\frac{d}{2}-1\right)\right) \Rightarrow m^{d/2-1} \leq \exp\left(\frac{1}{e}\left(\frac{d}{2}-1\right)\right)^m \leq c_d^m$. Hence, we have

$$\gamma_T(\mathcal{X}) \leq \left[1 + (d+1)e^{d-1}M^d\right] \ln\left(1 + \frac{T}{\sigma^2}\right) + \frac{C_{d,\theta}T}{\sigma^2}(d+1)e^{d-1} \sum_{m=M+1}^{\infty} \left(\frac{ec_d}{\theta m}\right)^m. \tag{216}$$

In the remaining proof, we consider the upper bound of $\left(\frac{ec_d}{\theta m}\right)^m$ separately based on $\theta > 0$. If $\theta \leq e^2 c_d$, we have

$$\left(\frac{ec_d}{\theta m}\right)^m = \exp\left(-m \ln\left(\frac{\theta m}{ec_d}\right)\right) \leq \exp\left(-m\right) \tag{217}$$

for any m such that $\frac{\theta m}{ec_d} \geq e \Leftrightarrow m \geq \frac{e^2 c_d}{\theta}$. Then, by noting that the condition $T/(e-1) \geq \sigma^2$ implies $\ln\left(1 + \frac{T}{\sigma^2}\right) \geq 1$, we have the following inequalities by setting $M = \left\lfloor \frac{e^2 c_d}{\theta} \ln\left(1 + \frac{T}{\sigma^2}\right) \right\rfloor$:

$$\begin{aligned}
\sum_{m=M+1}^{\infty} \left(\frac{ec_d}{\theta m}\right)^m &\leq \sum_{m=M+1}^{\infty} \exp(-m) && (218)\\
&\leq \int_M^{\infty} \exp(-m)\mathrm{d}m && (219)\\
&\leq \exp(-M) && (220)\\
&\leq \exp\left(-\frac{e^2 c_d}{\theta} \ln\left(1 + \frac{T}{\sigma^2}\right) + 1\right) && (221)\\
&\leq e\left(1 + \frac{T}{\sigma^2}\right)^{-\frac{e^2 c_d}{\theta}} && (222)\\
&\leq e\left(1 + \frac{T}{\sigma^2}\right)^{-1} && (223)\\
&\leq e\frac{\sigma^2}{T}. && (224)
\end{aligned}$$

Therefore, for $\theta \leq e^2 c_d$, the following inequality holds from Eqs. (216) and (224), and the definition of M:

$$\gamma_T(\mathcal{X}) \leq \left[1 + (d+1)e^{d-1}\left(\frac{e^2 c_d}{\theta} \ln\left(1 + \frac{T}{\sigma^2}\right)\right)^d\right] \ln\left(1 + \frac{T}{\sigma^2}\right) + eC_{d,\theta}(d+1)e^{d-1}. \tag{225}$$

Next, if $\theta > e^2 c_d$, we have

$$\left(\frac{ec_d}{\theta m}\right)^m = \exp\left(-m \ln\left(\frac{\theta m}{ec_d}\right)\right) \leq \exp\left(-m \ln\left(\frac{\theta}{ec_d}\right)\right) \tag{226}$$

for any $m \in \mathbb{N}_+$. Then, similarly to the proof under $\theta \leq e^2 c_d$, we have the following inequalities for any $M \in \mathbb{N}$:

$$\begin{aligned}
\sum_{m=M+1}^{\infty} \left(\frac{ec_d}{\theta m}\right)^m &\leq \sum_{m=M+1}^{\infty} \exp\left(-m \ln\left(\frac{\theta}{ec_d}\right)\right) && (227)\\
&\leq \int_M^{\infty} \exp\left(-m \ln\left(\frac{\theta}{ec_d}\right)\right) \mathrm{d}m && (228)\\
&= \frac{1}{\ln\left(\frac{\theta}{ec_d}\right)} \exp\left(-M \ln\left(\frac{\theta}{ec_d}\right)\right) && (229)\\
&< \exp\left(-M \ln\left(\frac{\theta}{ec_d}\right)\right), && (230)
\end{aligned}$$

where the last inequality follows from $\theta/ec_d > e \Leftrightarrow \theta > e^2 c_d$. By setting $M = \left\lceil \frac{1}{\ln\left(\frac{\theta}{ec_d}\right)} \ln\left(1 + \frac{T}{\sigma^2}\right) \right\rceil$, we have

$$\sum_{m=M+1}^{\infty} \left(\frac{ec_d}{\theta m}\right)^m \leq \exp\left(-\ln\left(1 + \frac{T}{\sigma^2}\right)\right) = \left(1 + \frac{T}{\sigma^2}\right)^{-1} \leq \frac{\sigma^2}{T}. \tag{231}$$

Hence, for $\theta > e^2 c_d$, we have

$$\gamma_T(\mathcal{X}) \leq \left[1 + (d+1)e^{d-1}\left(\frac{1}{\ln\left(\frac{\theta}{ec_d}\right)} \ln\left(1 + \frac{T}{\sigma^2}\right) + 1\right)^d\right] \ln\left(1 + \frac{T}{\sigma^2}\right) + C_{d,\theta}(d+1)e^{d-1} \tag{232}$$

$$\leq \left[1 + (d+1)e^{d-1}2^{d-1}\left(\frac{1}{\ln^d\left(\frac{\theta}{ec_d}\right)} \ln^d\left(1 + \frac{T}{\sigma^2}\right) + 1\right)\right] \ln\left(1 + \frac{T}{\sigma^2}\right) + C_{d,\theta}(d+1)e^{d-1}. \tag{233}$$

Finally, combining the bounds in Eqs. (225) and (232) by focusing on the dependence on T, σ^2, and θ, we obtain the desired result. Note that $C_{d,\theta}$ can be further upper bounded by a d-dependent constant due to the condition $\theta > e^2 c_d$.

Matérn kernel. Similarly to the proof for the SE kernel, for any $M \geq 2\nu$, we have

$$\frac{1}{2} \max_{\boldsymbol{x}_1,\ldots,\boldsymbol{x}_T \in \mathbb{S}^d} \ln\det(\boldsymbol{I}_T + \sigma^{-2}\mathbf{K}(\mathbf{X}_T, \mathbf{X}_T)) \tag{234}$$

$$\leq N_M \ln\left(1 + \frac{T}{\sigma^2}\right) + \frac{T}{|\mathbb{S}^d|\sigma^2} \sum_{m=M+1}^{\infty} \lambda_m N_{d+1,m} \tag{235}$$

$$\leq \left[1 + (d+1)e^{d-1}M^d\right] \ln\left(1 + \frac{T}{\sigma^2}\right) + \frac{T(d+1)e^{d-1}}{|\mathbb{S}^d|\sigma^2} \sum_{m=M+1}^{\infty} \lambda_m m^{d-1} \tag{236}$$

$$\leq \left[1 + (d+1)e^{d-1}M^d\right] \ln\left(1 + \frac{T}{\sigma^2}\right) + \frac{T(d+1)\widetilde{C}_{d,\nu}e^{d-1}}{|\mathbb{S}^d|\sigma^2\ell^{2\nu}} \sum_{m=M+1}^{\infty} m^{-2\nu-1} \tag{237}$$

$$= \left[1 + (d+1)e^{d-1}M^d\right] \ln\left(1 + \frac{T}{\sigma^2}\right) + \frac{T\overline{C}_{d,\nu}}{\sigma^2\ell^{2\nu}} \sum_{m=M+1}^{\infty} m^{-2\nu-1}, \tag{238}$$

where the second inequality follows from Lemma 16, and the third inequality follows from $M \geq 2\nu$ and Lemma 19. In the last equation, we set $\overline{C}_{d,\nu} = \frac{(d+1)\widetilde{C}_{d,\nu}e^{d-1}}{|\mathbb{S}^d|}$. Furthermore,

$$\sum_{m=M+1}^{\infty} m^{-2\nu-1} \leq \int_M^{\infty} m^{-2\nu-1}\mathrm{d}m = \frac{M^{-2\nu}}{2\nu}. \tag{239}$$

By balancing $M^d \ln(1 + T/\sigma^2)$ and $\frac{TM^{-2\nu}}{\sigma^2\ell^{2\nu}}$ under the condition $M \geq 2\nu$, we set $M = \left\lceil \max\left\{2\nu, \left\lceil \frac{T}{\sigma^2\ell^{2\nu}} \ln^{-1}\left(1 + \frac{T}{\sigma^2}\right)\right\rceil^{1/(2\nu+d)}\right\}\right\rceil$. Then,

$$\frac{1}{2} \max_{\boldsymbol{x}_1,\ldots,\boldsymbol{x}_T \in \mathbb{S}^d} \ln\det(\boldsymbol{I}_T + \sigma^{-2}\mathbf{K}(\mathbf{X}_T, \mathbf{X}_T)) \tag{240}$$

$$\leq \left[1 + (d+1)e^{d-1}M^d\right] \ln\left(1 + \frac{T}{\sigma^2}\right) + \frac{T\overline{C}_{d,\nu}}{\sigma^2\ell^{2\nu}} \frac{M^{-2\nu}}{2\nu} \tag{241}$$

$$\leq \left[1 + (d+1)e^{d-1}M^d\right] \ln\left(1 + \frac{T}{\sigma^2}\right) + \frac{\overline{C}_{d,\nu}}{2\nu} M^d \ln\left(1 + \frac{T}{\sigma^2}\right) \tag{242}$$

$$= \ln\left(1 + \frac{T}{\sigma^2}\right) + \left[\frac{\overline{C}_{d,\nu}}{2\nu} + (d+1)e^{d-1}\right] M^d \ln\left(1 + \frac{T}{\sigma^2}\right) \tag{243}$$

$$\leq \ln\left(1 + \frac{T}{\sigma^2}\right) + C'_{d,\nu}\left((2\nu)^d + \left(\frac{T}{\sigma^2\ell^{2\nu}}\right)^{\frac{d}{2\nu+d}} \ln^{-\frac{d}{2\nu+d}}\left(1 + \frac{T}{\sigma^2}\right)\right) \ln\left(1 + \frac{T}{\sigma^2}\right) \tag{244}$$

$$= \left(C'_{d,\nu}(2\nu)^d + 1\right) \ln\left(1 + \frac{T}{\sigma^2}\right) + C'_{d,\nu}\left(\frac{T}{\sigma^2\ell^{2\nu}}\right)^{\frac{d}{2\nu+d}} \ln^{\frac{2\nu}{2\nu+d}}\left(1 + \frac{T}{\sigma^2}\right), \tag{245}$$

where we set $C'_{d,\nu} = 2^d(\frac{\overline{C}_{d,\nu}}{2\nu} + (d+1)e^{d-1})$. In the above equations, the second inequality follows from

$$M^d \ln(1 + T/\sigma^2) \geq \frac{TM^{-2\nu}}{\sigma^2\ell^{2\nu}} \Leftrightarrow \frac{\sigma^2\ell^{2\nu}}{T} \ln(1 + T/\sigma^2) \geq M^{-2\nu-d} \tag{246}$$

$$\Leftrightarrow \left[\frac{\sigma^2\ell^{2\nu}}{T} \ln(1 + T/\sigma^2)\right]^{-\frac{1}{2\nu+d}} \leq M \tag{247}$$

$$\Leftrightarrow \left[\frac{T}{\sigma^2\ell^{2\nu}} \ln^{-1}(1 + T/\sigma^2)\right]^{\frac{1}{2\nu+d}} \leq M. \tag{248}$$

Finally, combining Eq. (245) with Lemma 9, we obtain the desired result [13]. □

C Auxiliary Lemmas

Lemma 20 (Sub-optimality gap and the neighborhood around the maximizer)**.** *Suppose f is continuous. Then, under conditions 1 and 3 in Lemma 2, $\boldsymbol{x} \in \mathcal{B}_2(\rho_{\mathrm{quad}}; \boldsymbol{x}^*)$ holds for any $\boldsymbol{x} \in \mathcal{X}$ such that $f(\boldsymbol{x}^*) - f(\boldsymbol{x}) \leq \varepsilon$ with $\varepsilon = \min\{c_{\mathrm{gap}}, c_{\mathrm{quad}}\rho_{\mathrm{quad}}^2\}$.*

Proof. When $\mathcal{B}_2(\rho_{\mathrm{quad}}; \boldsymbol{x}^*) = \mathcal{X}$, the statement is trivial. Hereafter, we assume $\mathcal{B}_2(\rho_{\mathrm{quad}}; \boldsymbol{x}^*) \neq \mathcal{X}$. Here, note that $f(\boldsymbol{x}^*) - f(\widetilde{\boldsymbol{x}}) \geq c_{\mathrm{quad}}\rho_{\mathrm{quad}}^2$ holds for any $\widetilde{\boldsymbol{x}} \in \mathcal{B}_2^b(\rho_{\mathrm{quad}}; \boldsymbol{x}^*)$ from condition 3 in Lemma 2, where $\mathcal{B}_2^b(\rho_{\mathrm{quad}}; \boldsymbol{x}^*) = \{\boldsymbol{x} \in \mathcal{X} \mid \|\boldsymbol{x} - \boldsymbol{x}^*\|_2 = \rho_{\mathrm{quad}}\}$. Furthermore, from the continuity of f and the compactness of $(\mathcal{X} \setminus \mathcal{B}_2(\rho_{\mathrm{quad}}; \boldsymbol{x}^*)) \cup \mathcal{B}_2^b(\rho_{\mathrm{quad}}; \boldsymbol{x}^*)$, there exists $\widetilde{\boldsymbol{x}}_* \in \operatorname{argmax}_{\boldsymbol{x} \in (\mathcal{X} \setminus \mathcal{B}_2(\rho_{\mathrm{quad}}; \boldsymbol{x}^*)) \cup \mathcal{B}_2^b(\rho_{\mathrm{quad}}; \boldsymbol{x}^*)} f(\boldsymbol{x})$. Then, we consider the following two cases separately.

- When $c_{\mathrm{gap}} \geq c_{\mathrm{quad}}\rho_{\mathrm{quad}}^2$, $\varepsilon = c_{\mathrm{quad}}\rho_{\mathrm{quad}}^2$ holds. If there exists $\boldsymbol{x} \in \mathcal{X} \setminus \mathcal{B}_2(\rho_{\mathrm{quad}}; \boldsymbol{x}^*)$ such that $f(\boldsymbol{x}^*) - f(\boldsymbol{x}) \leq \varepsilon = c_{\mathrm{quad}}\rho_{\mathrm{quad}}^2$, we can choose $\widetilde{\boldsymbol{x}}_*$ such that $\widetilde{\boldsymbol{x}}_* \in \mathcal{X} \setminus \mathcal{B}_2(\rho_{\mathrm{quad}}; \boldsymbol{x}^*)$ since $f(\boldsymbol{x}^*) - f(\widetilde{\boldsymbol{x}}) \geq c_{\mathrm{quad}}\rho_{\mathrm{quad}}^2$ holds for any $\widetilde{\boldsymbol{x}} \in \mathcal{B}_2^b(\rho_{\mathrm{quad}}; \boldsymbol{x}^*)$. Furthermore, such $\widetilde{\boldsymbol{x}}_*$ is the local maximizer on $\mathcal{X}$, which satisfies $f(\boldsymbol{x}^*) - f(\widetilde{\boldsymbol{x}}_*) \leq f(\boldsymbol{x}^*) - f(\boldsymbol{x}) \leq \varepsilon \leq c_{\mathrm{gap}}$. This contradicts condition 1 in Lemma 2.

[13] Note that we need to adjust the noise variance parameter by a factor $1/2$ from Lemma 9.

- When $c_{\mathrm{gap}} < c_{\mathrm{quad}}\rho_{\mathrm{quad}}^2$, $\varepsilon = c_{\mathrm{gap}}$ holds. If there exists $\boldsymbol{x} \in \mathcal{X} \setminus \mathcal{B}_2(\rho_{\mathrm{quad}}; \boldsymbol{x}^*)$ such that $f(\boldsymbol{x}^*) - f(\boldsymbol{x}) \leq \varepsilon = c_{\mathrm{gap}}$, we can choose $\widetilde{\boldsymbol{x}}_*$ such that $\widetilde{\boldsymbol{x}}_* \in \mathcal{X} \setminus \mathcal{B}_2(\rho_{\mathrm{quad}}; \boldsymbol{x}^*)$ since $f(\boldsymbol{x}^*) - f(\widetilde{\boldsymbol{x}}) \geq c_{\mathrm{quad}}\rho_{\mathrm{quad}}^2 > c_{\mathrm{gap}}$ holds for any $\widetilde{\boldsymbol{x}} \in \mathcal{B}_2^b(\rho_{\mathrm{quad}}; \boldsymbol{x}^*)$. Furthermore, such $\widetilde{\boldsymbol{x}}_*$ is the local maximizer on $\mathcal{X}$, which satisfies $f(\boldsymbol{x}^*) - f(\widetilde{\boldsymbol{x}}_*) \leq f(\boldsymbol{x}^*) - f(\boldsymbol{x}) \leq \varepsilon = c_{\mathrm{gap}}$. This contradicts condition 1 in Lemma 2.

From the above two arguments, we have $\forall \boldsymbol{x} \in \mathcal{X} \setminus \mathcal{B}_2(\rho_{\mathrm{quad}}; \boldsymbol{x}^*)$, $f(\boldsymbol{x}^*) - f(\boldsymbol{x}) > \varepsilon$. This implies that it is necessary to satisfy $\boldsymbol{x} \in \mathcal{B}_2(\rho_{\mathrm{quad}}; \boldsymbol{x}^*)$ under $f(\boldsymbol{x}^*) - f(\boldsymbol{x}) \leq \varepsilon$. □

Lemma 21 (Upper bound of regret of GP-UCB for any index subset)**.** *Fix any index set $\mathcal{T} \subset [T]$. Then, when running GP-UCB, we have the following inequality under $\mathcal{A}$:*

$$\sum_{t \in \mathcal{T}} f(\boldsymbol{x}^*) - f(\boldsymbol{x}_t) \leq 2\sqrt{C\beta_T |\mathcal{T}| I(\mathbf{X}_{\mathcal{T}})} + \frac{\pi^2}{6} \leq 2\sqrt{C\beta_T |\mathcal{T}| \gamma_{|\mathcal{T}|}(\mathcal{X})} + \frac{\pi^2}{6}, \tag{249}$$

where $C = 2/\ln(1 + \sigma^{-2})$ and $\mathbf{X}_{\mathcal{T}} = (\boldsymbol{x}_t)_{t \in \mathcal{T}}$.

Proof. By following the proof strategy of GP-UCB, we have

$$\sum_{t \in \mathcal{T}} f(\boldsymbol{x}^*) - f(\boldsymbol{x}_t) \leq \sum_{t \in \mathcal{T}} \frac{1}{t^2} + \sum_{t \in \mathcal{T}} f([\boldsymbol{x}^*]_t) - f(\boldsymbol{x}_t) \leq 2\beta_T^{1/2} \sum_{t \in \mathcal{T}} \sigma(\boldsymbol{x}_t; \mathbf{X}_{t-1}) + \frac{\pi^2}{6} \tag{250}$$

due to event $\mathcal{A}$. Here, we define a new input sequence $(\widetilde{\boldsymbol{x}}_t)_{t \leq |\mathcal{T}|}$ as $\widetilde{\boldsymbol{x}}_t = \boldsymbol{x}_{j_t}$, where j_t is the t-th element of $\mathcal{T}$ in increasing order. Furthermore, we define $\widetilde{\mathbf{X}}_t = (\widetilde{\boldsymbol{x}}_1, \ldots, \widetilde{\boldsymbol{x}}_t)$. Then, since $\widetilde{\mathbf{X}}_{t-1}$ is a subsequence of $\mathbf{X}_{j_t - 1}$, by the monotonicity of the posterior variance against the input data, we have

$$\begin{aligned}
\sum_{t \in \mathcal{T}} \sigma(\boldsymbol{x}_t; \mathbf{X}_{t-1}) &= \sum_{t=1}^{|\mathcal{T}|} \sigma(\widetilde{\boldsymbol{x}}_t; \mathbf{X}_{j_t - 1}) && (251)\\
&\leq \sum_{t=1}^{|\mathcal{T}|} \sigma(\widetilde{\boldsymbol{x}}_t; \widetilde{\mathbf{X}}_{t-1}) && (252)\\
&\leq \sqrt{C|\mathcal{T}| I(\widetilde{\mathbf{X}}_{|\mathcal{T}|})} && (253)\\
&= \sqrt{C|\mathcal{T}| I(\mathbf{X}_{\mathcal{T}})} && (254)\\
&\leq \sqrt{C|\mathcal{T}| \gamma_{|\mathcal{T}|}(\mathcal{X})}, && (255)
\end{aligned}$$

where the second inequality follows from Lemmas 5.3 and 5.4 in [53]. □

D Numerical Simulation for Information Gain

In addition to the simple example provided in Figure 1, we empirically confirm the gap between the worst-case (MVR) and GP-UCB's information gain under the Bayesian assumption. In Figures 2 and 3, we report the mean and quantiles, respectively, of the realized information gain of GP-UCB over 20 different sample paths generated by changing the random seed. We conduct experiments under the same settings as in Figure 1 of the main text. We also report the information gain corresponding to the sequence of maximum variance reduction (MVR), following the same setup as Figure 1. In all cases, consistent with Figure 1 in the main text, we observe a noticeable gap in information gain between GP-UCB and MVR.

E Additional Remarks about Local Quadratic Condition in Lemma 2

The local quadratic condition of the GP-sample path (property 3 in Lemma 2) is claimed in [14] and is used in the following works [47, 50]. Since de Freitas et al. [14] provide only a concise discussion, a more detailed verification is desired. In this section, we provide the detailed, general, and rigorous proof of the local quadratic condition, which is applicable beyond the SE and Matérn kernels.

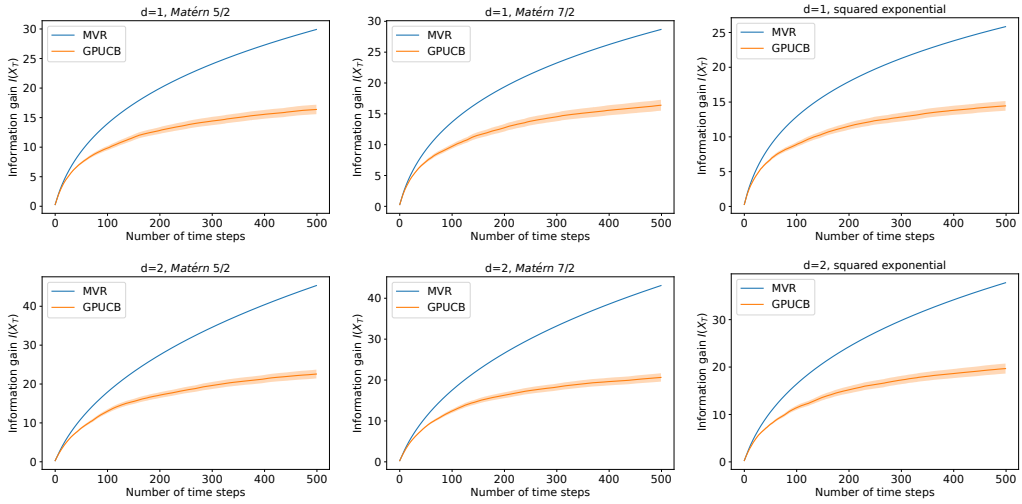


Figure 2: Average results of information gain under 20 different sample paths. The top row shows the results for the input space $\mathcal{X} = [0, 1]$ with lengthscale parameter $\ell = 0.1$. The bottom row corresponds to the input space $\mathcal{X} = [0, 1]^2$ with lengthscale $\ell = 0.25$, and from left to right, we use the Matérn 5/2 kernel, the Matérn 7/2 kernel, and the SE kernel. The shaded regions indicate one standard error.

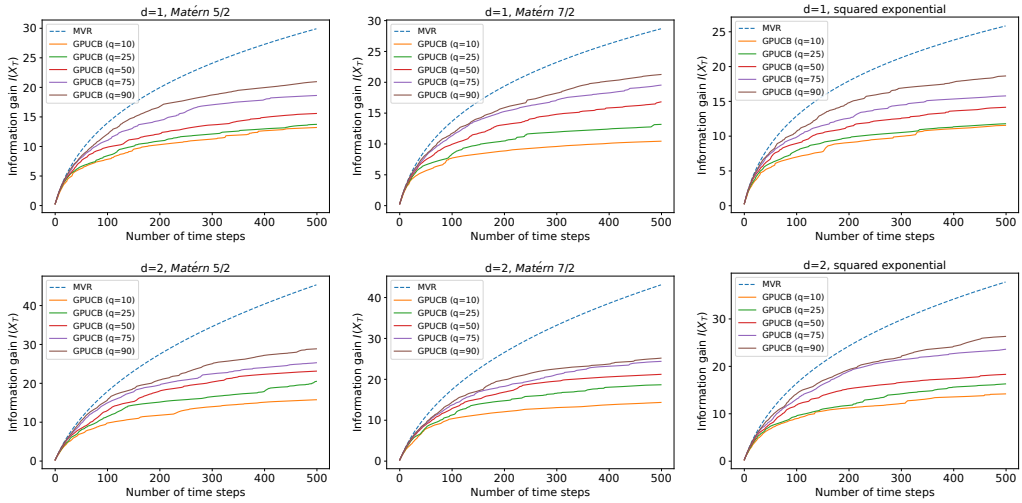


Figure 3: Quantiles of information gain over 20 different sample paths. We report the 10%, 25%, 50%, 75%, and 90% quantiles of the information gain of GP-UCB.

E.1 Proof for Almost-Sure Statement

Firstly, we show the following almost-sure version of the local quadratic condition for a general kernel.

Lemma 22. *Fix $\mathcal{X} \coloneqq [0, r]^d$. Suppose that Assumption 1 holds. Furthermore, assume that the positive definite kernel function $k : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}$ satisfies the following conditions:*

1. *The function $k(\cdot, \cdot)$ has mixed partial derivatives up to fourth order on $\mathcal{X}$.*
2. *$f \sim \mathcal{GP}(0, k)$ is almost-surely C^2 on $\mathcal{X}$.*
3. *For any fixed $\boldsymbol{x} \in \mathcal{X}$, the Hessian matrix $\nabla\nabla^\top f(\boldsymbol{x})$ is non-degenerate almost surely.*

4. *For any $\boldsymbol{x} \in \mathcal{X}$, the probability density function of the gradient $\nabla f(\boldsymbol{x})$ is bounded around some neighborhood of $\mathbf{0}$.*

5. *The kernel k is stationary. Namely, $k(\boldsymbol{x}, \boldsymbol{z}) = \Phi(\boldsymbol{x} - \boldsymbol{z})$ holds for some function $\Phi : \mathbb{R}^d \to \mathbb{R}$.*

Then, with probability 1*, there exist constants $c, \rho > 0$ such that $f(\boldsymbol{x}^*) - c\|\boldsymbol{x}^* - \boldsymbol{x}\|_2^2 \geq f(\boldsymbol{x})$ for all $\boldsymbol{x} \in \mathcal{B}_2(\rho; \boldsymbol{x}^*)$. Here, $\nabla \coloneqq \left(\frac{\partial}{\partial x_1}, \ldots, \frac{\partial}{\partial x_d}\right)^\top$ represents the gradient operator.*

We can easily confirm that the SE kernel and Matérn kernel with $\nu > 2$ satisfy the conditions in the above lemma (see the descriptions following Lemma 2 in Section 2).

Proof. We use Proposition 6.5 in [2], which provides a sufficient condition for guaranteeing that any points with zero gradient have a non-degenerate Hessian almost surely. Specifically, by applying Proposition 6.5 of [2] to the gradient field, we confirm that conditions 1, 2, and 4 imply the conditions that are necessary for applying Proposition 6.5 in [2]. Thus, we can confirm that, if $\sup_{\boldsymbol{x} \in \mathcal{X}, \boldsymbol{g} \in N(\mathbf{0})} \mathbb{P}(|\det(\nabla\nabla^\top f(\boldsymbol{x}))| < \zeta \mid \nabla f(\boldsymbol{x}) = \boldsymbol{g}) \to 0$ as $\zeta \to 0$, we have

$$\mathbb{P}\left(\exists \boldsymbol{x} \in \mathcal{X},\ \nabla f(\boldsymbol{x}) = \mathbf{0} \text{ and } \det(\nabla\nabla^\top f(\boldsymbol{x})) = 0\right) = 0. \tag{256}$$

Here, $N(\mathbf{0})$ is some neighborhood of $\mathbf{0} \in \mathbb{R}^d$. Since the kernel is stationary, $\nabla f(\boldsymbol{x})$ and $\nabla\nabla^\top f(\boldsymbol{x})$ are independent for any fixed $\boldsymbol{x} \in \mathcal{X}$; therefore, $\mathbb{P}(|\det(\nabla\nabla^\top f(\boldsymbol{x}))| < \zeta \mid \nabla f(\boldsymbol{x}) = \boldsymbol{g}) = \mathbb{P}(|\det(\nabla\nabla^\top f(\boldsymbol{x}))| < \zeta)$. By combining this with condition 3, we can confirm that the condition $\sup_{\boldsymbol{x} \in \mathcal{X}, \boldsymbol{g} \in N(\mathbf{0})} \mathbb{P}(|\det(\nabla\nabla^\top f(\boldsymbol{x}))| < \zeta \mid \nabla f(\boldsymbol{x}) = \boldsymbol{g}) \to 0$ is valid. Therefore, under the conditions of the lemma, we have Eq. (256).

When the maximizer $\boldsymbol{x}^*$ is in the interior. In this case, $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$ and $\nabla\nabla^\top f(\boldsymbol{x}^*) \preceq 0$ hold from the fact that f is C^2. Thus, from Eq. (256), $\nabla\nabla^\top f(\boldsymbol{x}^*) \prec 0$ holds almost surely. Since f is C^2, with probability 1, there exist constants $\rho > 0, c > 0$ such that $\lambda_{\min}(-\nabla\nabla^\top f(\boldsymbol{x})) \geq 2c$ for any $\boldsymbol{x} \in \mathcal{B}_2(\rho; \boldsymbol{x}^*)$. Here, $\lambda_{\min}(A)$ denotes the minimum eigenvalue for a matrix A. By applying Taylor's theorem, for any $\boldsymbol{x} \in \mathcal{B}_2(\rho; \boldsymbol{x}^*)$, we have

$$f(\boldsymbol{x}) = f(\boldsymbol{x}^*) + \langle \nabla f(\boldsymbol{x}^*),\ \boldsymbol{x} - \boldsymbol{x}^* \rangle + \frac{1}{2}(\boldsymbol{x} - \boldsymbol{x}^*)^\top (\nabla\nabla^\top f(\theta\boldsymbol{x} + (1-\theta)\boldsymbol{x}^*))(\boldsymbol{x} - \boldsymbol{x}^*) \tag{257}$$

for some $\theta \in (0, 1)$. Since $\nabla f(\boldsymbol{x}^*) = \mathbf{0}$ and $\lambda_{\min}(-\nabla\nabla^\top f(\theta\boldsymbol{x} + (1-\theta)\boldsymbol{x}^*)) \geq 2c$, Eq. (257) implies

$$f(\boldsymbol{x}) \leq f(\boldsymbol{x}^*) - c\|\boldsymbol{x}^* - \boldsymbol{x}\|_2^2. \tag{258}$$

When the maximizer $\boldsymbol{x}^*$ is on the boundary. To show the case where $\boldsymbol{x}^*$ is on the boundary, we repeatedly apply the same argument for the (relative) interior of each facet of $\mathcal{X}$. Let $\mathcal{X}_1 \coloneqq \{\boldsymbol{x} \in \mathcal{X} \mid x_m = \frac{r}{2}(u+1)\}$ for some index $m \in [d]$ and $u \in \{-1, +1\}$. We prove Eq. (258) when the maximizer is in the relative interior of $\mathcal{X}_1$. Here, by applying the same arguments as those for $\mathcal{X}$, we have

$$\mathbb{P}\left(\exists \boldsymbol{x} \in \mathcal{X}_1,\ \nabla_{(-m)} f(\boldsymbol{x}) = \mathbf{0} \text{ and } \det(\nabla_{(-m)} \nabla_{(-m)}^\top f(\boldsymbol{x})) = 0\right) = 0, \tag{259}$$

where:

$$\nabla_{(-m)} = \left(\frac{\partial}{\partial x_1}, \ldots, \frac{\partial}{\partial x_{m-1}}, \frac{\partial}{\partial x_{m+1}}, \ldots, \frac{\partial}{\partial x_d}\right)^\top. \tag{260}$$

We consider the following two cases separately, depending on the value of $\frac{\partial}{\partial x_m} f(\boldsymbol{x}^*)$:

- If $\frac{\partial}{\partial x_m} f(\boldsymbol{x}^*) \neq 0$, then, we have $u\frac{\partial}{\partial x_m} f(\boldsymbol{x}^*) > 0$ since $\boldsymbol{x}^*$ is the maximizer. Furthermore, since f is C^2, there exist constants $\rho \in (0, 1)$ and $c > 0$ such that both (i)$\lambda_{\min}(-\nabla_{(-m)} \nabla_{(-m)}^\top f(\boldsymbol{x})) \geq 2c$ and (ii)$u\frac{\partial}{\partial x_m} f(\boldsymbol{x}) \geq c$ hold for any $\boldsymbol{x} \in \mathcal{B}_2(\rho; \boldsymbol{x}^*)$

almost surely. Thus, from Taylor's theorem, for any $\boldsymbol{x} \in \mathcal{B}_2(\rho; \boldsymbol{x}^*)$, we have

$$f(\boldsymbol{x}) = f(\boldsymbol{z}) + \sum_{j=1}^{d} (x_j - z_j) \frac{\partial}{\partial x_j} f(\theta_1 \boldsymbol{z} + (1-\theta_1)\boldsymbol{x}) \tag{261}$$

$$= f(\boldsymbol{z}) + (x_m - x_m^*) \frac{\partial}{\partial x_m} f(\theta_1 \boldsymbol{z} + (1-\theta_1)\boldsymbol{x}) \tag{262}$$

$$\leq f(\boldsymbol{z}) - c|x_m - x_m^*| \tag{263}$$

$$\leq f(\boldsymbol{z}) - c(x_m - x_m^*)^2 \tag{264}$$

for some constant $\theta_1 \in (0,1)$. In the above expression, we set $\boldsymbol{z} = (x_1, \ldots, x_{m-1}, x_m^*, x_{m+1}, \ldots, x_d)^\top$. Furthermore, for some constant $\theta_2 \in (0,1)$, we have

$$f(\boldsymbol{z}) = f(\boldsymbol{x}^*) + \frac{1}{2}(\boldsymbol{x}_{-m} - \boldsymbol{x}_{-m}^*)^\top (\nabla_{(-m)} \nabla_{(-m)}^\top f(\theta_2 \boldsymbol{z} + (1-\theta_2)\boldsymbol{x}^*))(\boldsymbol{x}_{-m} - \boldsymbol{x}_{-m}^*) \tag{265}$$

$$\leq f(\boldsymbol{x}^*) - c\|\boldsymbol{x}_{-m} - \boldsymbol{x}_{-m}^*\|_2^2, \tag{266}$$

where $\boldsymbol{x}_{-m} = (x_1, \ldots, x_{m-1}, x_{m+1}, \ldots, x_d)^\top \in \mathbb{R}^{d-1}$ and $\boldsymbol{x}_{-m}^* = (x_1^*, \ldots, x_{m-1}^*, x_{m+1}^*, \ldots, x_d^*)^\top \in \mathbb{R}^{d-1}$. By combining Eq. (261) with Eq. (265), we obtain Eq. (258).

- If $\frac{\partial}{\partial x_m} f(\boldsymbol{x}^*) = 0$, we have $\nabla f(\boldsymbol{x}^*) = \boldsymbol{0}$. In this case, Eq. (258) clearly holds by applying the same arguments for the case where $\boldsymbol{x}^*$ is in the interior of $\mathcal{X}$.

In summary, if $\boldsymbol{x}^*$ is in the relative interior of the input domain with the form $\{\boldsymbol{x} \in \mathcal{X} \mid x_m = \frac{r}{2}(u+1)\}$, Eq. (258) holds almost surely. By following the same arguments for the input domain with the form $\mathcal{X}_2 \coloneqq \{\boldsymbol{x} \in \mathcal{X} \mid x_{m_1} = \frac{r}{2}(u_1+1), x_{m_2} = \frac{r}{2}(u_2+1)\}$ for some $m_1, m_2 \in [d]$ and $u_1, u_2 \in \{-1, +1\}$, we confirm that Eq. (258) holds almost surely if $\boldsymbol{x}^*$ is in the relative interior of $\mathcal{X}_2$. By repeatedly applying the same arguments for $\mathcal{X}_i \coloneqq \{\boldsymbol{x} \in \mathcal{X} \mid x_{m_1} = \frac{r}{2}(u_1+1), \ldots, x_{m_i} = \frac{r}{2}(u_i+1)\}$ until we reach $\mathcal{X}_d$ (i.e., the corner of $\mathcal{X}$), we conclude that Eq. (258) holds for some constants $c, \rho > 0$ almost surely when $\boldsymbol{x}^*$ is on the boundary of $\mathcal{X}$. □

E.2 From Almost-Sure Statement to High-Probability Statement

We give the following rigorous proof of Lemma 2 by translating an almost-sure statement into a high-probability statement.

Proof of Lemma 2. Let $(\Omega, \mathcal{B}, \mathbb{P})$ be the probability space, where the GP is defined.

Regarding property 1. Since property 1 in Lemma 2 holds almost surely, there exists $B \in \mathcal{B}$ such that $\mathbb{P}(B) = 1$ and[14]

$$\forall w \in B,\ \exists c(w) > 0,\ \forall \widetilde{\boldsymbol{x}}^* \in \mathcal{X}_{\mathrm{local}}(w),\ f(\boldsymbol{x}^*) > f(\widetilde{\boldsymbol{x}}^*) + c(w), \tag{267}$$

where $\mathcal{X}_{\mathrm{local}}(w) \subset \mathcal{X}$ represents the set of all the local (but non-global) maximizers under the realization $w \in \Omega$. Then, we can confirm that the following statement holds[15]:

$$\forall \delta \in (0,1),\ \exists c_\delta > 0,\ \mathbb{P}(B_{c_\delta}) \geq 1 - \delta, \tag{268}$$

where $B_c = \{w \in B \mid c(w) \geq c\}$. Indeed, if the above statement does not hold, for some $\delta \in (0,1)$, we have

$$\forall c > 0, \mathbb{P}(B_c) < 1 - \delta. \tag{269}$$

However, from the continuity of the probability, we have $\lim_{c \downarrow 0} \mathbb{P}(B_c) = 1$, which contradicts $\forall c > 0, \mathbb{P}(B_c) < 1 - \delta$. Therefore, statement (268) holds. From the definition of c_δ, we have

$$\forall w \in B_{c_\delta},\ \forall \widetilde{\boldsymbol{x}}^* \in \mathcal{X}_{\mathrm{local}}(w),\ f(\boldsymbol{x}^*) > f(\widetilde{\boldsymbol{x}}^*) + c_\delta, \tag{270}$$

which implies

$$\mathbb{P}(\forall \widetilde{\boldsymbol{x}}^* \in \mathcal{X}_{\mathrm{local}},\ f(\boldsymbol{x}^*) > f(\widetilde{\boldsymbol{x}}^*) + c_\delta) \geq 1 - \delta. \tag{271}$$

Therefore, by setting $c_{\mathrm{gap}} = c_{\delta_{\mathrm{GP}}/3}$, property 1 in Lemma 2 holds with probability at least $1 - \delta_{\mathrm{GP}}/3$.

[14] We omit the dependence of w in $f(\boldsymbol{x}^*)$, $f(\widetilde{\boldsymbol{x}}^*)$ for notational simplicity.

[15] Strictly speaking, here, we require to verify the condition that the number of local maximizers is finite almost surely. This clearly holds for the SE and Matérn kernels with $\nu > 2$ from the Kac-Rice formula [2].

Regarding property 2. Since property 2 in Lemma 2 holds almost surely, there exists $B \in \mathcal{B}$ such that $\mathbb{P}(B) = 1$ and

$$\forall w \in B,\ \exists c(w) \geq 0,\ \|f\|_\infty \leq c(w). \tag{272}$$

Based on the same argument for the proof regarding property 1, we have

$$\forall \delta \in (0,1),\ \exists c_\delta \geq 0,\ \mathbb{P}(B_{c_\delta}) \geq 1 - \delta, \tag{273}$$

where $B_c = \{w \in B \mid c(w) \leq c\}$. From the definition of c_δ, we have

$$\forall w \in B_{c_\delta},\ \|f\|_\infty \leq c_\delta, \tag{274}$$

which implies

$$\mathbb{P}(\|f\|_\infty \leq c_\delta) \geq 1 - \delta. \tag{275}$$

Therefore, by setting $c_{\sup} = c_{\delta_{\mathrm{GP}}/3}$, property 2 in Lemma 2 holds with probability at least $1 - \delta_{\mathrm{GP}}/3$.

Regarding property 3. Since property 3 in Lemma 2 holds almost surely (as shown in Lemma 22), there exists $B \in \mathcal{B}$ such that $\mathbb{P}(B) = 1$ and

$$\forall w \in B,\ \exists c(w), \rho(w) > 0,\ \forall \boldsymbol{x} \in \mathcal{B}_2(\rho(w); \boldsymbol{x}^*),\ f(\boldsymbol{x}^*) - c(w)\|\boldsymbol{x}^* - \boldsymbol{x}\|_2^2 \geq f(\boldsymbol{x}) \tag{276}$$

Based on the same argument for the proof regarding properties 1 and 2, we have

$$\forall \delta \in (0,1),\ \exists c_\delta, \rho_\delta > 0,\ \mathbb{P}(B_{c_\delta, \rho_\delta}) \geq 1 - \delta, \tag{277}$$

where $B_{c,\rho} = \{w \in B \mid c(w) \geq c, \rho(w) \geq \rho\}$. From the definition of c_δ, ρ_δ, we have

$$\forall w \in B_{c_\delta, \rho_\delta},\ \forall \boldsymbol{x} \in \mathcal{B}_2(\rho_\delta; \boldsymbol{x}^*),\ f(\boldsymbol{x}^*) - c_\delta\|\boldsymbol{x}^* - \boldsymbol{x}\|_2^2 \geq f(\boldsymbol{x}), \tag{278}$$

which implies

$$\mathbb{P}(\forall \boldsymbol{x} \in \mathcal{B}_2(\rho_\delta; \boldsymbol{x}^*),\ f(\boldsymbol{x}^*) - c_\delta\|\boldsymbol{x}^* - \boldsymbol{x}\|_2^2 \geq f(\boldsymbol{x})) \geq 1 - \delta. \tag{279}$$

Therefore, by setting $c_{\mathrm{quad}} = c_{\delta_{\mathrm{GP}}/3}, \rho_{\mathrm{quad}} = \rho_{\delta_{\mathrm{GP}}/3}$, property 3 in Lemma 2 holds with probability at least $1 - \delta_{\mathrm{GP}}/3$.

Finally, by taking the union bound, we obtain the desired result. □